

SEMESTER 3

POLYMER ENGINEERING

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hr. 30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9

3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	9
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3

POLYMER SCIENCE

Course Code	PCPOT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To establish fundamental principles of Polymer Science in a simple and up-front manner and to provide the broad background
2. Introduces students to basic concepts of polymer microstructures, polymer solutions, chemical reactions and properties of polymer materials which acts as the fundamental to the core areas of Polymer Engineering course.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Glass transition temperature - Amorphous state, molecular motion, first order and second order transitions, T _g , T _m , secondary transitions, free volume, kinetic and thermodynamic views of glass transition, factors influencing glass transition temperature, Crystalline state, polymorphism, polymer single crystals, lamellae, spherulites, super molecular structures, fringed micelle model, degree of crystallinity, factors affecting crystallinity, X-ray diffraction	11
2	Polymer solutions- terms and definitions: - types of solutions, Hildebrand approach, Flory-Huggin's theory, thermodynamic view of miscibility, upper critical solution temperature (UCST), lower critical solution temperature (LCST), concentration regimes in polymer solutions, theta conditions	11

3	Optical and Electrical Properties of polymers: - Effect of polymer structure on dielectric constant, capacitance, dielectric loss, power factor, dissipation factor and loss factor, effect of additives on electrical properties of polymers, effect of polymer structure on optical properties, clarity, transparency, haze, transmittance, reflectance, gloss. Chemical reactivity of linear and cross-linked polymers, hydrolysis, acidolysis, aminolysis, hydrogenation, addition and substitution reactions, cross linking reactions, reactivity of functional groups.	11
4	Polymer analogous reactions: - polymer bound reagents, chain end and random degradation, degradation by oxygen, ozone, heat, UV light, microorganism, crazing, weathering, stabilisation to prevent degradation.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of Tg, crystallinity and polymer morphology.	K2
CO2	Estimate the properties of polymer solutions	K6
CO3	Relationship of the structure of polymers with their electrical and optical properties.	K4
CO4	Define the reactions and degradation of polymers	K1
CO5	Develop polymer systems capable of resistance to various degrading agents	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3		1								
CO2	3	3	2									
CO3	3	1			2							
CO4	3	2		1								
CO5	3	2		3								

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Essentials of Polymer Science and Engineering	Paul C. Painter, M.M. Coleman	DES tech Publications	2008
2	Textbook of Polymer Science	F W Billmeyer	John Wiley & Sons	Third Edition, 2007
3	Physical Properties of Polymers Handbook	J. E. Mark (Ed.)	Springer	2007
4	An Introduction to the Mechanical Properties of Solid Polymers.	I. M. Ward, J. Sweeney	John Wiley & Sons	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Properties of Polymers	D.W. V. Krevelen, K. Nijenhuis	Elsevier	2009
2	Introduction to Polymer Science & Technology	Manas Chanda	CRC Press	2 nd Edition, 2013
3	Material Science of Polymers for Engineers	TA Osswald and G Menges	Hanser	2012
4	Electrical Properties of Polymers	D. A. Seanor	Elsevier	2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/noc24_ch50
2	https://archive.nptel.ac.in/noc24_cy37
3	https://archive.nptel.ac.in/noc24_ch50
4	https://archive.nptel.ac.in/noc24_ch55

SEMESTER S3

LATEX TECHNOLOGY

Course Code	PCPOT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in Polymer materials and Polymer science.	Course Type	Theory

Course Objectives:

1. To impart knowledge on processing and product development of polymer latices
2. To familiarize various compounding ingredients

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Natural rubber latex: - General composition, preservation, fundamental latex characteristics, particle size and distribution, stability and destabilisation, coacervation, viscosity-concentration relationship, surface free energy and wetting behaviour, zeta potential, electrical properties of colloidal system, thermal movement of molecules, Brownian motion, synthetic lattices and their blends, latex concentration methods, concentrated latex, significance of specification limits.	11
2	Test methods:- total solids, dry rubber content, total alkalinity, coagulum content, sludge content, pH, KOH number, mechanical, chemical and thermal stability, VFA number, zinc oxide stability principles of latex compounding, de-ammoniation of latex, vulcanising agents, accelerators, antioxidants, fillers, dispersing and emulsifying agents, stabilisers, thickening agents, and other miscellaneous additives, special ingredients, preparation of dispersions and emulsions, latex compounding, pre-vulcanised latex.	11

3	Dipping methods- straight dipping, coagulant dipping, different types of formers, dipping process, after treatments, defects in dipped goods, Manufacture of dipped goods like rubber band, medical gloves, household gloves, industrial gloves, dipped fabric gloves, balloon, nipples, prophylactics, impregnation, spreading, fabric proofing and coating. Rubber-textile composite products, latex bonded fibrous structures, coir foam, latex treated rugs and carpet backing.	11
4	Latex foam processing methods: - compounding, mechanical frothing by beating, vulcanisation, washing and drying, gelling, gelling systems, merits and demerits of gelling systems, continuous foam production, typical latex compounds for foam production. Latex castings and mouldings- principles and production of hollow articles, solid articles, use of porous moulds in casting, manufacture of latex thread, latex cement and adhesives, latex paints, protective coatings, chewing gum, use of latex in road rubberisation	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the principles, instrumentation and applications of latex concentration.	K5
CO2	List the significance and methods of latex specification tests and various grades of NR latex.	K1
CO3	Identify the different latex product manufacture techniques and formulate compounds for the same.	K3
CO4	Compare and contrast between various latex product manufacture methods and the significant steps involved in each.	K4
CO5	Discuss the significance safety aspects in latex processing. Also, how production and processing of NR latex affects the economy of Kerala	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		1				2			
CO2	3	3	2		1							
CO3	3	3	2	2					2			
CO4	3	3	1	1								
CO5	2	2	1	2		1						

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer Latices: Science and Technology, Vol 1, 2 and 3	D.C. Blackley,	Springer Science	1997
2	Natural Rubber Latex Technology,	N.R.Peethambaran,		2016
3	Practical Guide to Latex Technology,	Rani Joseph	Smithers Rapra	2013
4	Vanderbilt Latex Handbook	Mausser	R.T.Vanderbilt Co. Inc., U.S.A	3rd edition,

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Latex Foam Rubber	E. W. Madge	Elsevier	1982
2	The Vanderbilt Latex Handbook,	R. F.Mausser	R.T. Vanderbilt Company	1987
3	Journal of Rubber Developments, Dipped Goods, Vol. 25			1972
4	High Polymer Lattices, Vol 1 and 2	D.C. Blackley,	Maclaren	1966

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/p5pk4Um6lsk
2	https://youtu.be/TMOqFXBMdVM?si=PfsDe6JBvq9riYJe
3	https://youtu.be/eo3K56MqELQ?si=V3pm3Jk9I7qOcrBh
4	https://youtu.be/f1V4POGaAsc?si=-ZUes7vpzrK2Efa0

SEMESTER S3

BIOMEDICAL & BIOPOLYMERS

Course Code	PBPOT304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Objective of this course is to explain the importance and applications of biopolymers and biomedical materials with its research and development.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Biomaterials – Definition, requirements, Classification, Bioactive, Bio inert and Bioresorbable biomaterial, sterilisation methods of biomaterials. Typical examples for polymers used as biomaterials: ultrahigh molecular weight polyethylene, poly (methyl methacrylate), hydrogels, silicone, polyurethanes, polylactic acid, polyglycolic acid and its copolymers, polycaprolactone, chitin and chitosan.	9
2	Biomedical applications of polymers- orthopaedics, cardio vascular, silicone implants, polymer membranes, polymer skin, contact lens types and materials, biodegradable polymers and its applications, polymeric tissue engineering scaffolds.	9
3	Dental applications: - denture base, crown and bridge resins, plastic teeth, mouth protectors, restorative materials, dental adhesives, dental impression and duplicating materials.	9
4	Chemistry of peptides , polypeptides and proteins, structural organisation in proteins. Nucleic acids, RNA, DNA, chemistry of polysaccharides, starch and cellulose, chemical modifications of cellulose, regenerated cellulose	9

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	- Each question carries 6 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain different types of biopolymer materials and their properties	K2
CO2	Analyze biomedical applications of polymers	K4
CO3	Analyze the use of biomaterial as a dental material	K4
CO4	Explain chemistry of biomaterials like protein, nucleic acids	K5
CO5	Explain chemistry of polysaccharides	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2			2						1
CO2	2	1	2			1	1					1
CO3	1	1	1	1	1	1	1					1
CO4	1		1	1	2							1
CO5	1		1	1	2							1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Biomaterials: An Introduction	Joon Park, R. S. Lakes	Springer Science, New York	3rd Edn, 2007
2	Biomaterials Science: An Introduction to Materials Science	Buddy D. Ratner, Allan S. Hoffman, Frederick J. Schoen, Jack E. Lemons	Elsevier Academic Press	3rd Edn, 2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Encyclopedia of polymer science and engineering	H.F. Mark (Ed)	John Wiley and Sons	1989
2	Soft compact Lenses clinical and applied Technology	Galini and M. Ruben Ed.	John Wiley and Sons	1978
3	Principles of protein structure	Shulz and Bhirmer	Academic Press	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/3CnNEiUrhZw?si=AfmuquqDHES35KlE
2	https://www.youtube.com/live/9dQRrMoykg8?si=gTa_0HLzaznqdVmc
3	https://youtu.be/g-FzzdfDDLs
4	https://youtu.be/COyTQmNleAM?si=TvK7vA92L_RRUsAk

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11
2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value	11

	Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	
3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis-linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	11
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								
CO5	3	3	3	3								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2 nd edition, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpcglclefndmkaj/https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. <table><tr><td>Year</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Cash flow</td><td>1000</td><td>900</td><td>800</td><td>700</td><td>600</td></tr></table> (In Crores)	Year	1	2	3	4	5	Cash flow	1000	900	800	700	600	CO4	(5)
	Year	1	2	3	4	5										
Cash flow	1000	900	800	700	600											
b)	Point out any three merits of NPV method.	CO4	(3)													

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered	6

	technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and	6

	regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12

3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks			50	

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.

- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

CHEMISTRY LAB

Course Code	PCPOL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Mins
Prerequisites (if any)	Basic Engineering Chemistry	Course Type	Lab

Course Objectives:

1. The chemistry lab is structured to teach skills in organic synthesis and analytical techniques.

Expt. No.	Experiments
A	Organic Synthesis:
1	Synthesis of ethyl n- butyl acetoacetate by the acetoacetic ester condensation
2	Synthesis of 3- 3-nitrobenzoic acid from benzoic acid
3	Nitration of aromatic hydrocarbons.
4	Side-chain oxidation of aromatic hydrocarbons.
5	Benzoylation of phenols.
6	Preparation of solid esters.
7	Bromination of amines
B.	Purification and characterization of organic compounds
	Purification of organic compounds by fractionation, crystallization, and chromatography.
C.	Separation of the components of a binary organic mixture
	Liquid-liquid, liquid-solid, and solid-solid
D.	Identify the components of the given binary mixture

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.

Endorsement by External Examiner: The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Examine and evaluate parameters for creating organic materials	K4
CO2	Utilize the analytical and chemical techniques to purify organic compounds.	K3
CO3	Develop precision in assessing organic compounds and mixtures	K6
CO4	Evaluate the characteristic features of organic mixtures	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3										
CO3	2			2					1			
CO4		2							1			

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Organic Chemistry: A Practical Guide	Michael C. Pirrung	Wiley	2016
2	Organic Chemistry Laboratory Manual	John W. Lehman	Prentice Hall	4 th edition, 2002
3	Vogel's Textbook of Practical Organic Chemistry	A.I. Vogel, B.S. Furniss, A. J. Hannaford, P.W.G. Smith, A. R. Tatchell	Longman Scientific & Technical	5 th edition, 1989
4	Organic Chemistry Laboratory Manual	R.K. Harris, J.R. Everett	Pergamon Press	1988

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Organic Chemistry: a Laboratory Manual	S. Mumtazuddin	Alpha Science International	2013
2	Practical Chemistry	A.O. Thomas	Scientific Book Centre	2003
3	Qualitative Organic Chemical Analysis	W.J. Criddle	Springer	2013
4	Comprehensive Practical Organic Chemistry: Qualitative Analysis	Universities Press	V.K. Ahluwalia, S. Dhingra	2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/KfWnDGDWWVI?si=0ShDF17241GMzfCb
2	https://onlinecourses.nptel.ac.in/noc22_cy42/preview
3	https://youtu.be/OPXs0u40GCY?si=C0YhuE6OzFfRjKW
4	https://youtu.be/99-LEzfCn5I?si=l5JoRVPj73l_zgS_

Continuous Assessment (25 Marks)

Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

COMPUTER AIDED DRAFTING LAB

Course Code	PCPOL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GCEST203	Course Type	Lab

Course Objectives:

1. Provide practical experience in using AutoCAD software for 2D and 3D drawing
2. Equip students with the skills necessary to create and modify 2D drawing for industrial applications
3. Establish a solid foundation in 2D drawing to facilitate the transition to AutoCAD 3D
4. Prepare students to master similar graphics software, enhancing their versatility and adaptability in various design and engineering environments.

Expt. No.	Experiments
1	Introduction to Computer Aided Drafting: Applications-Advantages, -Hardware & Software Requirements-Different drafting software
2	Familiarize the AutoCAD Basics: Main menu, starting a new drawing, open, save, exit etc. Limits of drawing, Units, Coordinates-absolute, relative, polar coordinate system, Drafting setting-grid, snap, osnap, ortho mode etc.
3	Drawing commands: lines, circle,arc,ellipse,rectangle,polygon, spline, polyline, helix, donut, block, table, hatch etc.
4	Modify commands: trim, copy, rotate, mirror, stretch, scale, trim, extend, fillet, chamfer, array, erase, offset, OOPS, explode etc
5	Practice Format tools: layer, colour, line type, line weight, text style
6	Perform Dimensioning: Linear, Angular dimension style
7	Draw orthographic views: Hexagonal bolt, dumb bell, angular tear etc.
8	Isometric drawing tools: isometric snap, isocircle, direction changing
9	Convert .dwg file into pdf/jpg format and take print out

10	Comprehend opening a various commands and operations required in AutoCAD for drawing 3D objects
11	Drawing simple figures to practice 3D drawing
12	Design and draft various polymer products

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Tota
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.

Endorsement by External Examiner: The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Build proficiency in computer aided drafting and various commands used in AutoCAD	K3
CO2	Develop 2D drawing, sectional views and dimensioning in AutoCAD	K6
CO3	Develop Isometric drawing of objects using AutoCAD	K6
CO4	Identify and use various commands and operations for drawing 3D objects in AutoCAD	K3
CO5	Design and draft various polymer products using AutoCAD 3D tools	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2			3									
CO3	3											
CO4			3									
CO5	3											

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	AutoCAD2024 For Dummies	Bill Fane and David Bymes		
2	AutoCAD Tutorial First Level 2D Fundamentals	Randy H Shih	SDC	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mastering CAD/CAM	Ibrahim Zeid	McGraw - HillEducation	2004
2	Mastering AutoCAD 2024 and AutoCADLT 2024	Brian C Benton	Kindle eTextbook	
3	AutoCAD 2024: A comprehensive Guide	CAD Artifex	Kindle eTextbook	
4	AutoCAD 2024 Instructor	James A Leach & Shawna Lockhart	SDC	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=EgKc9L7cbKc
2	https://help.autodesk.com/view/ACD/2023/ENU/?guid=GUID-39B76D69-225D-45EE-A9AC-59CCBF48AF64
3	https://www.youtube.com/watch?v=pvKVy-eMDYc

Continuous Assessment (25 Marks)

Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.

- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

POLYMER ENGINEERING

SEMESTER S4

MATHEMATICS FOR PHYSICAL SCIENCE – 4

Course Code	GCMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus.	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To provide the students with the basics of various numerical methods to develop problem solving skills used in various engineering disciplines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	9

3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and t-distribution), Hypotheses and Test Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Newton-Raphson Method, Gauss Elimination Method, Gauss - Jordan Method, Numerical solution of ordinary differential equations-Euler's method, Modified Euler's method, Runge - Kutta method of 2nd Order, Numerical solution of Laplace equation –Jacobi's Method, Curve Fitting by Method of Least Squares - Straight lines, Parabola. (Text 2: Relevant topics from sections 2.5, 4.2, 7.5, 8.4, 8.5, 9.4)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination -1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z-tests and the one-sample <i>t</i> -test.	K3
CO4	Apply numerical methods to find solutions of linear system of equations, ordinary differential equations and Laplace equations.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Introductory Methods of Numerical Analysis	S S Sastry	PHI Learning Pvt Limited	5 th edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.,	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Numerical methods for Engineers	Steven C. Chapra, Raymond P. Canale	McGraw Hill Education	8 th edition, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
3	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/111/107/111107105/

SEMESTER S4

POLYMER PHYSICS

Course Code	PCPOT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in Engineering Chemistry	Course Type	Theory

Course Objectives:

1. To understand the basics of polymer properties in elasticity and viscosity, and flow and deformation
2. Students will be able to describe physical properties and its implications on long term properties of polymers through time- temperature relations and WLF equations along with elasticity properties of rubbers. They also attain the analyze to solve various polymer material's long-term properties.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Elasticity & Viscosity: Stress – engineering stress, true stress. Strain – engineering strain, true strain. Modulus- Young's modulus, shear modulus, bulk modulus, compliance, Poisson's ratio, elastic solid, Hooke's Law, viscous fluid, Newton's law, Static loading and dynamic loading, differences between elastic solid and viscous fluid. Rheology: Newtonian fluids and non- Newtonian fluids, Power law, shear rate dependent fluids- pseudoplasticity, dilatancy. Time dependent fluids – thixotropy, rheopexy	11

2	Viscoelastic Materials: Short Term, Long Term and Dynamic Properties: Viscoelastic materials, properties of viscoelastic materials, factors affecting viscoelasticity. Short-term properties: stress-strain curves, stress-strain curve of a typical plastic, comparison of the stress-strain curves for plastic, rubber and fibre, hysteresis. Long term properties: creep, stress relaxation. Dynamic properties: behaviour of elastic solids, viscous fluids and viscoelastic materials under dynamic loading, dynamic mechanical analysis (DMA), storage modulus, loss modulus, $\tan \delta$, damping.	11
3	Mechanical Models and Superposition Principle: Spring, dashpot, Maxwell model, Voigt model, Derivation of the following equations: Governing equation for Maxwell model, Equation for Maxwell model under creep, Equation for Maxwell model under stress relaxation, Governing equation for Voigt model, Equation for Voigt model under creep, Equation for Voigt model under stress relaxation, Maxwell- Wiechert model, Burgers model, Deborah number. Basis of Time-Temperature superposition principle, WLF equation	11
4	Rubber Elasticity: Rubber elasticity, Molecular requirements of rubber-like elasticity, Gough-Joule effect, Thermoelastic experiment, difference in the elasticity of metals and rubbers, energy driven elasticity, entropy driven elasticity, thermodynamics of rubber elasticity	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the basic concepts of stress, strain, Hooke's law & Newton's law.	K2
CO2	Explain the time and shear dependent non-Newtonian fluids	K2
CO3	Analyze the concept of viscoelasticity and to distinguish between short term long term and dynamic properties	K4
CO4	Elaborate the representations and equations of Maxwell's, Voigt's and Kelvin's models and to illustrate WLF Knowledge and apply knowledge in time temperature properties.	K6
CO5	Determine the basic concepts of rubber elasticity and describe Gough-Joule effect.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3									1
CO2	3	3	2	2	2							1
CO3	3	3	3	2	2							1
CO4	3	3	2	3		3						1
CO5	3	3	2									1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Physical Polymer Science	L H Sperling	Wiley	4 th Edition, 2006
2	Principles of Physical Chemistry	B. R. Puri, L.R. Sharma, M.S. Pathania	Vishal Publishing Co.	2013
3	Polymer Science and Engineering.	David J. Williams	Maclaren & Sons	1978
4	Essentials of Physical Chemistry	A. Bahl, B. S Bhal and G D Tuli	S. Chand Publishing	2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Viscoelastic Properties of Polymers	John D. Ferry	John Wiley & Sons.	3 rd Edition
2	Structured Polymer Properties.	R. J. Samuels	John Wiley & Sons	1974
3	Flow Properties of Polymer Melts.	J. A. Brydson	George Godwin Ltd.	1981
4	Introduction to Polymer Science and Technology.	H. F. Haufman & J. J. Falcetta	John Wiley & Sons.	1997

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/103103146
2	https://onlinecourses.nptel.ac.in/noc24_ch45
3	https://nptel.ac.in/courses/103103139
4	https://nptel.ac.in/courses/103103139

SEMESTER S4

PLASTICS MATERIALS

Course Code	PCPOT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To make the students aware of types of plastic materials and its properties
2. To make the students capable to estimate the difference in thermal and physical behaviour of plastic materials.
3. To understand the processing behaviour of plastics.
4. To understand the applications of plastics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Brief history of plastics: - Advantages and disadvantages - thermoplastics and thermosets. Monomer preparation, polymerisation, properties and application of LDPE, HDPE, cross linked and chlorinated PE and PP, PVC, polyvinylidene chloride	11
2	Monomer preparation, polymerisation, properties and application of PVA, polyvinyl acetate, PMMA, PET PAN, PC, nylon-5,6,6-6,6-12, aromatic polyamide and polyacrylamide, regenerated cellulose, and epoxies	11
3	Monomer preparation, polymerisation, properties and application of PU, PTFE, polyvinyl fluoride, polyvinylidene fluoride, PS, HIPS, ABS, SAN, and polyacetal. Thermoplastic polyesters, unsaturated polyester laminating resins, polyester moulding compositions, fibre and film forming polyesters.	11

4	Monomer preparation, polymerisation, properties and application of PF, preparation of phenol formaldehyde moulding powders, PF laminates. novolac, resol, MF and UF resins	11
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Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the importance of plastic materials.	K2
CO2	Determine the structure, monomer synthesis and properties of chlorine containing and acrylics polymers.	K5
CO3	Evaluate the manufacturing processes and application of fluorine containing and styrene polymers	K5
CO4	Compare and contrast aliphatic and aromatic amide.	K4
CO5	Discuss the manufacturing processes, properties and applications of thermosets	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3				2	1				
CO2	3	2	3				1	1				
CO3	3	2	3				1	1				
CO4	3	2	3				1	1				
CO5	3	2	3				2	1				

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to plastics engineering.	Shrivastava, Anshuman	William Andrew	2018
2	Plastics materials.	Brydson, John Andrew.	Elsevier	1999

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Applied plastics engineering handbook: processing and materials.	Kutz, Myer, ed.	William Andrew	2011
2	Introduction to plastics engineering.	Stokes, Vijay K.	John Wiley & Sons	2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	Introduction - Polymers: Concepts, Properties, Uses, Sustainability - YouTube

SEMESTER S4
PAINTS & SURFACE COATINGS

Course Code	PBPOT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. After the completion of this course, students should achieve sufficient knowledge about the basics of paint technology including components, types, application methods and testing.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of paint science: reflection, refraction, diffraction Paint components, pigments, pigment properties, different types, pigment dispersion methods, solvents, different types, solvent properties, oils, driers, additives	9
2	Classification based on polymeric resin - emulsion, oil and alkyd paints, acrylic paints, epoxy coatings, polyurethanes, silicones, chlorinated rubbers, vinyl coatings, Classification based on application - appliance finishes, automotive finishes, coil coatings, can coatings, marine coatings, aircraft finishes.	9
3	Mechanism of film formation and drying: - factors affecting coating properties, film thickness, film density, internal stresses, pigment volume concentration (PVC). Mechanical and optical properties of coatings, ageing properties, effect of rheological behaviour on paint performance.	9
4	Adhesion properties of coatings: -factors affecting adhesive bond, thermodynamics of adhesion, testing of paint materials- destructive and non-destructive methods, paint film defects, surface cleaning treatments,	9

	paint application methods.	
Project based learning (8 hrs) 1. On the first class before starting the first module, direct the students to prepare a polymer-based paint formulation 2. Students shall refer water based and solvent based paint formulations of different companies. 3. Analyze the importance of different components in a paint formulation 4. Analyze the different additives and its importance based on their function. 5. Design a suitable equipment for pigment dispersion and compare different pigment dispersion methods. Manufacture water and solvent based paints and study different mechanical properties using various testing methods after the completion of all 4 modules. 6. Prepare a project report based on their project work.		

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze basics of paint science and paint components	K4
CO2	Analyze paint classifications based on binders and application sectors	K4
CO3	Explain drying mechanism of paint film and properties of coatings.	K2
CO4	Analyze testing methods of paint film, paint film defects and paint application methods	K4
CO5	Discuss the importance of adhesion properties of coatings.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2									1
CO2		3				2			2			1
CO3	2			2								1
CO4	2	2	2	2				2		2		1
CO5	2		2					2				1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Outline of Paint Technology,	W.M. Morgen,	CBS Publishers	3 rd Edn, 2000
2	Paint Coatings and Solvents	D. Stoye,	Wiley VCH-	2 nd Ed 1998
3	Surface Coatings	S. Paul,	Wiley	3 rd Edn 1996

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/kyjYsKfEz88?si=OdISkwZfHWSgmCYQ
2	https://youtu.be/cXPSk-3oAN8?si=ibqzSVQFCSCGVFf6
3	https://youtu.be/cXPSk-3oAN8?si=ibqzSVQFCSCGVFf6
4	https://youtu.be/tM4Uqs0PpTU?si=NivJ0itEVd8gONBm

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project

Creativity in solutions and approaches

SEMESTER S4

MATERIAL SCIENCE AND ENGINEERING

Course Code	PEPOT411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce the various types of solid structures and to develop methodologies for the properties of engineering materials.
2. To familiarize the properties of alloys and composite materials and their engineering applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Crystal Structure and Phase: Structure of atom-present concept of atom-Rutherford's and Bohr's model, Bonding in solids-Types of solids-crystalline and amorphous solids-crystal systems -Bravais lattices-miller indices-coordination number-crystal defects-determination of crystal structure-X-ray diffraction-electron diffraction methods. Equilibrium between phases, Solid solutions-types of solid solutions-Hume Rothery rules	9
2	Properties of Engineering Materials: -Properties of engineering materials-mechanical properties -isotropy and anisotropy-elasticity, plasticity, Poisson's ratio-stress-strain relation- electrical and magnetic properties-resistivity-conductivity-ionic and electrical conductivity, semiconductors, superconductivity, insulators, ferroelectricity, piezoelectricity, magnetization, paramagnetism, ferromagnetism, and diamagnetism	9

3	Alloys and Composites: - Non-ferrous metals and alloys-aluminium and its alloys-copper and its alloys-Nonferrous metals and alloys used for high temperature services and nuclear application Polymers and their properties-ceramics-classification-comparison of ceramic and non-ceramic structures-properties and application of ceramics Composite materials-classification-general characteristics, Introduction to Nanocomposites.	9
4	Selection of Materials: -Corrosion-different types, mechanism and factors influencing corrosion-corrosion prevention-inhibitors and their applications-oxidation-aging of rubber-oxidation of metals and radiation damage-factors affecting the selection of materials for engineering purposes selection of suitable materials for construction in chemical industry.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define qualitatively the structure and bonding schemes of solids.	K1
CO2	Determine the stability of materials with all sorts of environments using phase diagrams.	K5
CO3	Explain physical properties and possible applications of a given material.	K2
CO4	Identify and compare the properties of alloys, ceramics, polymers, composite materials and their engineering applications.	K3
CO5	Select suitable material for construction in chemical industries.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3						1					
CO2	3						1					
CO3	3						3					
CO4	3						3					
CO5	3						3					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Text Book of Material Science & Metallurgy	Khanna O.P.	Dhanpat Rai publishers	2010
2	Material Science & Processes	Hajra Choudhary	Dhanpat Rai publishers	1978
3	Material Science and Engineering	V Raghavan	Prentice Hall India	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Elements of Material Science	Van Vlack	Pearson publishers	Sixth Edition - 2002
2	Material Science and Engineering	William F. Smith, Javad Hashemi, Ravi Prakash	Mc Graw Hill	Fifth Edition 2017
3	Material Science and Engineering	Nehal Dash Kaushik Kumar, Apurba Kumar Roy	Dreamtech Press	2020
4	Callister's Material Science and Engineering	R. Balasubramaniam	Wiley	2014

**Video Links (NPTEL,
SWAYAM...)**

Module No.	Link ID
1	Introduction to crystal structure, Dr. Vivek Pancholi, IIT Roorkee https://youtu.be/5nBBUahtz-c?si=gPny3Jotr3ltMvdL
2	Material property, Prof. Ratna Kumar Annabattula, IIT Madras https://youtu.be/RoTlt1unOMw?si=co408c0sjli0AC8s
3	Composite materials, Dr. Inderdeep Singh, IIT Roorkee https://youtu.be/LHHAPJbakEc?si=IgcM96r3kdaJxJ0f
4	Corrosion, Prof. Kallol Mondal, IIT Kanpur https://youtube.com/playlist?list=PLFW6lRTa1g81kobA2tvZ84YHB2li6qwx7&si=FfgN9crK_ BWV- TRA

SEMESTER S4

CATALYST SCIENCE AND CATALYTIC PROCESSES

Course Code	PEPOT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To make the students aware of basic concepts & theory for characterization of catalysts
2. To make the students capable to understand various catalytic processes in industries
3. To understand the deactivation of catalysts.
4. To understand the modern trends in catalysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	General characteristics of catalysis: - Classification of Catalyst, Thermodynamics of adsorption, Physical adsorption and chemisorption. Catalyst preparative Methods-Precipitation and co precipitation, Sol gel process, Flame hydrolysis, Supported catalyst from CVD and related techniques, Synthesis of aluminosilicate zeolites.	9
2	Catalyst Characterisation: - surface area measurements, BET theory, Pore size distribution, Porosimetry, Chemisorption techniques, Static and dynamic methods, Crystallography and surface analysis techniques – XRD, NMR.	9

3	Deactivation -classification of catalyst deactivation processes: - poisoning of catalysts, poisoning of metallic catalysts, poisoning of non-metallic catalysts, poisoning of bifunctional catalysts, coke formation on catalysts, metal deposition on catalysts, sintering of catalysts. Regeneration of deactivated catalyst.	9
4	Industrial catalysis: - Homogeneous, Heterogeneous catalysis, Biocatalysts and their typical industrial applications. Transition metal catalyst, Organo-metallic catalyst, Dual function catalyst, Zeolite and their typical industrial applications. Modern trends in catalysis – Phase transfer catalysis, electro catalysis, Nano catalysis, Polymer supported catalysis, Bio catalysis, Photo catalysis.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the basic concepts & theory for characterization of catalysts	K2
CO2	Outline the preparation of catalysts for various unit processes	K2
CO3	Distinguish between various catalytic processes in industries.	K4
CO4	Explain the deactivation of catalysts.	K2
CO5	Elaborate modern trends in catalysis	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical engineering kinetics.	Smith, Joseph Mauk, and J. M. Smith.	McGraw-Hill	1970
2	Catalysis: concepts and green applications.	Rothenberg, Gadi.	. John Wiley & Sons	2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluidization Engineering	Diazo Kunii, and Octave Levenspiel	Butterworth Heinemann	2008
2	Introduction to chemical engineering kinetics and reactor design.	Hill, Charles G., and Thatcher W. Root.	John Wiley & Sons	2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	https://youtu.be/zpMwCc7q8Y0?si=UH6ULJGFfv2vKqjZ NPTEL: Chemical Engineering - Catalyst Science and Technology

SEMESTER S4

HEAT & MASS TRANSFER

Course Code	PEPOT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in heat transfer and various unit operations.	Course Type	Theory

Course Objectives:

1. To introduce the various modes of heat transfer and to develop methodologies for solving a wide variety of practical heat transfer problems
2. To familiarize about mass transfer and the various unit operations and its material balances.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Modes of heat transfer: - laws governing heat transfer, thermal conductivity, general heat conduction equation in Cartesian coordinates, Fourier heat conduction equation, thermal conductivity of solids, liquids and gases-comparison between them, effect of temperature on thermal conductivity, thermal diffusivity, conduction through systems of constant thermal conductivity - conduction through plane and composite walls. Critical thickness of insulation.	9

2	Convection thermal boundary layer concept: - individual and overall heat transfer coefficients, Dimensional analysis - Rayleigh and Buckingham's pi theorem, its limitations, application of dimensional analysis to forced convection. Introduction to heat exchangers: - types of heat exchangers-the overall heat transfer coefficient-LMTD analysis of heat exchangers- Physical mechanism of radiation heat transfer-Radiation properties-; Black body radiation Planck's law, Wein's displacement law, Stefan Boltzmann law, Kirchoff's law;	9
3	Molecular diffusion: - diffusion in liquids and gases, Ficks law, diffusivity of gases and liquids, steady state and equimolar counter diffusion in gases and liquids, analogies between heat and mass transfer, Daltons law, Henrys law, Raoult's law, ideal and non- ideal liquid solutions, various unit operations and material balances- drying, absorption, distillation, evaporation, leaching. Drying: - equilibrium moisture content, bound and unbound moisture, critical moisture content, time of drying, rate of drying, drying curves, equipment for drying.	9
4	Gas absorption: - tray, spray and packed column, types and properties of tower packings, solubility of gas in a liquid and choice of solvent. Distillation: - plate efficiency, rectification, construction of fractionating column, determination of theoretical plates by McCabe Thiele method, relative volatility. Evaporation: - single and multiple effect evaporators, performance of evaporators, boiling point elevation, Duhrings rule, methods of feeding.	9

**Course Assessment Method (CIE:
40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Heat and Mass Transfer	R. K Rajput	S. Chand	2007
2	Mass Transfer, Theory and Applications.	K.V.Narayanan and B. Lakshmikutty	CBS Publishers	2nd Edition
3	Heat and Mass Transfer.	D.S Kumar	Kataria& Sons	2009
4	Principles of Mass Transfer and Separation Processes	Binay K Dutta	PHI Learning Pvt. Ltd.	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mass Transfer Operations	Robert E	McGraw Hill	1980
2	Introduction to Chemical Engineering.	Badger	Tata Mc Graw-Hill	1984
3	Unit operations of chemical Engineering	Warren L McCabe, Julian C Smith, P Harriot	McGraw Hill	(7thed.) 2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Heat and Mass Transfer, Prof. S.P. Sukhatme, Prof. U.N. Gaitonde -IIT Bombay https://nptel.ac.in/courses/112101097
2	Convective Heat and Mass Transfer, Prof. A.W. Date- IIT Bombay https://nptel.ac.in/courses/112101002
3	Mass Transfer Operations -I, Dr. B. Mandal -IIT Guwahati https://nptel.ac.in/courses/103103034
4	Mass Transfer II, Prof. Nishith Verma -IIT Kanpur https://nptel.ac.in/courses/103104046

SEMESTER S4

INDUSTRIAL BIOTECHNOLOGY

Course Code	PEPOT414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To make the students aware of fermentation and its upstream and downstream processing.
2. To make the students capable to understand production of commercially important primary and secondary metabolites.
3. To understand the biopolymers, vaccines and monoclonal antibodies.
4. To understand the modern trends in use of enzymes, biofertilizers, bioremediation techniques.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Industrial Bioprocess, Fermentation- Solid State Fermentation and Submerged Fermentation, brief survey of organisms - bacteria, viruses, fungi, their products, fermenter, basic concepts of upstream and downstream process, process flow sheet, block diagrams.	9
2	Primary metabolites- production of commercially important primary metabolites like amino acids, organic acids and vitamins. Production of alcohol.	9
3	Secondary metabolites- production processes for various classes of secondary metabolites like antibiotics, Production of enzymes.	9

4	Biopolymer, SCP, biopesticides, biofertilizers, beer, wine, dairy products. Bio preservatives, bioremediation, production of vaccines, monoclonal antibodies, human insulin production, bioreactors.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To interpret on fermentation and its upstream and downstream processing.	K2
CO2	Compare the production of commercially important primary and secondary metabolites.	K4
CO3	Illustrate biopolymers, vaccines and monoclonal antibodies.	K2
CO4	Explain the use of biotechnology in human life like biofertilizers, bioremediation techniques	K2
CO5	Discuss the modern trends in the use of enzymes and its production for industrial applications	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2										
CO3	3	2						1	1			
CO4	3	2						1	1			
CO5	3	2						1				

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Bioprocess Technology	P T Kalaichelvan and I Arul Pandi	MJP Publishers	2007
2	<i>Biotechnology</i>	Satyanarayana, U	Books & Allied (P) Ltd	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Textbook of Biotechnology	Dubey, R.C	S.Chand& Co. Ltd.,	2006
2	Comprehensive Biotechnology	Moo-Young. M	Pergamon Press (Oxford)	2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://youtu.be/dKxxerfNJts?si=rwqIdnG1knigANqd https://youtu.be/EfcEOLQDQu0?si=bIjlmXGeh1OOUQ T https://youtu.be/fpGEUn18H4E?si=rBONiwmO-4g2ugSW

SEMESTER S4

FLUID MECHANICS

Course Code	PEPOT416	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in Fluid Mechanics	Course Type	Theory

Course Objectives:

1. To study the mechanics of fluid motion.
2. To establish fundamental knowledge of basic fluid mechanics and address specific topics relevant to simple applications involving fluids.
3. To familiarize students with the relevance of fluid dynamics to many engineering systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fluids: - properties and nature of fluids, ideal fluids, real fluids, physical properties of fluids, density, viscosity, effect of viscosity on temperature and pressure, specific volume, stress and strain, bulk modulus, specific weight, relative density, surface tension, compressibility, capillarity. Rheology of fluids Rheological classification of fluids: - Newtonian and non-Newtonian fluids, Newton's law of viscosity, classifications of non-Newtonian fluids. Fluid Pressure Measurements: Pascal's law, vacuum and gauge pressures, hydrostatic law, measurement of fluid pressure using manometers and its different classifications, problems based on manometer.	9

2	Introduction to fluid flow: - Fluid flow classifications, Flow field, Eulerian and Lagrangian approach, velocity potential, stream function, circulation and vorticity. Streamline, Path line, Streak line, Stream tube classification of flow, Reynolds experiment, Reynolds number, Mach number, Turbulence, Reynolds stress. Velocity and acceleration in fluid	9
3	Fluid dynamics: Continuity equation: - Euler's, and Bernoulli's equations (Derivation not required) –Applications - Orificemeter, Venturimeter and associated simple numerical problems, Hydraulic coefficients Velocity measurements: - Pitot tube and Pitot-static tube. Introduction to Rectangular and Triangular Notches	9
4	Shear stress and velocity distribution in a pipe: - law of fluid friction, head loss due to friction, Hagen Poiseuille equation. Turbulent flow: Darcy-Weisbach equation, Chezy's equation Moody's chart, Major and minor energy losses. Hydraulic gradient and total energy line, flow through long pipes: - pipes in series, pipes in parallel, equivalent pipe. Transmission of power through pipes, efficiency of transmission, Water hammer, Cavitation.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination -1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understanding of Fluid Behavior, Properties, Application and Measurement in Engineering Systems	K3
CO2	Application and Analysis of Fluid Flow Phenomena and Fluid Flow Dynamics.	K4
CO3	Application and Measurement Techniques in Fluid Flow	K3
CO4	Interpret Euler and Navier-Stokes equations and Solve problems using Bernoulli's equation	K3
CO5	Evaluate energy losses in pipes and sketch energy gradient lines	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									
CO2	3	2	1									
CO3	3	3	2									
CO4	3	3	2									
CO5	3	3	2									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Textbook of Fluid Mechanics and Hydraulic Machines	Bansal R. K.,	Laxmi Publications,	2005
2	Engineering Fluid Mechanics	Balachandran.P	PHI	2012
3	Fluid Mechanics	Dr. D S Kumar	S K Kataria and Sons;	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid Mechanics	Cengel	McGraw Hill Education India	2014
2	Hydraulics & Fluid Mechanics	Modi P. N. and S. M. Seth,	S.B.H Publishers, New Delhi,	2002
3	Fluid Mechanics	White, F. M.	McGraw Hill Education India Private Limited	2017
4	Fluid Mechanics: An Introduction	Radhakrishnan E.	Prentice Hall India	3rd Edition 2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Fluid Mechanics, Prof. Suman Chakraborty, IIT Kharagpur https://nptel.ac.in/courses/112105171
2	Fluid Mechanics, Prof. Nishith Verma, IIT Kanpur https://nptel.ac.in/courses/103104043
3	Fluid Mechanics, Dr. T.I. Eldho, IIT Bombay https://nptel.ac.in/courses/105101082
4	Fluid Mechanics, Dr. N. Sahoo, Dr.Subhashisa Dutta, IIT Guwahati https://nptel.ac.in/courses/105103095

SEMESTER S4

OPERATIONS RESEARCH

Course Code	PEPOT417	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the role of Operation Research in decision making
2. To impact the various Operation Research techniques for effective problem solving

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Operations Research: - Concept of Optimization, Decision making through Operations Research, Models and Modelling in OR, General methods for solving OR models, Application and Scope of OR – Basic OR models. Linear programming: - Formulation of LPP, Graphical method, Simplex method, Maximization problems – Minimization problems, Problems involving artificial variables. Concepts of- Duality, Sensitivity analysis, Degeneracy in LPP	9
2	Transportation problems: Formulation, Methods of finding initial solution (North West Corner Rule, Least Cost Method and Vogel's Approximation Method), Test for optimality (MODI Method), Unbalanced Transportation Problems, Maximization Transportation Problems. Assignment Problems: Formulation, Methods of solution, Hungarian method, Unbalanced problems, Maximization problems.	9

3	Introduction to queuing theory: – Terminologies– Classification of queuing model 1, Single server problems 1, Multi server problems 1. Inventory control: – variables – deterministic inventory models – purchasing model without shortages 1, manufacturing model without shortages 1, purchasing model with shortages 1, manufacturing model with shortages 1	9
4	Network Analysis: CPM and PERT-Time Estimation-Critical Path, Basic Concepts of Crashing. Replacement Problems: Replacement of assets that deteriorate with time, Replacement of assets that fail completely.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain scope and limitations of OR and various methods to find the optimal solution.	K2
CO2	Determine the various methods in Transportation problem	K5
CO3	Explain Assignment problems	K2
CO4	Summarize various queueing models and inventories	K2
CO5	Compare and contrast CPM and PERT	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	1									
CO2	2	3							1			
CO3	2	2	3									
CO4	2	3										
CO5	2	3										

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Operations Research Theory and Applications	Sharma J K.	Laxmi Publications	2013
2	Operations Research	Kanti Swarup P K Gupta Manmohan	Sulthanchand & Sons	2019
3	Operations Research	S Kalavathy	VIKAS Publications	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Quantitative Techniques for Management	Vohra N D	Tata McGraw-Hill Education	2015
2	Operations Research: An Introduction	Taha, Hamdy A	Prentice Hall	2010
3	Operations Research: Principles and Practice.	Ravindran, A and Don T Phillips	John Wiley & Sons	1987

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/zTpmyvkNxp4?si=x1Xswk9Pt9_HOAdK
2	https://youtu.be/BDBhpxRzImI?si=Gg3eGmzwioZRTN3Y
3	https://youtu.be/BDBhpxRzImI?si=Gg3eGmzwioZRTN3Y
4	https://youtube.com/playlist?list=PLjc8ejfjpgTf0LaDEHgLB3gCHZYcNtsoX&si=d66td1vYNntU8eIR

SEMESTER S4

ENERGY ENGINEERING

Course Code	PEPOT 418	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To gain a thorough understanding of the different forms of energy and the various methods and technologies employed for harnessing these energy resources efficiently
2. To understand the significance of energy conservation and learn about various strategies and techniques for reducing energy consumption in both industrial settings and daily life, fostering a culture of sustainability and environmental responsibility.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Energy: - general classification of energy, Indian energy resources and energy consumption, energy Crisis, energy alternatives, thermal, hydel and nuclear power plants, merits and demerits of the above power plants. Solar energy: - solar thermal systems, flat plate collectors, focusing collectors, solar water heating, solar cooling, solar distillation, solar refrigeration, solar dryers, solar pond, solar thermal power generation, solar energy application in India, photo voltaic systems, energy plantations.	9

2	Ocean wave energy: - conversion, ocean thermal energy conversion, tidal energy conversion, Wind energy, types of windmills, types of wind rotors, Darrieus rotor and Savanius rotor, wind electric power generation, wind power plants in India. Biomass energy resources: - thermo-chemical and biochemical methods of biomass conversion, combustion, gasification, pyrolysis, fermentation, anaerobic digestion	9
3	Fuel cells: - alkaline fuel cell, phosphoric acid fuel cell, molten carbonate fuel cell, solid oxide fuel cell, solid polymer electrolyte fuel cell, Basic concepts of Microbial Fuel Cell Magneto hydro dynamics: open cycle and closed cycle systems, geothermal energy.	9
4	Energy audit and Energy conservation in chemical process plants: - energy saving in heat exchangers, distillation columns, furnaces and boilers, steam economy in chemical plants, energy conservation in petroleum, fertilizer and steel industry, cogeneration (CHP), pinch technology, recycling for energy saving, electrical energy conservation in chemical Process plants, environmental aspects of energy use.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination- 1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify different sources of energy, analyse the energy scenario and understand different conventional energy production systems	K3
CO2	Explain the concepts of solar and ocean energy conversion	K2
CO3	Explain the Wind and biomass energy conversion technologies	K2
CO4	Explain the working and types of fuel cells and MHD systems	K2
CO5	Explain the concepts of energy conservation and energy audit and apply the knowledge in process plants and daily life	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2				2	2					2
CO2	2	2										
CO3	2		3									
CO4			3									
CO5			3			3	2	2			2	

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Energy Technology,	Rao S. &Parulekar B.B.	Khanna Publishers	3rdedn1994
2	Renewable Energy Sources & Conversion Tech	Bansal N.K., Kleeman M. &Meliss M.	Tata McGraw Hill.	1988
3	Energy for a Sustainable World	Goldmberg J., Johansson, Reddy A.K.N. & Williams R.H.	John Wiley	1987
4	A Text Book on Energy System and Engineering	Pandey G.N.	Vikas Publishing	1995

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Non-Conventional Energy Sources	Rai G. D	Khanna Publishers.	1988
2	Solar Energy,	Sukhatme S.P	Tata McGraw Hill	Fourth Edition 2017
3	Non-Conventional Energy Systems	Mittal K.M	Wheeler Publications	1999
4	Energy conservation and management	S.S.Thipse,	Narosa Publishing House	2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Renewable Energy Engineering: Solar, Wind and Biomass Energy Systems, Prof. Vaibhav Vasant Goud, Prof. R. Anandalakshmi- IIT Guwahati https://nptel.ac.in/courses/103103206
2	Solar Energy Engineering and Technology, Prof. Pankaj Kalita -IIT Guwahati https://nptel.ac.in/courses/115103123
3	Fuel Cell Technology, Prof. S. Basu ,Dr. Anil Verma -IIT Delhi https://nptel.ac.in/courses/103102015
4	Environment Management, Prof. T.V. Ramachandra- IISc Bangalore https://nptel.ac.in/courses/120108004

SEMESTER S4

ADVANCED BIOPOLYMERS & BIOCOMPOSITES

Course Code	PEPOT415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course provides information about biodegradable polymer classes, natural, synthetic and modified naturally biodegradable polymer
2. This course will give a basic concept of biopolymers, bio composites different methods of preparation of Nano fillers and potential application in biomedical field.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Polymer biodegradation: -Biodegradable polymer classes, Natural biodegradable polymer, Synthetic biodegradable polymer and modified biodegradable polymer. Testing methods of biodegradability of biopolymers, Need for biodegradation of packaging materials	9
2	Biopolymer Films: – Biodegradable mulching, Advantages and Disadvantages, Chemical sensors, Biosensors, Functionalized Biopolymer Coatings and Films, Applications of biopolymers in horticulture Polymer based bio nanocomposites as advanced biomaterials: Basic concepts of biopolymers, Nano fillers, Preparation: solvent casting, electrospinning, freeze drying, 3D printing, properties, characterization and applications,	9

3	Green chemistry for polymers: - Raw materials for polymers – Sustainability of Petroleum resources, Need for Alternate Sources for Polymers -Polymer Recycling and Environmental Issues, Bio derived Polymers, Biodegradation and its Evaluation techniques, Standards for biodegradation – Need for biodegradation of packaging materials, Introduction to Life Cycle Assessment, Monomers from bio sources.	9
4	Bio composites- classification, Bio-based polymers and resins Thermoplastic Bio-based polymers, Thermosetting Bio resins Synthetic-Bio-based, Polymer Blends Bio-Based. Core Materials -Cores from Trees, Bio-based polymer Honeycombs, Bio-based polymer Foams, Nano Cellulose Foams Natural Fibre-Synthetic Polymer Composites, Fully Bio-based Composites, Current and Future Applications of bio composites - green bio composites for food packaging	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. Problem Definition (2 points)

- Check for clarity and completeness of the problem definition.
- Check for clear objectives and divide into sub objectives

2. Problem Analysis (4 points)

- Examine the problem and identify the variables and responses
- Divide the problem to manageable parts

3. Implementation of software (2 points)

- Use suitable software for getting the solution
- Check for the appropriateness of tools used

4. Evaluate (4 points)

- Evaluate the proposed solution.
- Assess the correctness of solution.

5. Validation of the results (4 points)

- Validate the results using available data.
- Do the error analysis of the results obtained.

6. Conclusion and presentation (4 points)

- Summarize the results obtained
- Do critical thinking of the results obtained and evaluate.
- Submit a technical report

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the classes, natural and synthetic and modified naturally biodegradable polymer. Non-biological and bio derived polymers are also included	K2
CO2	Analyze biodegradation and standards for biodegradation of packaging materials	K4
CO3	Evaluate green chemistry of polymers	K5
CO4	Analyze bio composites classifications	K4
CO5	Discuss importance of green bio composite in food packaging	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2		2		1		2					1
CO2	2	2		2	2				2			1
CO3	2											1
CO4	2			2				2				1
CO5	2	2	2				2					1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hand Book of Biodegradable polymers	Catia Bastioli	Rapra Tech	1987
2	Hand Book of Bioplastics & Bio composites for Engineering Applications	Srikanth Pillai	Wiley	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Surface modification of biomaterials: Methods analysis and applications	R Williams	Woodhead Publishing Series in Biomaterials	2010
2	Biopolymers	Steinbuechel Alexander	Wiley	2003
3	Biomaterials –novel materials from biological sources	D. Byrom	Stockton press.	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/RI5kU3tIQzU?si=hkxnAnNYhDz6JXFu https://youtu.be/dsDv4Fma_AA?si=6YByGVucvSOK333Q
2	https://youtu.be/kmpk8xTGUYM?si=VTJXS2rVWgoQH3u
3	https://youtu.be/xLDwcLS0SAQ?si=kdjTTTASqe7pl9I5
4	https://youtu.be/HEipr6CMp-g?si=tbvedCTNU7SMbtHl

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP_{fc} from GDP_{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP_{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, $NFIA=(-200)$ and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives.	6

2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	6

Course Assessment Method
(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample

activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessmen	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzingler,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).
- Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

POLYMER PREPARATION AND ANALYSIS LAB

Course Code	PCPOL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To familiarise the students about various polymer preparation and monomer characterisation methods
2. To familiarise the analysis of polymers and polymer compounds

Expt. No.	Experiments
1	Synthesis of the following Polymers: Polymethyl methacrylate, Polyacrylamide, Regenerated Cellulose, Phenol-Formaldehyde Resin (Novolac and Resol), Polystyrene, Polyurethanes and Glyptal resins, Urea-Formaldehyde and Melamine Formaldehyde.
2	Quantitative estimation of the following monomers: Aniline, Phenol, Acetone, Ethyl Acetate, Formaldehyde, Acrylonitrile, Urea, Glycol, Methyl methacrylate
3	Determination of molecular weight by viscosity method and end group analysis.
4	Estimation of Polymers: Acrylonitrile content of NBR, Chlorine content of CR, Rubber hydrocarbon content of NR.
5	Analysis of Polymer Compounds: Iodine value of rubber compounds, Carbon black content, Free sulphur content, Total inorganic content, Silica content
6	Identification of Rubbers: NR, SBR, BR, IR, IIR, EPDM, CR, NBR, Hypalon, Thiokol, Silicone.
7	Identification of Plastics: PE, PP, PS, PVC, PVA, PF, UF, MF, Polyester
8	Identification of Thermoplastic Elastomers: SIS, SBS, SEBS, Hytrel.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design /Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop skill to prepare polymers	K6
CO2	Determine the molecular weight and constituents of polymer materials.	K5
CO3	Define polymer by chemical analysis.	K1
CO4	Distinguish between the analysis of polymer compound	K4
CO5	Compare the practical knowledge on various polymeric materials	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		3										
CO2		3										
CO3		3		3					3			
CO4				3					3			3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer Synthesis and Characterization: A Laboratory Manual,	S. R. Sandler, W.Karo, J. Bonesteel and E.M. Pearce,	Elsevier,	1998
2	Polymer Synthesis: Theory and Practice,	D. Braun, H. Cherdon and H.Ritter,	Springer Science,	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Organic Chemistry	Morrison & Boyd	Prentice Hall. New Delhi,	6th edition,1992
2	Advanced Organic Chemistry	B.S.Bahl and Arun Bhal,	S. Chand & Co. Ltd., New Delhi	15th edition,1998
3	Textbook of Organic Chemistry	I.L.Finar	ELBS,	5th edition, 1996
4	Advanced Organic Chemistry,	Jerry March	John Wiley & Sons, New york,	1992

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=XaY63SGPoDo
2	https://www.youtube.com/watch?v=q8IMKft663I
3	https://www.youtube.com/watch?v=CW7qJOTbyXw
4	https://www.youtube.com/watch?v=lHt1aBH_6X8

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4
FLUID MECHANICS LAB

Course Code	PCPOL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. Understand fluid properties and relate it to fluid mechanics.
2. Correlate fluid behaviour with different types of pumps.

Expt. No.	Experiments
1	Experiments on Reynolds apparatus for determination of flow regime and construction of fanning friction factor vs. Reynolds No. plot
2	Determination of co efficient of discharge for orifice meter
3	Determination of co efficient of discharge for venturimeter
4	Determination of co-efficient of Pitot tube and construction of velocity profile across the cross section of pipe.
5	Determination of co-efficient of discharge for different types of weirs.
6	Determination of efficiency of a centrifugal pump.
7	Determination of viscosity of Newtonian & non-Newtonian fluid by Falling Sphere method
8	Verification of Bernoulli's Theorem
9	Determination of efficiency of a reciprocating pump.
10	Determination of coefficient of discharge C_d , coefficient of velocity C_v , and coefficient of contraction C_c for open orifice.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Plan and perform experiments in fluid moving machinery and analyse the principle involved	K6
CO2	Plan and perform experiments in flow measuring equipment	K6
CO3	Examine working of centrifugal pump	K4
CO4	Examine working of reciprocating pump	K4
CO5	Evaluate flow properties and its applications of fluids	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2							1	2		
CO2	2	2							1	2		
CO3	2	2							1			
CO4	2	2							1			
CO5	2	2							1	2		

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Text Book of Fluid Mechanics and Hydraulic Machines	R. K. Bansal	Laxmi Publications	2005
2	Introduction to fluid mechanics.	Nakayama, Yasuki	Butterworth-Heinemann	2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fluid mechanics for engineers: a graduate textbook.	Schobeiri, Meinhard T.	Springer Science & Business Media	2010
2	Engineering fluid mechanics	Yamaguchi, Hiroshi	Springer Science & Business Media	2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Fluid Mechanics [Introduction Video] (youtube.com)
2	https://onlinecourses.swayam2.ac.in/nou24_me12
3	https://youtu.be/X3Pf6eIeg0I
4	Transport Phenomena of Non-Newtonian Fluids [Intro Video] (youtube.com)

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

POLYMER ENGINEERING

SEMESTER: S5

RUBBERS-SCIENCE & TECHNOLOGY

Course Code	PCPOT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Knowledge in Polymer Materials and Polymer Science	Course Type	Theory

Course Objectives:

1. To understand the structure property relationship in rubbers
2. To know about the properties and application of general purpose and special purpose rubbers
3. To understand the role of compounding ingredients and mechanism of vulcanization in rubbers
4. To get familiarized with rubber processing techniques and equipment

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Natural rubber: - Natural rubber, source, chemical formula, molecular weight distribution, crystallisation, structure-property relationship, chemical reactivity, electrical and oxidation properties, Various forms of natural rubber- crumb, sheet, crepe, SP rubber, DPNR, LNR, liquid rubber, powdered rubber	11
2	Vulcanisation:- Chemical and physical aspects, curing characteristics such as scorch time, induction time, cure time. Mechanism of crosslinking by different crosslinking agents- sulphur, sulphur monochloride, phenolic resins, metal oxide, diamines, peroxides, oximes. Different curing systems, EV, semi-EV, conventional and sulphur less cure, Vulcanisation techniques.	11
3	Classification of elastomers: -Manufacture, structure, properties, processing, vulcanisation and applications of SBR, polybutadiene and polyisoprene rubber, IIR, NBR, CR, PU, and silicone rubbers.	11

4	Elastomers: - Monomer preparation, polymerization, structure, properties and application of the following rubbers- chlorosulphonated Rubbers, fluorine containing rubbers, acrylic rubbers, polysulphide rubber, polyalkenamers, polynorbornene, thermoplastic elastomers.	11
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze structure, properties of NR.	K4
CO2	Analyze rubber vulcanisation and its importance.	K4
CO3	Design rubber formulation based on end – use requirement.	K6
CO4	Explain various rubbers and its properties and applications.	K2
CO5	Select an appropriate rubber for a given application.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	2			1		2					1
CO2		1		3		1			1			1
CO3	1		2		2		2					1
CO4		1		2				2			2	1
CO5	2			1		1			2			1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Science and Technology of Rubber	James E. Mark and BurakErman	Elsevier	Ed 3 2011
2	Rubber Technology	I M Morton	Springer Science &Business Media	Ed 3 2013
3	Rubber Compounding- Chemistry and Applications	Brendan Rodger	CRC press	Ed 3 2021
4	Rubber Technology Handbook	Wemer Hofman	Hanser Gardner Publications	2001

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rubber Technology Compounding and testing for performance	J.S.	Hanser Publisher	,2001
2	Rubber Technology and Manufacture	C M Blow	Butterworths	Ed3
3	Rubber Technology	Maurice Morton	Van Nostrand Reinhold Co., New York	3rd edition
4	Science and Technology of Rubbers	Frederick R Eirich	Applied Science Publishers	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/t8mdOkXSPfc?si=4j4DcC46-7RHZj2J
2	https://youtu.be/98TSZFkvhbQ?si=Ma4kt_ZLcF42S9ib https://youtu.be/8wMTuZqId2s?si=5XJiBl0IK9IcCC2Q
3	https://youtu.be/BXI9ltRk3rA?si=rZzwlpmkz7SmEzn8
4	https://youtu.be/4tUGd9x1QlA?si=2ny1kOwgnryqyJA3

SEMESTER S5

POLYMER PROCESSING

Course Code	PCPOT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Knowledge in Polymer Materials and Polymer Science	Course Type	Theory

Course Objectives:

- 1 To understand the compounding ingredients used in rubbers and plastics.
- 2 To know about the machineries used for polymer compounding

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Compound development: Factors to be considered for compound development, formulation of a mix, Compounding ingredients stabilizers, fillers, antioxidants, antiozonants, UV absorbers, flame retardants, peptiser, retarder, curing agents. plasticisers, process aids, extenders, factice, mould release agents, tackifying agents, blowing agents, bonding agents, fragrances, antistatic agents, colorants, and other special additives.	11
2	Compounding methods: Different methods, principles of mixing, dispersive and distributive mixing, mastication, two-roll mill mixing, internal mixers, comparison between open mill and internal mixer, Banbury mixing, continuous mixing, master batching Compounding for specific applications- ozone resistance, heat resistance, weather resistance, oil resistance, electrical, optical and medical applications.	11

3	Compression moulding: Processing techniques, compression moulding, moulding presses, flash, positive and semi positive moulds, compression moulding cycle, trouble-shooting, automatic compression moulding.	11
4	Transfer moulding: Transfer moulding cycle, advantages, limitations, theoretical and design considerations, general mould design consideration, troubleshooting. Calenders and accessories, machineries for specific products- tyre and tubes, footwear, hoses and belting, wires and cables, latex and fibre foam.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the main compounding ingredients in plastics	K2
CO2	Analyze the significance and importance of compounding ingredients used in rubbers.	K4
CO3	Illustrate different rubber compounding methods for rubbers and compound formulations for specific applications.	K2
CO 4	Compare and contrast between various processing techniques for polymers.	K2
CO 5	Analyze the significant machineries for specific products.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	2			1		2					1
CO2		1		3		1			1			1
CO3	1		2		2		2					1
CO4		1		2				2			2	1
CO5	2			1		1			2			1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Science and Technology of Rubber	James E. Mark and Burak Erman	Elsevier	Ed 3 2011
2	Polymer Engineering Principles	Polymer Engineering Principles	Hanser	1993
3	Rubber Technology	M. Morton	Springer Science	2013
4	Rubber Technology and Manufacture	C. M. Blow, C.Hepburn,	Butterworths	1971

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Applied Polymer Processing Tech	N. P. Charemisinoff and P.N Chere	Marcel Dekker Inc, NY,	1996
2	Rubber Compounding: Principles: Materials, and Techniques	Barlow	CRC Press	1993
3	An Introduction to Rubber Technology	Ciesielski,	Smithers Rapra Publishing	1999

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/cg7f3loKlmU?si=csnCvm0UxFbVD4k9
2	https://youtu.be/p27Hy9Ms_PI?si=flffzjzWk8xDV-iw
3	https://youtu.be/yK6-QNB1roo?si=XSsq6h43g7sjNZi6 https://youtu.be/pOGpXZ-UMfo?si=qGBYbHjedMoDypSa
4	https://youtu.be/2DUB9DoIoi8?si=awk-oKjE0XptMM0t

SEMESTER S5

POLYMER MACHINERY & PRODUCT MANUFACTURING

Course Code	PCPOT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Plastic materials, Rubber Science	Course Type	Theory

Course Objectives:

1. To provide in depth knowledge about the product manufacturing processes used in polymer industries.
2. Student will be able to select the best processing options available and also to troubleshoot.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Injection moulding , terminology, process description, moulding cycle, classification of moulds, 2-plate and 3-plate moulds, different types of gates, cavity lay-out, setting up of mould, types of injection unit, classification of screw, screw design, process control, clamping unit, classification of machine hydraulics, ancillary equipment, computer operation, trouble shooting of injection moulding, reaction injection moulding.	9
2	Extrusion , principle, types of extruders, single screw and twin-screw extruders, metering, screw design, process control variables, types of dies, die design, typical extrusion dies straight through dies, offset dies, crosshead dies, wire coating, dies for tubular film, flat film dies, sheet dies, tube dies, elastic properties and die swell, manufacturing of pipes, cables, wire coating, extrusion profiles, blown films, flat film, sheets, filaments, lamination, extrusion of elastomers.	9

3	Blow moulding , terminology, basis, process variables, injection & stretch blow moulding, single and multi-layer, extrusion blow moulding, extrusion heads, process controls for blow moulding machine, process and product controls, trouble shooting in blow moulding. Thermoforming, definition, methods of forming, thermoforming machinery, heating of sheet, stretching, concept, hot strength, blistering, sags, cooling and trimming the parts, heat balance, shrinkage, trimming operations.	9
4	Rotational moulding , types of machines, moulds, materials, part design, calendering, types of calenders and strainer, embosser, winder, crowning, machinery powder coating, manufacturing methods, application methods, types of powder coating, finishing and machining of plastics, joining, welding and assembling of plastics.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Examine the basic concepts, machinery and troubleshooting of injection moulding.	K4
CO2	Define the importance of die design and process of Extrusion.	K1
CO3	Inspect the correlations of processing parameters and product manufacture by Blow moulding.	K4
CO4	Distinguish between various thermoforming and rotational moulding process for different products.	K4
CO5	Summarize the decoration of plastic products and methods of joining plastic products.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	1										
CO3	3	2										
CO4	3	1										
CO5	3	1										

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced thermoforming: methods, machines and materials, applications and automation.	Engelmann, Sven.	John Wiley & Sons	Vol. 8., 2012.
2	Extrusion processing technology.	Bouvier, Jean-Marie, and Osvaldo H. Campanella.	Hoboken: Wiley	2014
3	Practical guide to blow moulding.	Lee, Norman C.	iSmithersRapra Publishing	2006
4	Advanced thermoforming: methods, machines and materials, applications and automation.	Engelmann, Sven.	John Wiley & Sons	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Plastics engineering handbook of the society of the plastics industry.	Berins, M. (Ed.).	Springer Science & Business Media.	1991
2	Technology of thermoforming	Throne, J. L.	Hanser	1996
3	Extruder principles and operation.	Stevens, M. J., & Covas, J. A.	Springer Science & Business Media.	2012
4	Blow molding handbook: technology, performance, markets, economics: the complete blow molding operation.	Rosato, D. V., Rosato, A. V., & Di Mattia	Hanser Verlag.	2004
5	Injection mould design: a design manual for the thermoplastics Industry.	Pye, R. G. W.	Godwin Books.	1983

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/64Wy9EI77rk?si=TXccl_fiZEjK3I5G
2	https://youtu.be/T-m045Rm6G0?si=WeG4kFtkJHYaq3b3
3	https://youtu.be/we_kWA8KKgc?si=10OCQymK4pHGXNG5
4	https://youtu.be/l_90bXS3DYY?si=tIP82pIC6Rk1g1sE

SEMESTER: S5

TYRE TECHNOLOGY

Course Code	PBPOT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Knowledge in Rubbers	Course Type	Theory

Course Objectives:

1. To explain the Industrial manufacturing of tyres along with its research and development
2. To introduce students about the machineries used to manufacture various types of tyres and also the formulation for various tyre components, its processing, building and curing process.
3. After this course the student will be able to explain the tyre grip mechanism, tyre design features and tyre testing methods

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to tyres: Tyre functions, History on the design and development of tyres, current status of tyre industry in India and its future prospects, tyre sizing, different components of a tyre, its geometry, basic functions, functions of a pneumatic tyre, load carrying, vibration and noise reduction, the tyre function as a spring, contribution to road adhesion, tyre friction contribution to driving control, steering control and self-aligning torque. Solid tyres, concave tyres, winter tyres, OTR tyre, bicycle tyre, different types of tyres, bias, bias belted, radial, relative merits and demerits, their components, tube and tubeless tyres basic features.	11
2	Manufacturing techniques of various tyres - two-wheeler, car tyres, truck tyres, cycle tyres, cycle tubes, OTR tyres, aircraft tyres, automotive tubes, manufacture of tyre treads, beads, sidewalls, compounding techniques, principles of designing formulations for various rubber components, tyre	11

	reinforcement materials (textile, steel, glass, aramid), criteria of selection, different styles and construction, textile treatment (RFL dip).	
3	Tyre mould design- green tyre design principles, methods of building green tyres for bias, bias belted, radial and tubeless tyres, green tyre treatments, tyre curing methods, curing bags, bladders, diaphragms, autoclave, airbag, bagomatic, autoform, post cure inflation, different types of tyre building machines, bead winding machine, bias cutters, curing presses.	11
4	Tyre Testing and Quality Control- Measurement of tyre properties, dimension and size-static and loaded, tyre construction analysis, endurance test, wheel and plunger tests, traction, cornering coefficient aligning torque coefficient, rolling resistance, non-uniformity dimensional variations, force variations, radial force variation, lateral force variation concentricity and ply steer, tyre balance, mileage, X-ray holography, foot print pressure distribution, BIS standards for tyres, tubes and flaps, quality control tests, failure mechanism of cord reinforced rubber, tyre forces on dry and wet road surface, traction forces on dry, wet, ice, snow and irregular pavements, tyre noise, mechanism of noise generation, flatspotting, fatigue resistance, pantographing, pneumatic resilience effect.	11

Suggestion on Project Topics

- After teaching the students the fundamentals of Tyre Technology (Module 1)
- Faculty can direct the students to design a tyre according to a particular need.
- Ensure that it is suitable for the course.
- Students should prepare the plan which includes the appropriate compounding ingredients and formulations for various tyre components in the due time of Module II.
- Students can refer the appropriate code of practices.
- Students can fabricate different components of tyre with the help of suitable machineries.
- Assembling of tyres and curing of tyres should also be done according to the tyre type in the due period of Module III.
- Any tyre testing methods can also be performed by the student on the tyre he/she has designed during the time of Module IV.

- Students should submit the report to the faculty.

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the parts and functions of tyre.	K2
CO2	Define tyre sizing, tyre design and tyre grip mechanisms	K1
CO3	Design formulations of tyre with the property required from each part of the tyre	K6
CO4	Evaluate the industrial manufacture of tyre including building, curing and post cure treatments	K5
CO5	Discover various raw material, in process and product tests performed in tyre companies for quality control	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

1	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	1										
CO3	3	2	2				1					
CO4	3	2			1							
CO5	3	2			1							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rubber technologist's handbook. Vol. 1.,	De, Sadhan K., and Jim R. White, eds.	iSmithersRapra Publishing	2001.
2	"The little book of tire pavement friction." Pavement Surface Properties Consortium 1	Flintsch, Gerardo W., Kevin K. McGhee, Edgar de León Izeppi, and ShahriarNajafi		2012.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechanics of Pneumatic Tires	Clark, S. K. (Ed.).	US Government Printing Office	1981
2	Tyre technology	French, T.		1989
3	Tyre technology	Kovac, F. J		1973
4	Tire engineering. In Science and Technology of Rubber	Rodgers, B., & Waddell, W		3 rd Edition 2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=fVmBXtSKDw8&t=33s NPTEL-NOC IITM mod 11 lec73- Tyres Part 1
2	https://www.youtube.com/watch?v=RJ0Ck736qrw NPTEL-NOC IITM mod 12 lec74- Tyres Part 2
3	https://www.youtube.com/watch?v=CX7_6mL7ksA
4	https://www.youtube.com/watch?v=oqtZRLUDDtQnp telhrd MOD-01 LEC-09

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER: S5

ENGINEERING STATISTICS AND QUALITY CONTROL

Course Code	PEPOT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce various statistical testing methods
2. To familiarize about the process of quality checking

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Population and sampling: - large and small samples, random sampling, stratified sampling, estimating the mean and variance, confidence intervals, choice of sample size for estimation.	9
2	Testing of hypotheses for large samples: - means, proportions, difference of means and standard deviations. Testing of hypothesis for small samples, t-distribution, test of hypothesis for mean, difference of means, test for paired data, F-distribution, f-tests and properties, contingency table, χ^2 distribution, goodness of fit and independence of attributes.	9
3	Meaning of quality, variables and attributes data: - benefits of control charts, control charts for variables, check list for $\bar{x}$ and R charts, calculation of 3-sigma limits for $\bar{x}$, control charts for range, OC curve for $\bar{x}$ chart, OC curve for R chart, average run length (ARL) for the $\bar{x}$ chart. Control charts for fraction rejected, control chart for attributes,	9

	control limit for the p-chart, control chart for non-conformities, c-chart in statistical process control, applications of c chart, limits for c-chart	
4	Acceptance sample: - types of acceptance, sampling plans, determination of probability of acceptance by these sampling plans, sampling risks, design of sampling for stipulated producers risk and consumers risk. Concepts of AQL, LTPD, AOQL in sampling, standard sample continuous and sequential sampling plans.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To summarize effectiveness of testing of hypothesis	K2
CO2	To apply the concept of variable control chart	K3
CO3	Outline the concept of attribute control chart	K2
CO4	To determine acceptance sampling.	K5
CO5	To utilize the concepts of sampling plans.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

Course Articulation Matrix (Mapping of course outcomes with program outcomes):

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3	1									
CO2	2	2					1					
CO3	2	2										
CO4	2	3	3									
CO5	2	3										

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistical Quality Control,	R.C.Gupta	Khanna Publishers	8th edition, 2008
2	Engineering Statics and Quality Control	I.W.Burr,	Mc-Graw Hill	1975
3	Quality Control and Industrial Statistics	A.J.Duncon,	Richard. Irwin, Inc.	1975
4	. Statistical Quality Control	Granth and Leavenworth,	TMH	7th edition, 1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistical Quality Control	Douglas C Montgomery	Wiley	Sixth edition 2010
2	Quality Control Handbook (TMH)	J M Juran	Mc Graw Hill	1985
3	Fundamentals of Mathematical Statistics	Gupta and Kapoor,	Sultan chand&sons	2017
4	Introduction to statistical quality control	M Mahajan	Dhanpat Rai &Co	6 th Edition 2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtube.com/playlist?list=PLh4RfvBD1iYprnTqv809XHEQXQ5FjQJlp&si=eZhmdGKny70cyp9G
2	https://youtu.be/03nSqAkIYOA?si=viwJ1t43N28N9NQ1
3	https://youtube.com/watch?v=5iQm2G8Zekk&feature=shared
4	https://youtu.be/5w2ALtjWqGQ?si=dT1Hjm3QNAoJbpwK

SEMESTER: S5

POLYMER NANOCOMPOSITES

Course Code	PEPOT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Polymer Science	Course Type	Theory

Course Objectives:

1. Study of fundamentals of Characterization and testing of polymers and polymer composites
2. Understand the methodology for preparation of composite, various methods used for incorporation of nano fillers in polymers.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to composite material: - Introduction to polymer composites – Nano, micro and macro scales. Reinforcements – Short fibre, long fibre and particulate fillers. Matrices – thermoplastics-thermosets and rubbers, nano and micro composites. short and long fibre composites- Treatment of reinforcements.	9
2	Polymer-filler interactions: - use of coupling and bonding agents. Incorporation of reinforcements in polymer solution mixing- latex stage mixing-melt mixing and in-situ polymerization and precipitation – Dispersion and nucleating effects-Intercalation and exfoliation.	9
3	Application of layered and nonlayered nano and micro particles in polymer modification: - Preparation of long fibre and short fibre composites Different methods of preparation of composites. Mechanism of film formation and drying, factors affecting coating properties, film thickness, film density, internal stresses, pigment volume concentration (PVC). different methods	9

	used for film preparation, Mechanical and optical properties of coatings, ageing properties, effect of rheological behaviour on paint performance.	
4	Adhesion properties of coatings: - factors affecting adhesive bond, thermodynamics of adhesion, testing of paint materials- destructive and non-destructive methods, paint film defects, surface cleaning treatments, paint application methods.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain fundamentals of composite	K2
CO2	Discuss the significance of the composite technology by using carbon nanotubes.	K6
CO3	Outline the special properties and application of polymers.	K2
CO4	Analyze the improvement in the product by nanotechnology	K4
CO5	Explain the incorporation of carbon nanotubes.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	1					1					
CO3	3	2								1		
CO4	3	1										
CO5	3	1										1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Composites.	Chiao, C. C., Chiao, T. T., & Lubin, G	Reinhold, New York.	1982
2	Physical properties and applications of polymer nanocomposites.	Tjong, Sie Chin, and Y-W. Mai, eds	Elsevier.	2010
3	Synthesis techniques for polymer nanocomposites.	Mittal, Vikas, ed.	John Wiley & Sons	2015
4	Polymer nanocomposites handbook.	Gupta, Rakesh K., Elliot Kennel, and Kwang-Jea Kim, eds.	CRC press	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Short fibre-polymer composites.	De, S. K., & White, J. R. (Eds.).	Elsevier	1996
2	Handbook of Composites.	Chiao, C. C., Chiao, T. T., & Lubin, G.	Reinhold, New York.	1982
3	SPI handbook of technology and engineering of reinforced plastics composites.	Mohr, J. G.	SPI Publishing.	1984
4	Composite materials: engineering and science.	Matthews, F. L., & Rawlings, R. D.	Elsevier	1999

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/ebO38bbq0_4
2	https://youtu.be/ssRNdNh5pCA
3	https://youtu.be/0MzIh7wkgMs?si=rtUISUfkfBXYau3l
4	https://youtu.be/mvtTqCrm_zY?si=eZRF2a2zTtdsn7I5

SEMESTER: S5

CORROSION ENGINEERING

Course Code	PEPOT523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart knowledge on the importance of corrosion and its prevention and control in process industries.
2. To develop plans for fighting corrosion and develop anti corrosive systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Principles of corrosion phenomenon: Corrosion rate expressions, Electrochemical aspects, Environmental effects, Metallurgical and other aspects.	9
2	Galvanic or two metal corrosion: Crevice corrosion, Pitting, Intergranular corrosion, Selective leaching, Erosion corrosion, Stress corrosion, Hydrogen damage.	9
3	Non-electrochemical and electrochemical methods: potentiostat, Tafel extrapolation, linear polarization, galvanostat, impedance spectroscopy, thermogravimetric technique, salt spray test, weight change measurements.	9
4	Design and coatings: inhibitors and surface engineering, cathodic protection and anodic protection. Corrosion and its control in industries: Power, Process, Petrochemical, ship building, marine and fertilizer industries. Corrosion auditing in industries, Corrosion map of India.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify and explain different types of corrosion	K3
CO2	Demonstrate various corrosion testing techniques	K2
CO3	Explain various corrosion prevention techniques	K2
CO4	Select the appropriate corrosion control technique for different engineering materials	K1
CO5	Identify the corrosion control techniques used in various industries	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3					2					
CO2	3	3				2	2					
CO3	3					1	1					
CO4	3	3				1	1					
CO5	3	3				2	2					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Corrosion Engineering	Fontana M. G.	Tata McGraw Hill, 3rd Edition, 2005	2005
2	Principles and Prevention of Corrosion	Jones D. A	Prentice-Hall, Inc., 2nd Edition, 1996.	1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Fundamentals of Corrosion	Scully J. C	Pergamon Press.	2nd Ed., 2006
2	Fundamentals of Electrochemical Corrosion.	Stansbury E. E. and Buchanan, R. A,	ASM International.	1998
3	Corrosion and Corrosion Control.	Uhlig H. H. and Revie R. W.	3rd Ed., John Wiley & Sons.	2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/zS0eaJRYgIM
2	https://youtu.be/fFDytTIM5lw
3	https://youtu.be/OdNyVqCJwxI
4	NPTEL :: Metallurgy and Material Science - NOC:Corrosion - Part I

SEMESTER: S5

AIR POLLUTION CONTROL

Course Code	PEPOT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To make students aware of air pollution and methods for controlling different pollutants.
2. Understanding the importance of measurement of air contents and quality control and the different methods employed for purification of air in industry and day to day life

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction - Significance of air pollution studies, factors that contribute to air pollution - possibilities to air pollution abatement - air pollution legislation - Techno - administrative aspects of air pollution - Emission and noise standards of Kerala State Pollution Control board. Gaseous pollutants -source, chemistry, adverse effects on plants, animals and human beings, properties - tolerance levels - carbon monoxide, carbon dioxide, aldehydes, hydrocarbons - compounds of sulphur, compounds of Nitrogen, Oxidants, Hydrogen fluoride - Control of gaseous pollutants - Automobile pollution control.	9
2	Particulates in the air - source, nature and adverse effects - control of particulates - settling, filtration, collection in fluids, electrostatic precipitation, conversion to harmless and useful products. Meteorology related to atmosphere - pressure, temperature, lapse rates - humidity - condensation -	9

	wind direction and velocity. Effects of meteorological parameters on transport and diffusion. Atmospheric electricity.	
3	Optics of the atmosphere - Effects of air pollutants on atmospheric visibility - methods of measurement of visibility - Introduction to noise pollution. Photochemical reactions of the atmosphere. Purpose and principles of measurement of (1) High volume sampler (2) Exhaust gas analyser (petrol and diesel) (3) Stack sampler (4) Sound level meter - industrial hygiene and in plant safety to workers.	9
4	Ventilation and Indoor Air Quality Control- An Overview of Indoor Air Quality, Basics of HVAC Systems; IAQ Issues and Impacts on Occupants; Application of Audits to Developing an IAQ Profile, Developing Management Plans, IAQ Problems; Control, Quantification and Measurement, Air Pollution Dispersion-Dispersion Theory Basics- Air Quality Impact of Stationary Sources- Models and Resources	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify different sources of air pollution	K3
CO2	Examine the concepts of air purification.	K4
CO3	Explain the chemistry of pollutants and its effects	K2
CO4	Analyze the effects of air pollution on environment and human beings	K4
CO5	Explain the concepts of air quality control and audit.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental Pollution Control Engineering	C.S.Rao	, New Age International, New Delhi,	1999.
2	Air Pollution Control and Design for Industry,	Paul N. Cheremisinoff,	Taylor and Francis, New York,	1993

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Air pollution control: A design approach Vol. 65	Cooper, C. D., & Alley, F. C.	Long Grove, Ill: Waveland press.	2002
2	Fundamentals of air pollution.	Stern, A. C.	Elsevier	2014
3	Air pollution control engineering	De Nevers, N	Waveland press.	2010
4	Air Pollution Control	Rao, M. N., & Rao, H. V. N.	Tata Mc Graw Hill NewDelhi.	1996

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://www.youtube.com/playlist?list=PLLy_2iUCG87BwOQUbs7WSdMVWHDXByk-w Prof. Bhola ram Gurjar, IIT Roorkee

SEMESTER: S5
BIOREACTOR DESIGN

Course Code	PEPOT526	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3: 0:0: 0	ESE Marks	40
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart basic knowledge on bioreactors.
2. To impart knowledge on different modes of operation bioreactor.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Bioreactors - Function, description, working, advantages and limitations of stirred tank, airlift, airlift pressure cycle bioreactor, packed bed, fluidized bed, trickle bed and flocculated cell bioreactors. Novel Bioreactors-inverse fluid flow units, hollow fiber reactors, centrifugal field reactors, rotating drum bioreactor, spin filter bioreactor, disposable culture systems and wave bioreactor.	9
2	Batch bioreactor, cell death in batch reactor, endogenous metabolism, maintenance, calculation of batch reaction time from ideal system for enzyme reaction and cell culture, batch reaction time with enzyme deactivation, calculation of total batchtime. Ideal continuous flow stirred tank bioreactor (CFSTBR)-chemostat, substrate conversion and biomass productivity, mean residence time	9

3	comparison of batch bioreactor and single stage CFSTBR, wash out condition, stability of the chemostat, chemostat with cell recycle, comparison of steady-state biomass concentration and volumetric biomass productivity for a chemostat with and without cell recycle.: reactors, calculation Plug flow tubular reactor (PFTR), comparison of ideal mixed flow (batch and CFSTBR) and plug flow tubular of reactor length and residence time, recycling in PFTRs, analysis of recycle reactors.	9
4	Design aspects of bioreactors, bioreactor geometry, guidelines for bioreactor design, bioreactor vessels, agitator assembly, mass transfer aspects, rheology and mixing, design, operation and types of agitators, power requirements for agitation, effects of agitation on mass transfer, oxygen delivery system - spargers, foam control system, factors affecting antifoam requirements, mass transfer between phases–factors affecting mass transfer between phases, oxygen uptake in fermenters	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain how different bioreactors work	K2
CO2	Estimate substrate conversion and biomass productivity in a chemostat	K5
CO3	Define the reactor length and residence time	K1
CO4	Develop the design aspects of bioreactors	K6
CO5	Analyze the basic knowledge in bioprocess industries	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Bio process Engineering Principles	Pauline M Doran	Academic Press,	2013
2	Bio chemical Engineering	Douglas S Clark,Harvey W Blanch	Marcel Dekker,	1997

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Basic Bioreactor Design	Klaasvan't Riet ,Johannes Tramper	Marcel Dekker,	1991.
2	Bioreactors: Analysis and Design	Tapobrata Panda	Tata McGraw-Hill	2011.
3	Bioreactors in Biotechnology A Practical Approach	Alan H Scragg	Ellis Horwood,	1991

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/uooShNgPhIQ
2	https://youtu.be/azdVSr7DBlg
3	https://youtu.be/xq9HTNCsWXM
4	https://youtu.be/1LwT7C2bxhU

SEMESTER: S5

PROCESS MODELLING & SIMULATION

Course Code	PEPOT527	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Knowledge of Heat Transfer, Mass Transfer, And Fluid Flow Operations.	Course Type	Theory

Course Objectives:

1. To understand the important physical phenomena from the problem statement.
2. To develop model equations for the given system.
3. To apply suitable numerical simulation methods for the solution of models.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Modelling and Fundamental Laws Definitions of Modelling, Uses of Mathematical Modelling, Classification of Modelling Techniques-Basic Modelling Principles. Fundamental Laws of Chemical Engineering: -Energy Equations-Continuity Equation-Equation of Motion - Transport Equations -Equations of State Equilibrium States -Chemical Kinetics	9
2	Reactor Modelling: Models of Reactors-Mixing with Reaction-Reversible Reaction-Steam Jacketed Vessel-Isothermal Constant and Variable Holdup CSTR in Series	9
3	Separation Process Modelling: Models of Separation Processes: - Multicomponent Flash Drum-Ideal Binary Distillation Column - Multicomponent Distillation Column, Batch Distillation-Condensation	9

4	Distributed System Modelling and Numerical Simulation Techniques: Distributed System Modelling: Jacketed Tubular Reactor-Laminar Flow in a Pipe - Counter Current Liquid-Liquid Heat Exchanger Numerical Simulation Techniques: Finite Difference Method -Method of Weighted Residuals. Simulation of Gravity Flow Tank -CSTR in Series -Non-Isothermal CSTR	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define the fundamental principles of mathematical modelling in chemical engineering.	K1
CO2	Develop and formulate mathematical models for various chemical processes, including reactors and separation processes.	K3,K6
CO3	Analyze and interpret the results obtained from model equations to understand the behaviour of chemical processes.	K4
CO4	Explain the numerical simulation techniques to solve complex process models and validate the results against theoretical and experimental data.	K2
CO5	Determine the Design and optimize chemical processes to improve efficiency and performance.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1											
CO2	1	2		2								
CO3	2	2	1									
CO4	2											
CO5	1	2			3							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical Process Modelling and Computer Simulation	Amiya K.Jana	Prentice Hall of India.	2008
2	Process Dynamics - Modelling Analysis and Simulation	Biquette W.B.	Prentice Hall of India.	1st Edition
3	Mathematical Modelling in Chemical Engineering,	Franks R.G. E	John Wiley.	1967
4	Chemical Engineering Dynamics - Modelling with PC Simulation	John Ingham et.al	VCH Publishers.	2nd Edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals and Modelling of Separation Processes	Holland C. D	Prentice Hall,	1975
2	Process Modelling Simulation and Control for Chemical Engineers	Luyben W. L	2nd Ed., McGraw Hill,	1990
3	Mathematical Modelling, Vol. 1: A Chemical Engineering Perspective (Process System Engineering),	Aris R	Academic Press,	1999
4	Process Modelling and Control in Chemical Engineering	Najim K	CRC,	1990

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Mathematical Modelling and Simulation of Chemical Engineering Processes by Prof. Sourav Mondal, IIT Kharagpur. This course covers mathematical modelling approaches, theoretical frameworks for heat, mass, and fluid flow, and includes simulation techniques https://onlinecourses.nptel.ac.in/noc24_ch76/preview https://onlinecourses.nptel.ac.in/noc22_ch47/preview.
2	Chemical Process Control by Prof. S. Pushpavanam, IIT Madras. This course includes fundamental principles such as energy and continuity equations, equations of motion, transport equations, and equilibrium states
3	Chemical Reaction Engineering by Prof. Kannan Moudgalya, IIT Bombay. This course covers models of reactors, including mixing with reaction, reversible reactions, and steam jacketed vessels
4	Process Equipment Design by Prof. M. K. Jha, IIT Roorkee. This course includes models of separation processes like multicomponent flash drums and distillation columns

SEMESTER: S5

OIL AND NATURAL GAS ENGINEERING

Course Code	PEPOT528	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. The field of natural gas engineering is very much important for petroleum engineers specializing in gas processing technology.
2. The course outlines an optimal balance between natural gas production, natural gas processing and gas transportation. An extensive treatise on natural gas engineering, both upstream and gas refining processes with key equipment and facility design will be covered.
3. This course will also highlight the current status of production of natural gas through unconventional sources/ techniques and the applications of natural gas.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of petroleum geology and exploration: Introduction, origin, formation, geological occurrence and characteristics of oil and natural gas. Source Rocks, Reservoir Rocks, and Cap rocks: Definition, Characteristics, Classification and nomenclature, Concept of Shale oil, Reservoir Properties, Hydrocarbon migration, Petroleum Exploration	9
2	Conventional natural gas: Associated gas, non-associated gas, Gas condensate. Unconventional production of natural gas: Coal bed methane, Natural gas hydrate, shale gas and tight gas sands. Coal bed methane - Introduction, present status, formation and properties of coal bed methane. Natural Gas Hydrate - Concepts and Structures,	9

	Estimation and Production of Natural Gas: Estimation of gas reserves by volumetric method – Production of natural gas –Pressure decline method	
3	Dehydration of Natural gas: Water content of natural gas streams,hydrate control in gas production, dehydration systems - Glycol dehydration and Solid desiccant dehydration. Acid gas removal: Metaloxide process, Slurry process, Amine process, Carbonate washing process, Methanol based process and Sulphur recovery process. Compression of Natural Gas: Types of Compressors, Selection of reciprocating and centrifugal compressors, Thermodynamics of Compressors, Compression calculations.	9
4	Gas flow measurement: Fundamentals, Methods of measurements, Orifice meters equation, turbine meters, Selection, Uncertainties in flow. Liquefied Natural Gas: Gas treating before liquefaction- Liquefaction cycles- Storage of LNG, Transportation-Regasification and cold utilization of LNG. Safety measures and environmental considerations.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify different source rocks from which hydrocarbons are generated.	K3
CO2	To perceive knowledge of how and why fluid hydrocarbons migrate from a source rock to reservoir rock, entrapment and accumulation of hydrocarbons.	K5
CO3	Identify sources of natural gas and its applications in variety of fields.	K3
CO4	Explain various methods of natural gas processing for dehydration and sweetening.	K2
CO5	Compare and contrast gas compression, gas gathering and transport installation.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Elements of Petroleum	Geology, Richard,C.Selley	Elsevier	1997
2	Gas Production Operations	Beggs D.H.,	OGCI Publications	1984
3	Natural Gas:ABasic Handbook	JamesG.Speight,	Gulf Publishing Company	2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamental of Natural Gas Processing,	Arthur J. Kidnay, William R. Parrish,	Taylor and Francis	2006.
2	Gas Conditioning and Processing	John M. Campbell,	Campbell Petroleum Series	Volume 2, 7th Edition, 1992.
3	Gas Production Engineering	Kumar Sanjay	Gulf Publishing Company, TX, USA,	1987.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/m8qCtIwquDc?si=BaMl6-Ko82WCSWli
2	https://youtu.be/nrx4l3rX-3g?si=pHyFZVjzBge1x1Yj
3	https://youtu.be/5nTkArHe4bY?si=KP5jy499jsiakMQz
4	https://youtu.be/IE0-mj_2Nis?si=zqCEE4KpI9NQ8zgQ

SEMESTER: S5

POLYMER ELECTROLYTES & GELS

Course Code	PEPOT525	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course is designed to provide students the fundamental concepts of polymer gels and electrolytes
2. To understand the preparation of polymer gels and polymer electrolytes as it is important to fabricate various products for various applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to polymer Electrolytes- Definition, composition, types of polymer electrolytes- solid polymer electrolytes, liquid polymer electrolytes, gel polymer electrolytes, ionic polymer electrolytes. Common polymers included as polymer electrolytes; different interactions seen in polymer electrolytes. Different synthesis method of polymer electrolytes, advantages and disadvantages of polymer electrolytes	9
2	Properties & Applications of Polymer Electrolytes- ionic conductivity, ion-transfer mechanism, chemical properties, mechanical & thermal properties, electro-chemical properties, electrochemical characterization by cyclic voltammetry and electrochemical impedance spectroscopy, optical properties, dielectric properties, solvation and desolvation kinetics, adhesion & cohesion, interfacial tension, contact angle.Applications in energy storage device, electrochemical sensors, optoelectronic devices and biomedical applications	9

3	Polymer Gels: Polymer hydrogels, aerogels, macro, micro and nanogels, organogels. Cellulose based polymer gels, polysaccharide-based polymer gels, graphene oxide-based polymer gels, Functional polymer gels. Importance of crosslinking in gels. Examples of crosslinking agents. Various synthesis methods for polymeric gels. Factors influencing network formation in gels, different interactions in gels.	9
4	Properties & Gelation- rheological properties, swelling study of gels, mechanical properties and its characterisation, role of gelation, factors influencing gelation. Biocompatibility of gels and its applications in various biomedical fields. Gels as shape memory materials, polymer gels for energy storage devices.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. Problem Definition (2 points)

- Check for clarity and completeness of the problem definition.
- Check for clear objectives and divide into sub objectives

2. Problem Analysis (4 points)

- Examine the problem and identify the variables and responses
- Divide the problem to manageable parts

3. Implementation of software (2 points)

- Use suitable software for getting the solution
- Check for the appropriateness of tools used

4. Evaluate (4 points)

- Evaluate the proposed solution.
 - Assess the correctness of solution.
- 5. Validation of the results (4 points)**
- Validate the results using available data.
 - Do the error analysis of the results obtained.
- 6. Conclusion and presentation (4 points)**
- Summarize the results obtained
 - Do critical thinking of the results obtained and evaluate.
 - Submit a technical report

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer Gels	Vijay Kumar Thakur, ManjuKumari Thakur	Springer	2018
2	Polymer Electrolytes: Fundamentals and Applications"	Michel B. Armand and Patrick G. Bruce		2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Gels Handbook: Fundamentals, Properties, and Applications	Yoshiki Chujo	Wiley-VCH	(2019)
2	Polymer Electrolytes: Principles and Applications"	Rameshwar Singh and Shailendra Kumar		2020
3	Polymer Gels: Advances in Synthesis, Characterization, and Applications"	A. K. Bajpai and A. K. Sharma	Smithers Rapra	(2019)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=uP2ebL_RGGc
2	https://www.youtube.com/watch?v=BXpfGWhR0Zs Abhijit P Deshpande
3-4	https://www.youtube.com/watch?v=VWJiqzz2coU Abhijit P Deshpande lec82

SEMESTER S5
SPECIFICATION TEST LAB

Course Code	PCPOL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Engineering Chemistry Lab	Course Type	Lab

Course Objectives:

1. To create practical knowledge on specification tests for latex and dry rubber
2. To impart knowledge on properties of latex and dry rubber

Details of Experiment

Expt. No	Experiment
1	Specification tests for field latex Viscosity
2	Specification tests for field latex density
3	Specification tests for field latex pH
4	Specification tests for content of Ammonia preserved latex
5	Specification tests for Magnesium, Copper and manganese content of preserved latex
6	Specification tests for Dry rubber content, Total solid content, KOH number and Volatile fatty acid number of preserved latex
7	Specification tests for Sludge content and Coagulum content of preserved latex
8	Specification tests for Mechanical stability time, Heat stability time and ZnO stability of preserved latex
9	Specification tests for Volatile matter, Ash content and Dirt content of dry rubber
10	Specification tests for Nitrogen content of dry rubber
11	Specification tests for Estimation of Cu and Fe of dry rubber
12	Specification tests for Estimation of Mn, P ₀ , PRI of dry rubber

Course Assessment Method

(CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop skill to conduct the specification tests for field and preserved latex	K6
CO2	Determine the skill on the specification tests for dry rubber	K5
CO3	Analyze skills to correlate latex specifications with properties of latex.	K4
CO4	Select teamwork skills through group activities	K1
CO5	Summarize the practical knowledge on properties of latex and dry rubber	K2

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1		3			1							
CO2	2	3			2							
CO3		3		3	2							
CO4	2		2		1							
CO5				3								

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	ASTM and BIS standards			1898
2	Hand book of Natural Rubber Production in India	Bhaskaran Nair V K	RRII, Rubber Board	1980
3	Polymer Latices and Their Applications	K O Calvert	Macmillan	1982

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Practical Guide to Latex Technology,	Rani Joseph,	Smithers Rapra,	2013
2	Polymer Latices: Science and technology	D.C. Blackley	Springer Science	1997
3	Vanderbuilt Latex Handbook	Mausser	R T Vanderblit	3 rd edition, 1987
4	Rubber Technologist's Handbook, Vol	S. K. De and J. R. White	Smithers Rapra Publishing,	2001

Video Links (NPTEL, SWAYAM...)	
Sl. No.	Link ID
1	https://www.assochem.in/blog-details/specifications-test-procedures-and-significance-of-60-natural-rubber-latex-cl-60-
2	https://byjus.com/chemistry/kjeldahl-method/
3	https://www.testing-instruments.com/blog/how-to-measure-the-presence-of-ash-content-in-rubbers/
4	https://www.scribd.com/document/617652232/ISO-248-Rubber-Raw-Determination-of-Volatile-Matter-Content-6-15-2005

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER: S5

PRODUCT MANUFACTURING LAB

Course Code	PCPOL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. Develop skill to operate different mixing and moulding equipment
2. Develop skill to manufacture various rubber and plastic products.

Details of Experiment

Expt. No	Experiment
1	Determination of cure time
2	Effect of mastication time on plasticity/ viscosity
3	Work practice in mastication, band formation, homogenisation and mixing using a laboratory mill
4	Preparation of micro cellular sheet
5	V-strap
6	Table mat
7	Injection bottle cap
8	Play ball
9	Solid tyre
10	Sponge
11	Man-made hose
12	Solvent based adhesives
13	Hand lay-up: Plates prepared using glass fibre reinforced polyester composite
14	Work Practice in calendaring, Injection moulding, Rotational moulding and Extrusion
15	Visit to factories manufacturing tyres and non-tyre products

Course Assessment Method

(CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop skill to operate two roll mills, hydraulic press, hand press, various moulds etc.	K6
CO2	Develop skill on measuring cure time and to correlate mastication time on plasticity/ viscosity	K6
CO3	Build skills to manufacture various rubber and plastic products	K3
CO4	Analyze troubleshooting in compression moulding	K4
CO5	Develop teamwork skills through group activities	K6

K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2		1						2		1
CO2	2		2			2						1
CO3	2	2							2			1
CO4	2			2						2		1

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rubber Chemistry	J.A. Brydson	Allied Science publishers, London	1978
2	Rubber Technology and manufacture	C M. Blow and C. hepburn	Butterworths	1971

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rubber Materials and their Compounds	J. A. Brydson	Elsevier	1988
2	Plastic materials	J.A. Brydson	Newnes Butterworths	1999

Video Links (NPTEL, SWAYAM...)	
Sl. No.	Link ID
1	https://youtu.be/1kWwawnHhbU?si=K5Be0Owvt4dkTuyL https://youtu.be/7BcDMZxYn2c?si=DCUA8qjqzAvLiukb
2	https://youtu.be/PtVHoTdKSZg?si=35lLRQlfNtWxkPK_

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. **Lab Reports and Record Keeping (6 Marks)**

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. **Viva Voce (5 Marks)**

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. **Procedure/Preliminary Work/Design/Algorithm (10 Marks)**

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. **Conduct of Experiment/Execution of Work/Programming (15 Marks)**

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. **Result with Valid Inference/Quality of Output (10 Marks)**

- Accuracy of Results: Precision and correctness of the obtained results.

- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. **Viva Voce (10 Marks)**

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. **Record (5 Marks)**

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

POLYMER ENGINEERING

SEMESTER: S6
POLYMER PRODUCTS - DESIGN & TESTING

Course Code	PCPOT601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Plastic Materials, Rubber Technology	Course Type	Theory

Course Objectives:

1. To provide in depth knowledge about steps involved in polymer product design with the basic rules and formulations.
2. Student will be able to design formulations of typical rubber products, its processing, and curing process.
3. Students will be able to design plastic products with the specifications as per the testing standards.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Design fundamentals: Introduction, design principles, steps in product design, functional design, aesthetic design, effect of fillers on properties and performance. Long term property analysis, creep, stress relaxation, stress in products- tension, compression and shear, effect of cyclic loading, structural design of products under static and dynamic loads.	11
2	Design features of products, inside and outside corners, wall thickness, ribs, tapers, draft angles, weld lines, gate size and location, tolerances, moulded-in-inserts, plastic threads, blind holes, undercuts, hinges, functional surfaces and lettering, snap fitting, welding	11
3	Design principles of plastic products- gears, plastic bearings, piping, films, sheets, bottles, water tanks, wires and cables, Standard test methods of plastic products- pipe, film, sheet, PU foam, plastic optical goods, food packing containers.	11

4	Design fundamentals: rubber product design features, compound design principles of natural Rubber, Styrene Butadiene Rubber, Butadiene Rubber, Acrylonitrile Butadiene Rubber, Chloroprene Rubber, Ethylene Propylene Diene Monomer, Isobutylene Isoprene Rubber. Testing of Rubber Products: Rubber bearings, bridge bearings, rubber seals, static and dynamic seals, hoses, couplings, beltings, cables, rubber to metal bonded products.	11
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Outline methods of polymer product design.	K2
CO2	Apply methodology of design principles in various polymer products.	K3
CO3	Relate various theoretical and practical aspects of polymer properties and product requirements.	K1
CO4	Develop methods for qualitative and quantitative testing of polymer products.	K6
CO5	Discover various raw material, in process and product tests performed in various polymer products.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	1										
CO3	3	2										
CO4	3	2										
CO5	3	2										

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Plastics products design Handbook: materials and components	Miller, E. (Ed.)	Marcel Dekker	(Vol. 8). 1981
2	Plastics Product Design Engineering Handbook,	Levy, S.& DuBois, J.	Springer, Boston	1984
3	Product design with plastics: A practical manual.	Dym, J. B	Industrial Press Inc.	1983
4	Theory and practice of engineering with rubber.	Freakley, P. K. & Payne, A. R	Applied Science Publishers	1978

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Plastics: product design and process engineering,	Belofsky, H	Hanser	1995
2	Handbook of plastics testing technology	Shah, V	John Wiley & Sons	1998
3	Plastics Product Design	P. F. Mastro	John Wiley & Sons	2016
4	Designing successful products with plastics: fundamentals of plastic part design	M.T.MacLean-Blevins	Elsevier	2017
5	Designing plastic parts for assembly	P. A. Tres	Carl Hanser Verlag GmbH & Co	2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/LsR9qOTSz4g?si=kNOB4CrnaBb4dsW-
2	https://youtu.be/dcup4kRxSEs?si=bEBUPHPhfr3OVKIg
3	https://youtu.be/qgVs8vskWl0?si=ppHcp2ncApTAOWFl
4	https://youtu.be/9WPZStQp03Q?si=7-iJAN50QS9kGt4X

SEMESTER: S6

POLYMER TESTING

Course Code	PCPOT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Polymer Materials and Properties	Course Type	Theory

Course Objectives:

1. To provide in depth knowledge about focuses on the testing, analysis and characterization of polymeric materials.
2. Student will be able to understand the international testing standards.
3. Student will be able to perform testing of polymer products as per technically specified standards.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction – Need for testing, need for standards and specifications, national and international standards BIS, ASTM, BS, DIN, FDA etc. Specimen preparation and conditioning, standard for test specimen preparation like moulding, machining, stamping and punching of specimens, conditioning of specimen.	9
2	Mechanical properties: a) Short term strengths: Tensile properties, compression properties, flexural properties, shear properties, impact resistance, toughness, Tear resistance, abrasion resistance and hardness. b) Long term strengths: Dynamic stress and strain properties and their measurements, creep, stress relaxation, fatigue properties, flexing, and resilience, heat build- up, ageing properties.	9

3	Structural evaluation of polymers: Fourier transform infrared spectrometry, Ultraviolet - visible spectrometry, Nuclear magnetic resonance spectrometry, Mass spectrometry, X-ray diffraction spectrometry, Gas chromatography, Scanning Electron Microscopy, atomic force microscopy. Electrical properties: volume resistivity, surface resistivity, breaks down voltage, dielectric strength, dielectric constant, arc resistance, power factor.	9
4	Thermal properties of polymers: Glass transition temperature, melting temperature, heat distortion temperature, etc. Principle and uses of DTA, TGA, DSC and TMA. Optical properties: Gloss, haze, refractive index, transmittance. Other test: MFI, MVI, specific gravity, weathering properties, toxicity, resistance to chemicals, VST, HDT.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the need for testing of polymers for different standards and specifications.	K3
CO2	Summarize the specimen preparation with condition, shape and size of test specimen and significance of long-term testing methods	K2
CO3	Classify various mechanical testing methods	K4
CO4	Explain various spectroscopic testing methods	K3
CO5	Inspect the significance of thermal, electrical and optical testing methods	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		1									
CO2	3		1				2					
CO3	3											
CO4	3											
CO5	3		1		2							

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Physical Testing of Rubber	Brown, R.	Springer Science & Business Media	2006
2	Polymer Testing: New Instrumental	M.N. Subramanian, N.S.Murali Srinivasan	Chand	2018
3	Testing and evaluation of plastics	Mathur, A. B.	Allied Publishers	2003
4	Physical Testing of Plastics: Correlation with End-use Performance.	Evans, R., and Robert E. Evans, eds	ASTM International	Vol. 736. 1981

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Spectroscopy of Polymers	Jack L. Koenig	Elsevier	1918
2	Rubber Technology	Morton, M. (Ed.)	Springer Science & Business Media	2013
3	Polymer Testing	Grellmann, W	Hanser	2013
4	Handbook of polymer testing: physical methods,	Brown, R. (Ed.)	CRC press	1999
5	Plastics testing. -Handbook of Materials Selection.	Shah, Vishu.	Rapra	2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/rjt9ITI6L_Q?si=fY_KJX3yA-yo8vuI
2	https://youtu.be/0JWa0-o3RLc?si=WroqC-oD6s3gK-vm
3	https://youtu.be/blvNJWuDtQY
4	https://youtu.be/S_WiucxnWlo?si=dn6QISWRkAGwK6z6

SEMESTER S6

PROJECT ENGINEERING

Course Code	PEPOT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. To Understand the Comprehensive Process of Project Development and Management.
2. To Develop proficiency in Planning, Scheduling, and Execution Techniques.
3. To Acquire Skills in Financial and Economic Evaluation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Project Development and Engineering Classification of Projects, Scope of Project Engineering, Role of the Project Engineer Development of Projects: R&D, TEFR Importance of Laboratory Development: Bench Scale Experiments, Pilot Plant Studies, Semi-commercial Plant, Plant Location and Site Selection Preliminary Data for Construction Projects Process Engineering: Flow Diagrams, Plot Plans, Engineering Design and Drafting	9
2	Project Planning, Scheduling, and Execution Planning and Scheduling of Projects: Bar Chart, PERT/CPM, Critical Path Calculations Site Development: Foundation, Erection, and Site Fabrication Construction: Alignment, Insulation Start-Up and Commissioning: Trial Runs, Guarantees, and Hand Over	9

	Contracts and Contractors: Introduction, Detailing of Scope of Work, Factors in Selecting Scope of Work Contract Types: Detailing and Selection, Contractor Selection	
3	Project Financing and Equipment Selection Selection of Process Equipment: Standard vs. Special Equipment, Selection Criteria, Specification Sheets Project Financing: Means of Finance, Financial Institutions, Special Schemes Financing in India: Financial Institutions, Key Financial Indicators, and Ratios Personnel Recruitment and Training Economic Evaluation of Projects: Capital Requirements, Cost of Production, Profitability, Break-Even Analysis, Minimum Cost Analysis	9
4	Piping Engineering and Safety Scope of Piping Engineering, Pipe Sizing Technique, Codes and Standards in Piping Design Piping Design: Thermal Insulation and Buildings Safety in Plant Design	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the interconnected elements of project management from initiation to completion.	K2
CO2	Applying various project planning and scheduling techniques in practical scenarios.	K3
CO3	Analyzing economic data and evaluating financial indicators to support decision-making in project financing.	K4
CO4	Applying engineering design principles to develop detailed plans and specifications for projects	K3
CO5	Evaluating safety protocols and compliance with industry standards to ensure safe project implementation	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2					3		3			
CO2	3	2										
CO3	3	2										
CO4	3	2						2				
CO5	3	2						2	2			

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Project Management: A Systems Approach to Planning, Scheduling, and Controlling	Harold Kerzner	Wiley	12th Edition, 2020
2	Engineering Project Management	N.P. Panchal, P.K. Dey	PHI Learning Pvt. Ltd	1st Edition, 2011
3	Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design	Gavin Towler, Ray Sinnott	Butterworth-Heinemann	2nd Edition, 2013
4	Piping Engineering Leadership for Process Plant Projects	James Pennock	Elsevier	1st Edition, 2010
Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Project Engineering: The Essential Toolbox for Young Engineers	Louis J. Goodman	CRC Press	1st Edition, 2017
2	Handbook of Project-Based Management	Rodney Turner	McGraw-Hill Education	4th Edition, 2014
3	Chemical Process Engineering: Design and Economics	Harry Silla	Marcel Dekker	2nd Edition, 2002
4	Safety and Reliability in the Oil and Gas Industry: A Practical Approach	B.S. Dhillon	CRC Press	1st Edition, 2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Project Development and Engineering NPTEL Course: [Project Engineering and Management] (https://nptel.ac.in/courses/112/104/112104051/) SWAYAM Course: [Project Management for Engineering and Construction] (https://onlinecourses.nptel.ac.in/noc20_ce06/)
2	Project Planning, Scheduling, and Execution NPTEL Course: [Project Management] (https://nptel.ac.in/courses/110/102/110102012/) SWAYAM Course: [Project Planning and Control] (https://onlinecourses.nptel.ac.in/noc20_mg23/)
3	Project Financing and Equipment Selection NPTEL Course: [Project Economics and Decision Analysis] (https://nptel.ac.in/courses/110/104/110104134/) SWAYAM Course: [Project Finance and Budgeting] (https://onlinecourses.nptel.ac.in/noc20_mg24/)
4	Piping Engineering and Safety NPTEL Course: [Piping Engineering and Pipeline Design] (https://nptel.ac.in/courses/103/104/103104075/) SWAYAM Course: [Safety in Process Plant Design] (https://onlinecourses.nptel.ac.in/noc20_ch11/)

SEMESTER S6

POLYMERS & ENVIRONMENT

Course Code	PEPOT632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Impact of polymers on environment
2. Remedial measures to reduce the harm caused by polymers

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Impact of polymers on environment: Introduction, resources of polymers, comparison of total energy costs for product manufacture based on polymers, ceramics and metals, polymer production and consumption, comparison of the impact of polymers and other materials on environment. Environmental pollution, non-biodegradability of polymers, drawbacks, burning of polymers, remedy for environmental problem by polymers, awareness program, need for recycling of plastics from urban solid wastes.	9
2	Plastic waste segregation: Sorting and segregation of waste, plastics identification, SPI coding. Primary recycling, equipments for primary recycling, specific recycling techniques, PE films, PP battery case, crushing and separation, PET films, Secondary recycling, plastics wastes containing paper, hydrolytic treatment	9
3	Rubber and mixed waste: Processing of mixed plastics waste, household waste, industrial sector, TPO based materials, use of recyclable plastics in motor vehicles, recoverable	9

	material, disposal of residuals, recycling of rubber, difficulty in recycling of rubber products, reclaimator process, WTR, advantages of reclaimed rubber in cost, mixing and processing	
4	Bio-degradation of polymers: Introduction, enzymes, enzyme nomenclature, enzyme specificity, physical factors affecting the activity of enzymes, enzyme mechanism, chemical degradation initiates biodegradation, hydrolysis of synthetic biodegradable polymers. Development of starch-based polymers, manufacture of master batch, biodegradation, assessment, bio polyesters, synthesis, isolation, solvent extraction, properties, degradation, intracellular biodegradation, extra cellular biodegradation, thermal degradation, hydrolytic degradation, environmental degradation	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the impact of polymers and other materials over environment	K2
CO2	Explain the impact of plastic waste on environment	K2
CO3	Outline different types plastic recycling methods	K2
CO4	Organize treatment methods for rubber recycling	K3
CO5	Appraise the prospects of biodegradable plastics	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		1									
CO2	2	2										
CO3	3		2									
CO4	3			3								
CO5	3		3									

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Recycling of plastic materials	La Mantia, F. P.,	Chem Tec Publishing	1993
2	Polymers: the environment and sustainable development.	Azapagic, A., Emsley, A., & Hamerton, I.,	John Wiley & Sons.,	2003
3	Polymers, Environmentally Degradable. Encyclopedia of Chemical Technology.	Swift, G., Brown, R.		2000

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Degradable polymers: principles and applications	Scott, G.	Springer Science & Business Media	2002
2	Chemical degradation methods for wastes and pollutants: environmental and industrial applications	Tarr, M. A.	CRC Press	2003
3	Plastics and the Environment	A. L. Andrady	John Wiley & Sons	2003
4	Biodegradable and biobased polymers for environmental and biomedical Applications	Kalia, Luc Avérous	John Wiley & Sons	2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1- 4	https://www.youtube.com/watch?v=SlEyf0aUEDE&t=52s NPTEL-NOC IITM Abhijit P Deshpande

SEMESTER: S6

CHEMICAL ENGINEERING THERMODYNAMICS

Course Code	PEPOT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic knowledge in Thermodynamics	Course Type	Theory

Course Objectives:

1. Students will grasp the fundamental principles and laws of thermodynamics.
2. Students will become proficient in using P-V and P-T diagrams for pure substances, and equations of state for real gases
3. To understand phase equilibria and chemical reaction equilibrium.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Thermodynamics Thermodynamic Systems: closed, open, and isolated systems, Intensive and Extensive Properties, Path and State Functions, Reversible and Irreversible Processes Zeroth Law of Thermodynamics- First Law of Thermodynamics: Energy Balance for Closed Systems, Limitations of First Law- Second Law of Thermodynamics: Carnot's Principles, Definition of Entropy, Calculation of Entropy Change in Processes Involving Ideal Gases Third Law of Thermodynamics, Energy Balance of Open Systems: Flow through Pipes, Nozzles, Compressors	9
2	Thermodynamic Properties and Equations P-V and P-T Diagrams of Pure Substances	9

	Equations of State for Real Gases: van der Waal's, Redlich-Kwong, Peng-Robinson, and Virial Equations, Principle of Corresponding States and Generalized Compressibility Chart, Fundamental Property Relations Maxwell's Equations, Clausius-Clapeyron Equation Entropy-Heat Capacity Relationships: Equations for Entropy, Internal Energy, and Enthalpy in Terms of Measurable Quantities Joule-Thomson Coefficient, Gibbs-Helmholtz Equation Fugacity and Activity of Pure Fluids: Selection of Standard State, Effect of Temperature and Pressure	
3	Solutions and Phase Equilibria Definition of Partial Molar Properties and Chemical Potential, Fugacity in Solution - Ideal Solution: Lewis-Randall Rule, Raoult's Law, Henry's Law, Activity in Solutions Activity Coefficient: Effect of Temperature and Pressure, Gibbs-Duhem Equations, Property Change of Mixing, Excess Properties Criterion of Phase Equilibria: Single Component and Multicomponent Systems, Criterion of Stability, Phase Rule for Non-reacting Systems,	9
4	Vapor-Liquid Equilibria and Chemical Reaction Equilibria Vapor-Liquid Equilibrium (VLE): Phase Diagram for Binary Solutions VLE in Ideal and Non-Ideal Solutions: VLE at Low Pressures Activity Coefficient Models: Wohl's Equation, Margules Equation, van Laar Equation, Wilson Equation (General Concepts Only) Equations of VLE at High Pressures, Definition of Vaporization Equilibrium Constant, Bubble Point, Dew Point, and Flash Calculations in Multicomponent Systems (Derivations Only) Chemical Reaction Equilibria: Extent of Reaction, Equilibrium Constant, Standard Free Energy Change, Feasibility of Reaction, Effect of Temperature on Equilibrium Constant, Evaluation of Equilibrium Constant	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply Thermodynamic Principles and laws	K3
CO2	Analyze Thermodynamic Systems and Processes	K4
CO3	Evaluate Thermodynamic Properties and Equations	K5
CO4	Design and Predict Phase Equilibria	K5
CO5	Create Solutions to Complex Thermodynamic Problems	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									3
CO2	3	2	2									3
CO3	3	2	2	3								3
CO4	3	3	3	3	2							3
CO5	3	2	2	3	2							3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Chemical Engineering Thermodynamics	Smith J. M. & Van Ness H.V.	McGraw Hill, 2018.	8th Edn, 2018
2	A Textbook of Chemical Engineering Thermodynamics	Narayanan K. V.	Prentice Hall of India,	2nd Edn., 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Chemical and Engineering Thermodynamics,	Stanley I. Sandler,	John Wiley Sons,USA	2nd Edn1989
2	Chemical and Process Thermodynamics	Kyle B.G.,	Prentice-Hall,	3rd Edn, 1999
3	Chemical Engineering Thermodynamics	Y.V.C. Rao	Universities Press,	1997
4	Engineering and Chemical Thermodynamics	Milo D. Koretsky	Wiley	2nd Edn, 2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Fundamentals of Thermodynamics NPTEL Video Link: [Fundamentals of Thermodynamics] (https://nptel.ac.in/courses/103/104/103104011/)
2	Thermodynamic Properties and Equations NPTEL Video Link: [Thermodynamic Properties and Equations] (https://nptel.ac.in/courses/103/104/103104018/)
3	Solutions and Phase Equilibria NPTEL Video Link: [Solutions and Phase Equilibria] (https://nptel.ac.in/courses/103/104/103104021/)
4	Vapor-Liquid Equilibria and Chemical Reaction Equilibria NPTEL Video Link: [Vapor-Liquid Equilibria and Chemical Reaction Equilibria] (https://nptel.ac.in/courses/103/104/103104022/)

SEMESTER: S6

ADDITIVE MANUFACTURING

Course Code	PEPOT634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To educate students with fundamental and advanced knowledge in the field of additive manufacturing technology

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction and basic principles: Need for Additive Manufacturing, Generic AM process, the benefits of AM, distinction between AM and CNC machining, classification of AM processes, metals systems, hybrid systems, the eight steps in additive manufacture, variations from one AM machine to another, materials handling issues, design for AM, and application areas.	9
2	Photo polymerization processes: Stereo lithography (SL), Materials, SL resin curing process, Benefits and Drawbacks, Applications of Photo polymerization Processes. Powder bed fusion processes: Selective laser Sintering (SLS), Materials, Powder fusion mechanism, SLS Metal and ceramic part creation, Electron Beam melting (EBM), Process Benefits and Drawbacks, Applications of Powder Bed Fusion Processes. Extrusion-based systems Fused Deposition Modelling (FDM), Principles, Materials, Bio-Extrusion, Process Benefits and Drawbacks, Applications of Extrusion-Based Processes. Sheet Lamination AM Processes: Bonding	9

	Mechanisms, Materials, Laminated Object Manufacturing (LOM), Ultrasonic Consolidation (UC), Gluing, Thermal bonding, LOM and UC applications.	
3	Additive manufacturing systems: Powder Bed Fusion AM Processes: Selective laser Sintering (SLS), Materials, Powder fusion mechanism and powder handling, SLS Metal and ceramic part creation, Electron Beam melting (EBM), Process Benefits and Drawbacks, Applications of Powder Bed Fusion Processes. Directed Energy Deposition AM Processes: Process Description, Material Delivery, Laser Engineered Net Shaping (LENS), Direct Metal Deposition (DMD), Electron Beam Based Metal Deposition, Benefits and drawbacks, Applications of Directed Energy Deposition Processes. Materials science for AM - Multifunctional and graded materials in AM, Role of solidification rate, microstructural studies, Structure property relationship, Three-Dimensional Printing– Principle, process, advantages and applications-	9
4	Choosing materials for manufacturing: Multiple materials, metal AM processes, Composite Materials, Biomaterials, Hierarchical Materials, Ceramics & Bio-ceramics, Shape-Memory Materials, 4D Printing, Advanced AM Materials. Medical and bio additive manufacturing, Customized implants and prosthesis: Design and production, Bio-Additive Manufacturing- Computer Aided Tissue Engineering (CATE), Case studies	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the knowledge of the broad range of AM processes, devices, capabilities and materials that are available.	K2
CO2	Analyze the various software tools, processes and techniques that enable advanced/additive manufacturing.	K4
CO3	Apply the concepts of additive manufacturing to design and create components that satisfy product development/prototyping requirements, using advanced/additive manufacturing devices and processes.	K3
CO4	Apply the latest trends and business opportunities in additive manufacturing.	K3
CO5	Build characterization techniques in additive manufacturing.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2		1		2					1
CO2	2	1	2	3		1			1			1
CO3	2		2		2		2					1
CO4	2	1		2				2	2		2	1
CO5	2	3		1		1			2			1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rapid Prototyping: Theory & Practice	Ali K. Kamrani	Springer	2006
2	Additive Manufacturing Technology	Hari Prasad	A.V.Suresh Cengage	2019
3	Rapid Prototyping: Laser-based and Other Technologies	Patri K. Venu Vinod and Weiyin M	Springer	2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Understanding additive manufacturing: rapid prototyping, rapid tooling, rapid manufacturing	Andreas Gebhardt	Hanser Publishers	2011
2	Rapid Prototyping: Principles & Applications	Chua Chee Kai, Leong Kah Fai	World Scientific	2003
3	Rapid Manufacturing: The Technologies and Applications of Rapid Prototyping and Rapid Tooling	D.T. Pham, S.S. Dimov	Springer	2001
4	Rapid Prototyping: Principles and Applications in Manufacturing	Rafiq Noorani	John Wiley & Sons	2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/EHvO-MlzAIM?si=2F_n-SPEwglhmwzq
2	https://youtu.be/_K1C_dkZK98?si=Ce1o4R4h9knKVnwd https://youtu.be/skAQavea3H4?si=meGJD079z2qb6hHy
3	https://youtu.be/_0dSVAElxk?si=RpT_-ya2rxgt934h
4	https://youtu.be/_pilQ7oNtuo?si=hV_mAJ4FAJ8gm0bm

SEMESTER: S6

ENVIRONMENTAL IMPACT ANALYSIS

Course Code	PEPOT636	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide in depth knowledge from methodology of impact analysis, socio impact analysis, air quality impact analysis and water quality impact analysis.
2. Understand the fundamentals of environmental impact and factors for consideration in assessing environmental impact analysis.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concepts of environmental impact analysis -Environmental protections, legislations, laws and Acts-air quality legislation-energy legislation-fish and wild life resources.	9
2	Legislation -historical preservation legislation Factors for consideration in assessing environmental impact concept short term vs. long term effects. Socio impact analysis-physical, social, aesthetic and economic environment-examples of types of socio impact analysis.	9
3	Air quality impact analysis - air pollutants-sources-atmospheric interactions-environmental impact-assessment methodology, case studies. Noise impact analysis-effects of noise on people-estimating transportation noise impact-examples.	9

4	Water quality impact analysis -water quality criteria and standards-modelling-water quality impact by projects like High ways, power plants, agriculture and irrigation, forest management, vegetation and wild life impact analysis. Assessment methodologies- impact on biota-summarisation of environmental impact-checklist method.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of impact analysis	K2
CO2	Summarize the significance of environmental impact analysis.	K2
CO3	Demonstrate Socio impact analysis	K2
CO4	Examine the Relationship between water quality criteria and standards.	K4
CO5	Compare Assessment methodologies	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3								2			1
CO2	3								2			1
CO3	3								2			1
CO4	3								2			1
CO5	3								2			1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Environmental impact analysis handbook.	Rau, J. G., & Wooten, D. C.	McGraw Hill	1980
2	The EIS book: Managing and preparing environmental impact statements.	Eccleston, Charles H.	CRC Press	2013
3	Strategic environmental assessment in action.	Therivel, Riki.	Routledge	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managing water quality: economics, technology, institutions.	Kneese, A. V., & Bower, B. T.	Rff Press.	2013
2	Introduction to environmental impact assessment.	Glasson, J., & Therivel, R.	Routledge	2014
3	Book of Environmental Science.	Thakur, Vidya.	Scientific Publishers	2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/E_oRwSRdgcQ?si=daLas9URNIaGv8tB
2	https://youtu.be/PrFskfLGHG8?si=aBDfbk5utvJrRR3k
3	https://youtu.be/_mLdjd7WqU4?si=wcQ6nzTwxLABgEKH
4	https://youtu.be/VlyqdbH-GbQ?si=lsjwkEbY9W-XTUi6

SEMESTER: S6

ELECTROCHEMICAL ENGINEERING

Course Code	PEPOT637	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Students should achieve sufficient knowledge about the basics of electrochemical engineering.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Electrochemical Engineering :- Fundamentals of Electrical Engineering, Electrostatics, Electrodynamics, Electrical Circuit Theory, Fundamentals of Electrochemical Systems, Elements of Electrochemical Systems, Electrolyte Transport processes within electrolytes, Empirical laws of electrolyte conductivity. The concept of pH and the idea of buffer solutions	9
2	Elements of Electrochemical Systems: - Electrode Electrochemical Thermodynamics, Nernst Equation, Reference Electrodes, Cell Potential, Electromotive Force, Electrode Kinetics	9
3	Electrical Double Layers Theory: - & Models Electro kinetic Phenomena, Electrochemical Energy Engineering Batteries, Fuel Cells, Electrical, & Electrochemical Capacitors	9
4	Industrial Electrochemical Processes: - Fundamentals, Reactor Design, & Parameter Chlor-Alkali Process, Electrochemical Extraction of Metals, Hall Heroult Process	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the fundamentals of electrochemistry.	K2
CO2	To interpret the fundamental knowledge electrochemical engineering	K2
CO3	Evaluate the potential of electrochemical systems based on thermodynamic data	K5
CO4	Utilize technical measures to characterize properties of galvanic elements and capacitors.	K3
CO5	Explain the significance of major industrial electrochemical processes and electrochemical reactor design	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											
CO5	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fuel Cell Engines	M. Mench	John Wiley, New York	2008.
2	"Electrochemical Supercapacitors: Scientific Fundamentals and Technological Applications"	B. E. Conway,	Kluwer Academic / Plenum publishers, New York	1999

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Lithium Batteries – Science and Technology”	Gholam Abbas Nazri	Springer, NewYork,	2009
2	“Lead – Acid Batteries: Science and Technology”	D.Pavlov	Elsevier, Amsterdam	2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://www.youtube.com/watch?v=FrIYUyxV4Nw&list=PLFppcO11RhKjvtKQ-eYKgDG0lNzLAILdq

SEMESTER: S6

PETROLEUM REFINERY ENGINEERING

Course Code	PEPOT638	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart knowledge of processes like cracking, reforming, alkylation, coking and asphalt technology.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Origin of Petroleum: - Origin- exploration and production of petroleum, testing of petroleum products, types of crudes, composition, characteristics, product pattern, crude heating, distillation principles, stripping, atmospheric and vacuum distillation, heat balance and material balance in distillation, types of trays, quality control	9
2	Refining processes: - Cracking, Thermal Cracking and Vis breaking, Hydro Cracking- principles, reactions in Hydro Cracking, Catalyst, Hydro Cracking reaction Conditions. Different Feed Stocks, Products Yields, Qualities and Recent Development. Catalytic Cracking, Commercial Catalyst, Feedstock and Catalytic Cracking Conditions, Types and Processes- Fixed Bed Cracker, Fluid Catalytic Cracking (FCC), Flexi Cracking.	9
3	Catalytic reforming &coking: - Theory, Reaction Conditions and Catalyst for Catalytic Reforming, Platforming, Hurd Forming, Rheine Forming, Power Forming, Selector Forming. Ultra-Forming and Rex Forming.,	9

	Feedstock Selection and Effect of Steam. Coking Methods of Petroleum, Coke Production, Delayed Coking, Fluid Coking and Contact Coking.	
4	Asphalt technology: - Source of Asphalt (Bitumen), Chemical Structure of Asphalt, Action of Heat on Asphalt, Types of Asphalts. Air Blowing of Bitumen and Upgradation of Heavy Crudes. Specialty Products: Industrial Grease- Manufacture of Calcium Grease, Liquid Paraffin and Petroleum Jellys.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Illustrate information about origin, exploration and production of petroleum crude and understand their properties with the help of standard testing protocols	K3
CO2	Analyze various primary crude processing techniques like distillation, de-asphalting, hydrocracking, hydro treating, and their supporting processes	K4
CO3	Analyze the process technologies for reforming, isomerization, alkylation and polymerization unit process	K4
CO4	Explain various primary crude processing techniques, various secondary thermal cracking	K2
CO5	Explain asphalt technology	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	2		1		2					1
CO2	2	1	2	3		1			1			1
CO3	2		2		2		2					1
CO4	2	1		2				2	2		2	1
CO5	2	3		1		1			2			1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Petroleum Refining Technology	Ram Prasad	Khanna Publishers,	2007
2	Handbook of petroleum processing	Jones, D.S.J. and Pujadó, P. R	Springer	2006
3	Petroleum Refinery Engineering	Nelson, W. L	McGraw Hill Publishing Company Limited	1985

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Petroleum Refinery Distillations	Watkins, R. N	Gulf Publishing Company, Texas	2nd Edition 1981
2	Modern Petroleum Refining Technology	Hobson, G. D	John wily Publishers	5th Edition 1984
3	Petroleum Refining Technology and Economics	Gary, J.H and Handwork G.E	Marcel Dekker	4th Ed 2001

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/lu3cthdVDqQ?si=js5pSMwXjzwwUOUo
2	https://youtu.be/vD0kbdIS6kE?si=Y69sNPrKKYkcMcV-
3	https://youtu.be/-BrhL4HB9z0?si=P-a8vEXSCnab240s
4	https://youtu.be/1t6Dy2qpWHA?si=_wX0xndJ22kf_HOp

SEMESTER: S6

ADVANCED NANOCOMPOSITES

Course Code	PEPOT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Knowledge in Polymer Science	Course Type	Theory

Course Objectives:

1. Aims to impart knowledge on advanced nanocomposite materials.
2. To gain ability to design nanocomposites based on the need and specifications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Nanocomposites & Nanofillers: Overview of polymer nanocomposites; Classification of nanofillers: Nano clay, nano silica, nanoparticles, carbon based nanofillers; Synthesis and properties of fillers. Synthesis of Nanocomposites: Direct mixing, solution mixing, in-situ polymerization, ex-situ processing; Ceramic/polymer composites; Metal/polymer nanocomposites; Modification of interfaces; Modification of nanofillers.	9
2	Particulate polymer composites Particulate polymer composites: Characteristics for a particulate reinforcer, its selection and its surface coatings, mineral, metallic and organic particulate reinforcers, processing of particulate polymer composite and product development, mechanics, models and equations for Young's Modulus of the composites, applications of particulate composites	9
3	FRP Composites Synthetic FRP composites, natural FRP composites; Short fiber composites; processing of short FRP composite, product development and manufacturing, collimated fiber compounds, fiber length distribution on the composites, short	9

	natural FRP composites, processes and manufacturing, applications, hybrid composites, Mechanical properties of short FRP composites. Continues FRP composites; Natural, inorganic and synthetic continuous FRP composites, processing and manufacturing of thermoset FRP composites by Vacuum Bag Molding, Vacuum infusion Molding, Resin Transfer Molding, Pultrusion and Filament Winding, structural and other applications of thermoset composites. Limitations of Processing and manufacturing, Mechanical and thermal properties of continues FRP composites, Applications	
4	Properties, Characterisation & Applications – Thermal, electrical & optical properties, mechanical properties, morphological & barrier properties. Applications of polymer nanocomposites in space, automobiles, defence, electronics, packaging, bio-medical engineering, filtration, coatings and cosmetic. Microstructure evaluation by scanning electron and optical microscopes. Structure evaluation by FTIR, NMR, C-13 NMR, UV. Elemental analysis – qualitative and quantitative, Dielectric measurements, Diffusion properties, Thermo-mechanical properties, DMA creep, thermal relaxations., contact angle measurements.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. Problem Definition (2 points)

- Check for clarity and completeness of the problem definition.
- Check for clear objectives and divide into sub objectives

2. Problem Analysis (4 points)

- Examine the problem and identify the variables and responses
- Divide the problem to manageable parts

3. Implementation of software (2 points)

- Use suitable software for getting the solution
- Check for the appropriateness of tools used

4. Evaluate (4 points)

- Evaluate the proposed solution.
- Assess the correctness of solution.

5. Validation of the results (4 points)

- Validate the results using available data.
- Do the error analysis of the results obtained.

6. Conclusion and presentation (4 points)

- Summarize the results obtained
- Do critical thinking of the results obtained and evaluate.
- Submit a technical report

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the various synthetic methods of preparing nanocomposites	K2
CO2	Distinguish between the various types of nanofillers used in making nanocomposites	K4
CO3	Explain the concept of polymer particulate composites	K2
CO4	Categorize short and continuous polymer composites, FRP systems, various manufacturing process and characterization techniques	K4
CO5	Recommend various properties, applications and characterisation techniques of nanocomposites.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1		1								
CO2	3											
CO3	3	2		2			1					
CO4	3	3	3									
CO5	3	1		2								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fiber-reinforced Composites	P.K. Mallick	CRC Press	3rd Edn, 2008.
2	Polymer Nanocomposites: Processing, Characterization and Applications	Joseph H. Koo		2021.
3	“Nanocomposite Science and Technology”	Ajayan P.M., Schadler, L.S., Braun, P.V.	Wiley-VCH Verlag GmbH Co. KGaA, Weinheim.	2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer Nanocomposites	Mai Y.W. and Yu Z.Z.,	Woodhead Publishing Limited and CRC Press LLC, USA.	2006
2	Handbook Of Technology and Engineering of Reinforced Plastics Composites.	Mohr, J.	SPI Publishing	1984
3	Hand Book of Polymer-Fibre Composites	F.R. Jones	Longman Scientific and Technical	1994.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://www.youtube.com/watch?v=5Zlx6FCbues Prof.Prathap Haridoss, IIT Madras https://www.youtube.com/watch?v=649flwvIvRc&list=PLwdnzlV3ogoVE2AIC-G4Uew8XsaINwJGo Prof. Debabrata Chakraborty, IIT Guwahatti

SEMESTER: S6

POLYMER BLENDS & COMPOSITES

Course Code	PBPOT604	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart basic knowledge on blends and composite materials.
2. To impart knowledge on various composite manufacturing processes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: preparation of polymer blends, thermodynamic criteria for polymer miscibility, specific interactions, copolymer effect, phase separation, spinodal decomposition, nucleation and growth, phase diagram, morphology, blend characterisation techniques, Structure-property relationship, rubber plastic blends, phase morphology, properties of blends prepared by dynamic vulcanization, technological application,	11
2	Thermoplastic styrene block copolymers: polyester thermoplastic elastomers, thermoplastic polyurethane elastomers, basic structure, manufacture, morphology, commercial grades, applications, thermoplastic 1,2- polybutadiene, trans 1,4-polyisoprene, ionic thermoplastic elastomers, silicone based thermoplastic elastomers, polyamide 1,2-elastomers. Introduction to particulate and fibre filled composites, applications, function of matrix, function of fibres, polymer-fibre interface, factors influencing the performance of composite	11

3	Coupling agents, bonding agents, short fibre composites, continuous fibre composites, analysis of long fibre composites, analysis of short fibre composites, critical fibre length, rule of mixtures preparation and properties of glass fibre, carbon fibre and aramid fibre, polymer concrete, polymer impregnated concrete	11
4	Polymeric binders for rocket propellants: composite manufacturing techniques, hand lay-up, spray-up, compression moulding, vacuum bag moulding, pressure bag moulding, filament winding, resin transfer moulding, pultrusion, Reinforced Reaction Injection Moulding	11

Suggestion on Project Topics

1. After completion of first module students are able to prepare blend by any one of the methods
2. After completion of second module students are able to collect the data of miscible and immiscible blend and familiarize different types of fibres used in composites.
3. After completion of third module students are able to produce polymer concrete.
4. After completion of fourth module students are able to prepare one FRP product by hand lay-up process.

Course Assessment Method (CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Ex-1	Internal Ex-2	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Course Outcomes (COs)

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 6 marks. (4x6 = 24 marks)	40

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Categorize the miscibility of polymer blends.	K4
CO2	Define the structure property relationships of polymer blends.	K1
CO3	Utilize the formulation of long and short fibre composite.	K3
CO4	Explain the significance of composite manufacturing.	K5
CO5	Discuss the applications of various types of polymer blends and composites.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer Alloys and Blends	Leszek A Utracki	Hanser, Newyork	1989
2	Polymer-polymer Miscibility	O. Olabisi, I.W. Robeson, and M.T. Shaw	Academic Press	1979
3	Polymer Blends'	Paul S. Newman	Academic Press,	1978
4	Composite, materials engineering and science	F L Matthews and R D Rawlings	Chapman and Hall, London	1994
5	Hand book for Polymer Blends' Composite	A K Kulshreshtha, Comelia Vasile	Rapra Technology	2003
6	An introduction to Composite Materials	T W Clyne, D Hull	Cambridge University Press	2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer science and technology	Joel R, Fried	Prentice Hall	2014
2	Fundamentals of polymer engineering	Anilkumar Rajesh K Gupta	CRC Press	2018
3	Elements of polymer Science and engineering	Alfred Rudin, Philp Choi	Elsevier	2013
4	Polymer Blends' and Composite	Muralisrinivasan Natamai Subramanian	Scrivener	2017
5	Textbook of polymer Science	Fred W Billmeyer	Wiley-Interscience	1984

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/D0I7ukw18FE
2	https://youtu.be/uiz-6A27EZ8
3	https://youtu.be/xAS4NS9RuI4
4	https://youtu.be/yERdn82iDwQ

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER: S6
LATEX PRODUCTS LAB

Course Code	PCPOL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To develop practical skill for latex products manufacturing
2. To develop practical skill for latex compounding

Details of Experiment

Expt. No	Experiment
1	Creaming of NR latex
2	Preparations of dispersion, slurry and emulsions.
3	Manufacture of rubber bands, balloons, finger caps, household gloves, surgeons' gloves, latex threads and articles by casting
4	Heat sensitized dipping
5	Latex impregnation in textiles
6	Preparations of SP, CV and LV rubber
7	Preparations of latex-based adhesives
8	Work practices in the production latex foam
9	Work practices in the production of bonded coir

Course Assessment Method (CIE: 50 Marks, ESE 50 Marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record. (Continuous Assessment)	Internal Exam	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Mandatory requirements for ESE:

- Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- Endorsement by External Examiner: The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course the student will be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop skill to concentrate the latex by creaming.	K6
CO2	Determine the skill to prepare compounding ingredients	K5
CO3	Analyze skill in the preparation of latex compounds	K4
CO4	Illustrate the properties of latex with manufacturing methods	K2
CO5	Define teamwork skills through group activities	K1

K1- Remember, K2- Understand, K3- Apply, K4- Analyze, K5- Evaluate, K6- Create

CO-PO Mapping Table

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2		2									
CO2	2		2									
CO3	2		2									
CO4	2	3	2									
CO5	2		2									

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), : No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer Latices: Science and technology	D C Blackley	Springer Science,	1997
2	Latex foam rubber	E W Madge	Maxlauren&Sons	
3	High polymer latices Vol 1& II	D.C. Blackley	Applied Science	
4	Rubber Technology and Manufacture	C M Blow	Butterworth	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Practical Guide to Latex Technology	Rani Joseph,	Smithers Rapra,	2013
2	Polymer Latices: Science and technology	D.C. Blackley,	Springer Science,	1997
3	Rubber Technologist's Handbook, Vol 1,	S. K. De and J. R. White	Smithers Rapra,	2001
4	ASTM and BIS standards			1898

Video Links (NPTEL, SWAYAM...)	
Sl. No.	Link ID
1	https://youtu.be/DY9lmGdv40A
2	https://www.youtube.com/shorts/MVygZw-ucpM?feature=share

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

POLYMER ENGINEERING

SEMESTER: S⁷

FIBRE SCIENCE & TECHNOLOGY

Course Code	PEPOT 741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Develop knowledge on manufacturing, testing and applications of fibres
2. Impart knowledge on various kinds of fibres.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction, definition of fibre, fibre dimensions, units of measurements, tex,denier, conversion from one system to another, use of fibres in the rubber and plastic industry, polymer products containing fibres, fibre manufacturing industries in India, present status and future prospects Fibres used in polymer industry-types and sources, chemical composition, properties, concept of order in fibres and polymers, crystallinity and orientation, orientation and fibre properties	9
2	Methods of investigating fibre structure. use of animal and mineral fibres. Detailed study of fibre properties such as mechanical properties, electrical properties, moisture absorption, moisture regain optical properties and fibre friction, relation between fibre properties and structure, uses of vegetable fibres such as cotton, flax, linen, coir, sisal, pineapple, jute, silk, banana	9
3	Manmade fibres, physical structure of fibre forming polymers, production, chemical composition, properties and testing of viscose rayon, cellulose acetate,	9

	nylon66 and nylon6, polyester, acrylic, poly vinyl alcohol, spandex, carbon fibre, metallic fibres, saran, Kevlar, Nomex, steel wire, finishing and dyeing.	
4	Yarn and textile production from fibres, definition of various textile terms, fibre spinning, drawing of fibres, felting, knitting, lacing, bonding and weaving, properties of yarns, yarn numbering system, brief idea about spinning of cotton yarns and blends .Yarn properties such as count appearance, evenness, strength, abrasion, fatigue, friction, twist, dynamic mechanical properties and their importance, fabrics, brief idea about the construction, cover factor, geometry of weaves, fabric properties such as strength, abrasion resistance, air permeability, bursting strength, thermal properties	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define the fundamentals of fibre science.	K1
CO2	Explain the manufacturing, testing and applications of fibres.	K5,K2
CO3	Analyse the classification of fibres.	K4
CO4	Create different types of spinning operations.	K6
CO5	Apply the various operations of making fabric.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											
CO5	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Man Made Fibres	R W Moncrief	Newnes-Butterworths	1975
2	Textiles Fibre to Fabric	Bernard P Corbman	McGraw-hill	VI Edn, 1983
3	Principles of textile testing	J E Booth	Butterworths,Newness	III rd ,1961
4	Fabric Science	Seema Sekhri	McGraw-hill	IV Edn, 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Textile reinforcement of Elastomers,	Wake and D. B. Wooton	Applied Sciences	1982
2	Text Book of Polymer Science	F.W. Billmeyer	Wiley– Inter science,	1971
3	Chemical Technology of Fibres and Materials	Sadow	Mir publishers	1978
4	Textile Yarns	B C Goswami, J G Martindale, F L Scardino	Wiley India	2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/P7qirqjzNao?si=ATDtSnBAGIA4K4pu
2	https://youtu.be/HkDhXIrz2Bk?si=ynNZbKWcI-0esZjg
3	https://youtube.com/shorts/NKfkVt2MZ2Y?si=tbu0AH0bjy5w14Up
4	https://youtu.be/k2GKVG-2qeM?si=iGCjKf_Le9UPF

SEMESTER: S7

COMPUTER AIDED DESIGN & MANUFACTURE

Course Code	PEPOT 742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Understand the Foundations of CAD Systems.
2. Develop Skills in Geometric Modeling and Transformations.
3. Master Computerized Manufacturing and CNC Programming.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to CAD (Computer aided design): CAD system architecture- Packages for CAD/CAM/CAE/CAPP. Hardware in CAD components, user interaction devices, design database, graphic Standards, data Exchange Formats Geometric transformations: Transformation of points and line, 2-D rotation, reflection, scaling and combined transformation, homogeneous coordinates, 3-D scaling. Shearing, rotation, reflection and translation, combined transformations. Algebraic and geometric forms, tangents and normal, blending functions, re-parametrization, straight lines, conics, cubic splines, Bezier curves and B-spline curves.	9
2	Geometric Modeling: Plane surface, ruled surface, surface of revolution, tabulated cylinder, bi cubic surface, Bezier surface, B-spline surfaces and their modelling techniques.	9

	Solid models and representation scheme, boundary representation, constructive solid geometry. Sweep representation, cell decomposition, spatial occupancy enumeration, coordinate systems for solid modelling.	
3	Computerized Manufacture Planning and Control System: Computer Aided Process Planning, benefits of CAPP, Production Planning and Control Systems, typical activities of PPC System, computer integrated production management system, Material Requirement Planning, inputs to MRP system, working of MRP, outputs and benefits. Flexible Manufacturing Systems: Fundamentals of Group Technology and Flexible Manufacturing Systems, types of FMS, FMS components, Material handling and storage system, applications, benefits, computer control systems, Automated Storage and Retrieval Systems, AS/RS and Automatic parts identification systems and data capture.	9
4	Computer Numerical Control: Introduction, components of CNC, CNC programming, manual part programming, G Codes, M Codes, programming of simple components in turning, drilling and milling systems, programming with canned cycles. Cutter radius compensations. Robot Technology: Robot anatomy, joints and links, common robot configurations, robot control systems, accuracy and repeatability, end effectors, sensors in robotics. Robot programming methods: on-line and offline methods. Robot industrial applications: material handling, processing and assembly and inspection.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the Architecture and Components of CAD Systems	K2
CO2	Apply Geometric Transformations in 2D and 3D Modeling	K3
CO3	Create Complex Geometric Models Using Advanced Modeling Techniques	K6
CO4	Analyze and Develop Computerized Manufacturing Plans	K4
CO5	Program and Operate CNC Machines and Robots for Manufacturing Tasks	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			3							
CO2	3	2	3		3							
CO3	3		3		3							
CO4		3		3	2			2				
CO5	3		2		3	2	1					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	CAD/CAM: Computer-Aided Design and Manufacturing	Mikell P. Groover and Emory W. Zimmers Jr.	Prentice Hall	1st Edition, 1983
2	Principles of Computer-Aided Design and Manufacturing	Farid Amirouche	Prentice Hall	1st Edition, 2004
3	Automation, Production Systems, and Computer-Integrated Manufacturing	Mikell P. Groover	Prentice Hall	4th Edition, 2014
4	Computer Graphics and Geometric Modeling for Engineers	Vera B. Anand	John Wiley & Sons	1st Edition, 1993

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to CAD/CAM	Ibrahim Zeid and R. Siva subramanian	McGraw-Hill	1st Edition, 1995
2	Computer-Aided Manufacturing	Tien-Chien Chang, Richard A. Wysk, and Hsu-Pin Wang	Pearson	3rd Edition, 2005
3	Computer-Aided Engineering Design	Anupam Saxena and Birendra Sahay	Springer	1st Edition, 2005
4	Numerical Control and Computer-Aided Manufacturing	T. Kundra, P. N. Rao, and N. K. Tewari	McGraw-Hill	1st Edition, 1993

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Title: Introduction to CAD Professor: Dr. Praveen Kumar Institute: Indian Institute of Technology (IIT) Madras Link: https://nptel.ac.in/courses/112/106/112106256/
	Title: Geometric Transformations Professor: Dr. Praveen Kumar Institute: Indian Institute of Technology (IIT) Madras Link: https://nptel.ac.in/courses/112/107/112107105/
2	Title: Computer Graphics and Geometric Modeling Professor: Dr. S. S. Murthy Institute: Indian Institute of Technology (IIT) Kharagpur Link: https://nptel.ac.in/courses/106/106/106106092/
3	Title: Introduction to Computer Integrated Manufacturing Professor: Dr. P. V. Rao Institute: Indian Institute of Technology (IIT) Kharagpur Link: https://nptel.ac.in/courses/112/106/112106060/
4	Title: CNC Machines and Programming Professor: Dr. M. P. Rao Institute: Indian Institute of Technology (IIT) Hyderabad Link: https://nptel.ac.in/courses/112/106/112106097/
	Title: Robotics and Automation Professor: Dr. V. K. Jain Institute: Indian Institute of Technology (IIT) Delhi Link: https://nptel.ac.in/courses/112/105/112105292/

SEMESTER: S7

POLYMERS & FUEL CELL TECHNOLOGY

Course Code	PEPOT743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Engineering Chemistry, Polymer Materials.	Course Type	Theory

Course Objectives:

1. Study of fundamentals of developments made in fuel cell development and the challenges facing in its wide practical applications.
2. Understand the methodology for research on future prospects in the development and applications of fuel cell.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fuel Cell, Brief History of fuel cells, Fuel Cell and conventional processes – comparison, Types of Fuel Cells Application scenarios, Advantages and disadvantages, Energy & power relations, units, Working of a PEM fuel Cell, Polymer materials as components of fuel cell, Major Cell Components, Material Properties, Processes and Operating Conditions of PEMFC.	9
2	Gibb's free energy -Work potential of fuel, Reversible voltage, NERNST Equation, Voltage and P, T and concentration dependence examples, Faraday's Laws, Efficiency: thermodynamic, voltage and fuel. Electrochemical reaction fundamentals, electrode kinetics, Charge transfer and activations energy, Exchange current density - slow and fast reactions, Potential and equilibrium - galvanic potential, Reaction rate and potential - Butler Volmer equation & Tafel equation, Exchange Currents and	9

	Electrocatalysis: How to Improve Kinetic Performance, Electrode design basics.	
3	Charge transport resistances, voltage losses, Ionic and electronic conductivities, Ionic conduction in different FC electrolytes: Aqueous, polymeric and ceramic, Diffusive transport & voltage loss: Limiting current density, Nernstian and kinetic effect, Convective transport: flow channels, gas diffusion / porous layer, gas velocity, pressure.	9
4	Sizing of a Fuel Cell Stack, Stack Configuration, Uniform distribution of Reactants, Heat removal, Stack Clamping. Fuel Cell Diagnostics: Polarization Curve, Current Interrupt, AC Impedance Spectroscopy, Pressure drop as a diagnostic tool.Fuel Cell System Design: Hydrogen-Oxygen Systems, Hydrogen-Air Systems, Fuel Cell Systems with Fuel Processor, System Efficiency Fuel Cells and Hydrogen Economy: Hydrogen Energy Systems, Hydrogen Energy Technologies, Transition to Hydrogen Economy.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of electrochemistry, thermodynamics, fluid mechanics, and heat and mass transfer, components of fuel cells and fuel cell systems.	K5
CO2	Illustrate various types of fuel cells and compare it with conventional systems.	K2
CO3	Defend the significance of fuel cell technology in the new global energy scenario.	K5
CO4	Design and explain fuel cell stack and fuel cell systems.	K6
CO5	Distinguish the expectances of hydrogen as a fuel and energy vector in the context of renewable energy.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1										
CO2	3	1										
CO3	3	1										
CO4	3	1										
CO5	3	1										

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fuel Cell Fundamentals.	John Ryan P. O'Hayre, Suk-Won Cha, Whitney Colella & Fritz B. Prinz	Wiley & Sons, Inc.	2006
2	PEM Fuel Cells: Theory and Practice.	Frano Barbir	Elsevier	2005
3	Advances in fuel cells.	Zhao, Tim, K-D. Kreuer, and Trung Van Nguyen.	Elsevier	2007
4	Fuel cells: principles, design, and analysis.	Revankar, Shripad T., and Pradip Majumdar.	CRC press	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Fuel Cells Fundamentals, Technology and Applications.	Vielstich, W, Gasteiger, H. A. Lamm, A. (Eds)	John Wiley & Sons Ltd.	2003
2	Fuel Cell Handbook.	John E	EG & G Technical Services, Nov 2004	2004
3	Alternative Fuels: The Future of Hydrogen.	Hordeski, M. F.	The Fairmont Press	2007
4	Fuel Cells and Their Applications.	Kordesch, Simader, G.	VCH	1996

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/Eb7pv0oOf_k?si=1u8OE1WaqBEw2rTs
2	https://youtu.be/62363H_I_Qk?si=52g9sGkeHxUmXuur
3	https://youtu.be/L2VSOccUrSk?si=5i_298CRKDjaJPH8
4	https://youtu.be/anDF-nUHZW4?si=HTTctXbHxI23cLYW

SEMESTER: S7

COMPOSITE MATERIALS

Course Code	PEPOT744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Impart the basic concepts of composite materials.
2. Develop knowledge on Composite behaviour and application.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	General characteristics of composites, advantages and disadvantages, advantages of using composites in high performance structures application trends. Composite applications in aircraft, space, transportation, energy electronics, sports and medical industries. Characteristics of fibres, matrices, interface bonding, adhesives microstructure of composites, the function of the fibre and matrix in a composite. Thermoplastic and thermoset matrix composites. Composite products and their unique properties. The strengthening mechanism that makes composites stronger. Environmental effects to design of composite structures.	9
2	Traditional and novel approaches process fundamentals. Fundamental physics in composite manufacturing, manufacturing processes for polymeric composites. Typical defects introduced in manufacturing and the methods utilized to minimize these defects. Common terminology in composites manufacturing. Special tooling considerations required for composite manufacturing Basic concepts, stiffness, strength, thermal and moisture	9

	expansion. Anisotropic and isotropic materials, tailored specific strengths. The cause of discontinuous stresses in composites and how it differs from metals	
3	Laminate theory; micromechanical behaviour of a ply, out-of-plane effects. Hooke's Law to unidirectional composites. The stress-strain relations of a unidirectional composite subjected to mechanical, thermal and moisture loads. Stress/strain/curvature of a laminate under constant axial forces and bending moments. The unusual behaviours which may occur in laminates such as bending/stretching coupling and stretching-shear coupling. The use of a specific layup orientation based on the loading conditions and CLT. The role of lamina and their arrangement in laminate Failure criteria, Laminate Strength, Stress Concentrations.	9
4	The service life (fatigue) and environmental (damage/corrosion) effects on metallic vs. composite structures. Key damage mode for composites and composite damage tolerance capabilities How do actual composites for aerospace, automotive, sporting goods, high temperature applications behave? Problem areas, long-term performance, influence of structural geometries the Advantages and disadvantages of composites with respect to Product Lifecycle Management. General considerations and process involved in composite structural design. Typical in-service damage types for composites. Non-destructive inspection techniques for detecting damage in composites. Basic types of composite repair and their benefits	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define the fundamentals of composite technology	K1
CO2	Explain the significance of composite manufacturing	K2
CO3	Analyze the anisotropic and isotropic materials	K4
CO4	Explain the role of lamina and their arrangements in laminate	K5
CO5	Develop the general considerations and process involved in composite structural design	K3, K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

3Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	<i>Rapid prototyping: Principles and applications</i>	ChuaC. K., LeongK .F .,and LimC.S	World Scientific	3 rd Edn, 2010
2	<i>Rapid prototyping</i>	Gebhardt A	Hanser Gardener	2003
3	<i>Mechanics and Analysis of Composite Materials</i>	V. V. Vasiliev and E. V. Morozov	Elsevier Science	2001
4	<i>Polymer Engineering Composites</i>	M. O. W. Richardson	Applied Science Publishers	1977

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	<i>Advances in composite materials</i>	G. Piatti	Applied Science Publishers	1977
2	<i>An Introduction to Composite Materials</i>	D. Hull	Cambridge University Press	1996
3	<i>Handbook of composites</i>	Globin	Van Nostrand	1982
4	<i>Composite Materials</i>	K. K. Chawla	Springer-Verlag	1987

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/mp5lwolO-wM
2	https://youtu.be/WFDMGn5pVws
3	https://youtu.be/CMhwNTMe_mo
4	https://youtu.be/Rx3nJicpDxs

SEMESTER: S7

POLYMERS FOR SPACE APPLICATIONS

Course Code	PEPOT746	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Knowledge in Polymer Materials for Space Applications	Course Type	Theory

Course Objectives:

1. Students should achieve sufficient knowledge about the basics of polymers for space application.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Polymers for aerospace research- adhesives, coatings, ablatives. Synthesis and processing of advanced thermoplastics: PEEK, polycarbonates, polyethers, polyether sulphones. Synthesis and processing of thermal protection systems in space research. High temperature resistant resins such as epoxy, phenolic and polyimides	9
2	High temperature resistant polymers with metals in their back bone - Boron, Silicon and Phosphorous contain polymers for space applications, composites for satellites and launch vehicles: Composites- type of composites- fibre composites, particulate composites, foam composites. Desired properties of a matrix. Polymer matrix: Thermosets and thermoplastics. Fibre reinforced polymer (FRP) Types of fibres, mechanical properties of fibres, glass fibres, carbon fibres, aramid fibres, metal fibres, alumina fibres, boron fibres, silicon carbide fibres, silica fibres, etc	9

3	Role of reinforcement: Theory of reinforcement property dependence on resin and reinforcement, directional properties. Laminated composites- Lamina & Laminate lay-up, ply-orientation definition. Strength of laminates, laminate stiffness and ABD Matrices, special classification of laminates, symmetric, antisymmetric and non-symmetric laminates. Failure mechanics of composites. Propellant binders, classification of propellants- solid, liquid, hybrid and air breathing. Solid propellants – homogenous smokeless propellants, Heterogeneous (composite) propellants. High energy binders- synthesis, characteristics, applications. Glycidylazide polymer and its homologues, polynitrato methyl methyloxetane, poly glycidyl nitrates.	9
4	Testing of Composites: Mechanical testing of composites, tensile testing, compressive testing, intra laminar shear testing, inter laminar shear testing, fracture testing etc. Joining of composites: Adhesively bonded joints & mechanically fastened joints, environmental effects on composites. Composite characterization by mechanical, thermal and DMTA techniques.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the need for thermal protection in space research	K2
CO2	Explain the significance of different types of composites	K2
CO3	Compare propellant binders and their characteristics	K5
CO4	Analyze the properties of polymer laminates	K4
CO5	Design composite structures and joints.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Composites manufacturing; materials, product and process engineering,	S.K. Mazundar	CRC press, Boca Raton	2002
2	Chemistry and Technology of Explosives, Vol.2, Vol.3 and Vol.4	T. Urbanski	Pergamon Press, New York,	1984
3	Propellants manufacture, Hazards and Testing, Advances in Chemistry Series, Vol.88,	Boyars C and Klager K	American Chemical Society, Washington D.C	., 1969.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	High Performance Thermosets, Chemistry, Properties and Applications	S.C Lin, E.M. Pearce	Hanser Publications,	1994
2	Engineered Materials Handbook vol 3 Adhesives and sealants, Vol.3,	C.A. Dostaletal	ASM International,	1990
3	Polymer composite-from nano-to macro-scale,	K. Friedrich et al	Springer, NY	2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://www.youtube.com/watch?v=GN03dj47UIg https://www.youtube.com/watch?v=ph5p515aXeU https://www.youtube.com/watch?v=649fIwvIvRc&list=PLwdnzlV3ogoVE2AIC-G4Uew8XsaINwJGo

SEMESTER: S7

LIQUID CRYSTAL POLYMERS

Course Code	PEPOT747	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Understanding of Polymer Science and Engineering, Including Polymer Chemistry, Properties, And Processing.	Course Type	Theory

Course Objectives:

1. To achieve sufficient knowledge about the basics of liquid crystal polymers

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	INTRODUCTION- crystals, liquids, liquid crystals, polymers, history of liquid crystal polymers, structural requirements of LCP, types of liquid crystalline (LC) phases, MLC, PLC.	9
2	TYPES OF LIQUID CRYSTALLINE POLYMERS- main chain and sidechain LC polymers, dendrimers, anisotropic properties, nematic, cholesterics, smectic, mesophase lyotropic, thermotropic, calamitic, discotic, structural features of side chain LC polymers, spacer length, mesogenic groups	9
3	THEORIES OF LIQUID CRYSTALLINITY IN POLYMERS- , steric theory, Onsager approach, Flory's lattice model, Advantages of LCP, disadvantages of LCP, Effect of electric and magnetic field, surface effect, birefringence in LCLC block copolymers, polymer dispersed LC, LC blends	9

	composites, LC elastomers, LC ionomers. Characterization of LC phases, DSC, POM, XRD, NMR, FTIR, dielectric studies	
4	APPLICATIONS OF LC POLYMERS- photochromic LC polymers, photo responsive LC polymers as sensors, LCD devices, high strength fibers, inorganic LC polymers, photomechanical LC actuators	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain fundamentals of structural requirements of LCP and characteristics of liquid crystal polymers	K2
CO2	Illustrate various types of liquid crystal polymers	K2
CO3	Summarize the advantages and disadvantages of liquid crystal polymers	K2
CO4	Compare LC blends and LC composites.	K5
CO5	Analyze the applications of liquid crystal polymers	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		1						1	1
CO2	2	1	2	3		1						1
CO3	2		2		2							1
CO4	2	1		2							2	1
CO5	1	3		1		1	2					1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Liquid Crystalline Polymers	A. M. Donald, A. H. Windle, S. Hanna	Cambridge University Press,	2006
2	Liquid Crystalline Polymers	V. Shibaev	Elsevier,	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Liquid Crystal Polymers: From Structures to Applications	A.A. Collyer	Springer science	2012
2	Liquid Crystal Polymers: Synthesis, Properties and Applications	Ed. D. Coates	Rapra Publishing	2000
3	liquid Crystalline Polymers: Volume 1–Structure and Chemistry	V. K. Thakur, M.R. Kessler	Springer	2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/N7HCjnR7Iig?si=Uhr6srI0-aFUmGQv
2	https://youtu.be/jIw0h02TMzA?si=LWi9ukJHwo8SPY3B
3	https://youtu.be/IhMhUjRtlEo?si=w-Lar2aic_R8VEm5
4	https://youtu.be/w-Ecd5ciOCc?si=T_IPQmMMs19v9ZB6

SEMESTER: S7

PLASTICS PACKAGING TECHNOLOGY

Course Code	PEPOT748	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Plastic materials, Processing	Course Type	Theory

Course Objectives:

1. Study of fundamentals of plastic materials, its properties, processing techniques and applications for packaging applications.
2. Understand the methodology for selection of packaging equipments, packaging equipment checklist, wrapping, sealing methods

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to plastics packaging, functions of packaging, advantages of plastic packaging, distribution hazards, special requirements of food and medical packaging, packaging legislation and regulation, packaging as a system, elements, approach, package, design, relation criteria for packaging materials, packaging equipment checklist, case histories.	9
2	Major packaging plastics, introduction, PE, PP, PS, PVC, polyesters, PVDC, vinylacetate, PVA, EVA, PV Alcohol, PA, PC ionomers & fluoro polymers. Conversion process, compression & transfer moulding, injection moulding.	9

3	Blow moulding, extrusion, rotary thermoforming, lamination, metallizing, decoration process, shrink wrapping, pallet & stretch wrapping, sealing methods, plasma barrier coatings, energy requirement for conversion, extrusion, film and flexible packaging, extrusion, cast film & sheet, blown film, multi-layer film & sheet coatings, laminations & co extrusions, stretch and shrink wrap, pouching, sealing, evaluation of seals in flexible packages, advantages of flexible packaging, flexible packaging products, specialized packaging for food products.	9
4	Thermoformed, moulded and rigid packages, thermoforming packages, position & thermoforming & wrap forming, solid phase pressure forming, skin packaging, thermoforming moulds, advantages & disadvantages of moulding foams, other cushioning materials & distribution packaging, polystyrene & other foams systems cushioning, design of moulded cushioning systems, plastic pallets, drums & other shipping containers, testing plastic packages, barrier, printing, labelling & pigmentation, sterilization systems and health care products, packaging hazards and their controls, environmental considerations.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Outline significance of plastic packaging.	K2
CO2	Summarize major packaging plastics.	K2
CO3	Apply methodology of conversion process	K3
CO4	Discover the relationship between film and flexible packaging.	K4
CO5	Compare different thermoforming and testing methods for plastic packages.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

[illegible]

Text Book				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Understanding plastics packaging technology.	Selke, S. E.	Hanser Verlag.	1997

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Food packaging technology	Coles, R., McDowell, D., & Kirwan, M. J. (Eds.).	CRC Press	2003
2	Permeability properties of plastics and elastomers: a guide to packaging.	Massey, L. K.	William Andrew.	2003
3	Plastic packaging materials for food: barrier function, mass transport, quality assurance, and legislation.	Piringer, O. G., & Baner, A. L. (Eds.).	John Wiley & Sons.	2008

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/piGjS-7VRMs?si=ZXj7kXDYCMp4cSXl
2	https://youtu.be/RCnyrD5FPVs?si=8JYtKL1yRrs-Wd8W
3	https://youtu.be/EF1nYb3kTxI?si=q7J4YUB6jxlyNr2U
4	Biodegradable polymers, definition, classification, example and applications (youtube.com)

SEMESTER: S7
POLYMER REACTION ENGINEERING

Course Code	PEPOT745	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Knowledge in Polymer Science	Course Type	Theory

Course Objectives:

1. To impart knowledge of application of reaction engineering for polymer production

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction To Reactor Design: Design equation for single reaction system, parallel and series reactions, temperature dependant terms, interpretation of batch reactor data, conversion, concentration and mole balance, constant volume reaction system, rate of reaction, first and zero order reaction	9
2	Reaction Engineering of Free Radical Polymerisation: thermodynamics and kinetic aspects of free radical polymerisation MWD of free radical polymerisation: Standard free radical polymerisation; design of tubular reactors involving radical polymerisation, solution of equations describing isothermal radical polymerisation Reaction Engineering of Ionic, Ziggler Natta: Smith-Ewart and other models, average chain length estimation of stereoregular polymers, chain segment estimation and its distribution, classification of various ionic species, effect of solvent, ionophores and ionogens	9
3	Reaction Engineering of Step Growth Polymerisation: kinetics of step growth polymerisation, accessibility of functional groups, equilibrium	9

	considerations: closed and open systems, cyclisation versus linear polymerisation, MWD in linear and non-linear polymerisations	
4	Bulk, Solution and Precipitation Polymerisations: Modelling of bulk and solution polymerisation reactor, precipitation polymerization: reactor modelling to interpret MWD of industrial polymers. Suspension and Emulsion Polymerisation: Suspension Polymerization: Principles, of reactor modelling and MWD estimation; Modelling of emulsion reactors, prediction of MWD and solution viscosity, examples	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. **Problem Definition (2 points)**
 - Check for clarity and completeness of the problem definition.
 - Check for clear objectives and divide into sub objectives
2. **Problem Analysis (4 points)**
 - Examine the problem and identify the variables and responses
 - Divide the problem to manageable parts
3. **Implementation of software (2 points)**
 - Use suitable software for getting the solution
 - Check for the appropriateness of tools used

4. **Evaluate (4 points)**

- Evaluate the proposed solution.
- Assess the correctness of solution.

5. **Validation of the results (4 points)**

- Validate the results using available data.
- Do the error analysis of the results obtained.

6. **Conclusion and presentation (4 points)**

- Summarize the results obtained
- Do critical thinking of the results obtained and evaluate.
- Submit a technical report

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the design of tubular reactors involving radical polymerisation	K2
CO2	Explain the mechanism of reaction engineering of ionic, Zeigler Natta & copolymerisation	K2
CO3	Interpret the mechanism of reaction engineering of step growth polymerisation	K5
CO4	To propose the modelling of bulk and solution polymerisation reactor	K6
CO5	To propose the modelling of suspension and emulsion polymerisation reactor	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3		3				2					
CO2	3						2					
CO3	1		2				2					
CO4	3			2			2					
CO5	2		2				2					

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Principles of Polymerisation Engineering"	Beisenberger J. A. and Sebastian D.H.;	John Wiley& Sons.	1983
2	Text Book of Polymer Science	Billmayer, F. W	Willey Inter Science, New York	3rd Edition 1984

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Polymerization	Odian G.	John Wiley & Sons.	2002
2	Polymer reaction engineering	J M Asua	Blackwell publishing	2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://archive.nptel.ac.in/courses/103/107/103107206/

SEMESTER: S7

SPECIALITY POLYMERS

Course Code	PEPOT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Understanding of Polymer Science and Engineering, Including Polymer Chemistry, Properties, And Processing.	Course Type	Theory

Course Objectives:

1. To learn about the design, synthesis, and structure-property relation of high performance and specialty polymers.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	High Temperature and Fire-Resistant Polymers – fluoropolymers, aromatic polymers, hydrocarbon polymers, polyphenylene sulphide, polysulphones, polyesters, aromatic polyamides, polyketones, heterocyclic polymers, fire resistant polymers, flame retardants.	9
2	Ionic Polymers – Synthesis and physical properties: Ion-exchange, hydrophilicity, applications, ionomers based on polyethylene, elastomeric ionomers, ionomers based on polystyrene, PTFE, ionomers with polyaromatic backbones. Polyelectrolytes: ion exchangers, polyelectrolytes based on carboxylates, polyelectrolyte complexes.	9
3	Polymers With Novel Properties – Liquid crystalline polymers (LCPs): concept of liquid crystalline (LC) phase, liquid crystalline polymers and their classification, main chain LCPs and side chain LCPs, structure-property	9

	relationship, applications of LCPs. Conducting polymers: charge carriers, doping, synthesis of polyacetylene, polyaniline, polypyrrole, polythiophene. Photo conducting polymers.	
4	Inorganic Polymers- polymers containing phosphorous, nitrogen and boron, organometallic polymers, polysiloxanes. Polymers for special applications – Polymer concretes, polymer binders for propellants, polymer-bonded explosives. Polymeric materials used in telecommunication and power transmission. Polymers in agricultural applications: green houses, mulches	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Outline the different types of high temperature resistant polymers	K2
CO2	Analyze the synthesis and properties of ionic polymers	K4
CO3	Develop polymers possessing novel properties	K6
CO4	Summarize electrical and electronic application of polymers.	K2
CO5	Choose the polymers used in concretes, propellants and explosives	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		1						1	1
CO2	2	1	2	3		1						1
CO3	2		2		2							1
CO4	2	1		2							2	1
CO5	1	3		1		1	2					1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Polymers, Specialty Polymers, And Their Applications,	Manas Chanda, Salil K. Roy	CRC Press	2009
2	Specialty Polymers: Materials and Applications	Faiz Mohammad	K. International Pvt. Ltd	2008
3	<i>Doping in Conjugated Polymers</i>	Pardip Kar	John Wiley & Sons	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Specialty Polymers	Robert William Dyson	Springer	2012
2	Plastics Technology Hand book	Manas Chanda, Salil K.Roy	CRC press	5th Edn., 2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/IztesS8AVIA?si=IrIkQIeZTDY9res6 https://youtu.be/GV8e8RKN_tA?si=n2Obo4q_wWeVdya6
2	https://youtu.be/SNIOKHFO0JU?si=uBkFL03ArxMGicdi
3	https://youtu.be/2xmYYO_qbLs?si=YI46riJ7TOuM3pvs
4	https://youtu.be/K8cwhVTJbmg?si=9nKpj3Rq3bUQ_ISl

SEMESTER: S7

POLYMERS AND SEPARATION PROCESSES

Course Code	PEPOT752	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart the basic concepts of separation process.
2. Develop knowledge on recent advances in separation techniques

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Characteristics and selection of separation process: Importance and variety of separation, economic significance, inherent separation factor, selection, factors influencing the choice of separation process, solvent selection, selection of equipment. Recent advances in separation techniques based on size, surface properties, ionic properties and other special characteristics of substances, Rate based versus equilibrium separation processes, Selection of separation process, Energy requirements of separation processes. Polymer materials used as membranes, functional polymer membranes	9
2	Introduction, Type and choice of membranes, Plate and frame, tubular, spiral wound and hollow fibre membrane reactors and their relative merits, Membrane filtration, Microfiltration, Nanofiltration, Ultrafiltration, Reverse Osmosis, Dialysis, Models for membrane separations.	9

	Basic principles, classifications, Surfactants at Inter phases and in bulk, Foam fractionation, Foam flotation, Adsorptive bubble separations, Ion flotation, Microemulsion/Macro emulsions, Hydro topes, Solvent ablation	
3	Interacting solutes, Adiabatic adsorbers, velocity effects. Adsorption-Desorption operations: Thermal desorption of gases, activated carbon solute recovery, processing liquid using thermal regeneration, Pressure swing and vacuum swing adsorption, Regeneration with purge and desorbent. Basics of Ionex change, Ionex change resins, Binary ion exchange equilibrium, Ion movement theory, Applications, Applications without exchange: Ion exclusion, Mass transfer in ion exchange systems	9
4	Introduction, types of chromatography, Elution chromatography: Principles and Retention theory, Band broadening and separation efficiency, Types of chromatography, large scale elution (cyclic/batch) chromatography, Selective adsorption of biological macromolecules, Simulated counter current techniques, Comparison with other separation methods.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the factors influencing the choice of separation techniques.	K2
CO2	Develop models and the solutions for membrane separation processes	K6
CO3	Compare the surfactant and adsorption-based separation	K5
CO4	Categorise the treatment of process liquids by ion exchange process and its applications	K4
CO5	Define the treatment of process liquids by Chromatographic Separations and its applications	K1

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											
CO5	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Separation Process Principles	Seader, J D, and Ernest J Henley	New York Wiley	1998
2	Basic Principles of Membrane Technology	Marcel Mulder	Springer	2007
3	Separation Processes	King C.J.;	Tata McGraw–Hill	1982.
4	Rate-Controlled Separations	Wankat, P.C	Springer,	1994

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Membrane Technology in the chemical industry	Nunes S P, Peinemann K V	Wiley-VCH	2 nd , 2006
2	Membrane Process	Rautanbachand Albrecht R	John Wiley and Sons..	1989
3	Membrane Processes in separation and Purification	Crespo. J G, Bodekes KW	Kluwer Academic	1994.
4	Transport processes and Unit Operations	Geankoplis C J	PHI	4 th , 2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/svfyI6S0MDg
2	https://youtu.be/MT9MeYjG6ko
3	https://youtu.be/Q_eAIJLxo_g
4	https://youtu.be/iHrKsfw827c

SEMESTER: S7

MODERN METHODS OF INSTRUMENTATION

Course Code	PEPOT 753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Understand the principles and applications of modern instrumentation methods in polymer engineering.
2. Gain knowledge about different characterization and testing techniques specific to polymers.
3. Learn advanced measurement techniques and data acquisition methods.
4. Explore emerging technologies and their applications in polymer instrumentation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Instrumentation in Polymer Engineering Introduction to Instrumentation in Polymer Engineering: Importance and applications of instrumentation in polymer engineering. Overview of traditional vs. modern instrumentation methods. Measurement Systems: Elements of a measurement system. Static and dynamic characteristics of instruments. Sensors and Transducers Specific to Polymers: Principles of sensors and transducers. Types of sensors: temperature, pressure, strain, and humidity sensors used in polymer processing. Signal Conditioning and Data Acquisition: Amplifiers, filters, and converters. Analog to Digital and Digital to Analog	9

	Conversion (ADC/DAC). Components of Data Acquisition Systems (DAS) and their applications in polymer engineering.	
2	Characterization Techniques for Polymers Thermal Analysis: Differential Scanning Calorimetry (DSC). Thermogravimetric Analysis (TGA). Dynamic Mechanical Analysis (DMA). Spectroscopic Methods: Fourier Transform Infrared Spectroscopy (FTIR). Nuclear Magnetic Resonance (NMR) Spectroscopy. Raman Spectroscopy. Chromatographic Techniques: Gel Permeation Chromatography (GPC). High-Performance Liquid Chromatography (HPLC). Microscopy and Imaging: Scanning Electron Microscopy (SEM). Transmission electron microscopy (TEM). Atomic Force Microscopy (AFM).	9
3	Advanced Instrumentation in Polymer Processing and Testing Process Instrumentation: Instrumentation for extrusion, injection molding, and blow molding. Real-time monitoring and control in polymer processing. Mechanical Testing of Polymers: Tensile, compressive, and flexural testing. Impact and hardness testing. Rheological Measurements: Principles of rheology and viscoelastic measurements. Rheometers and their applications in polymer engineering. Surface and Interface Characterization: Contact angle measurements. Surface energy analysis. Atomic Force Microscopy (AFM) for surface characterization.	9
4	Emerging Technologies and Applications in Polymer Instrumentation Smart Sensors and IoT in Polymer Engineering: Integration of IoT in polymer instrumentation. Smart sensors and real-time data analytics. Polymer Nanocomposites and Instrumentation: Characterization techniques for nanocomposites. Challenges and solutions in nano composite instrumentation. Biomedical Applications of Polymers: Instrumentation in the development of polymer-based biomedical devices Characterization of biocompatible polymers. Sustainability and Environmental Monitoring: Instrumentation for recycling and degradation of polymers. Monitoring environmental impact of polymer materials.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamentals of instrumentation and measurement systems specific to polymer engineering.	K2
CO2	Utilize thermal analysis and spectroscopic methods to characterize polymer materials.	K3
CO3	Analyze data obtained from mechanical testing and rheological measurements of polymers.	K4
CO4	Design instrumentation setups for real-time monitoring and control in polymer processing.	K6
CO5	Evaluate the use of emerging technologies like smart sensors and IoT in enhancing polymer instrumentation.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2		3	2									
CO3		2	3	2								
CO4			3	3								
CO5				2	3							3

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Polymer Engineering	N.G. McCrum, C.P. Buckley, C.B. Bucknall	Oxford University Press	2nd Edition, 1997
2	Polymer Characterization: Physical Techniques	D. Campbell, J. R. White	CRC Press	1st Edition, 1989
3	Modern Electronic Instrumentation and Measurement Techniques	Albert D. Helfrick, William D. Cooper	Prentice Hall	2nd Edition, 2005
4	Polymer Analysis	Barbara H. Stuart	Wiley	1st Edition, 2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Polymer Testing	Roger Brown	CRC Press	1st Edition, 1999
2	Rheology: Principles, Measurements, and Applications	Christopher W. Macosko	Wiley-VCH	1st Edition, 1994
3	Differential Scanning Calorimetry: An Introduction for Practitioners	G.W.H. Höhne, W. Hemminger, H.-J. Flammersheim	Springer	2nd Edition, 2003
4	Spectroscopic Techniques for Polymer Characterization	Yukihiro Ozaki, Satoshi Kawata	American Chemical Society	1st Edition, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	"https://nptel.ac.in/courses/103/105/103105064/"
2	"https://nptel.ac.in/courses/113/106/113106064/"
3	"https://nptel.ac.in/courses/112/102/112102289/"
4	"https://nptel.ac.in/courses/112/107/112107239/"

SEMESTER: S7

SAFETY ENGINEERING OF PROCESS PLANTS

Course Code	PEPOT754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objective:

1. This course introduces students to the concepts of process safety, safety practices in industries and emergency procedures.
2. It also outlines the various types of hazards and risk.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concept and importance of process safety. Safety in the site selection and layout. Review of Industrial Accidents-Major Chemical Industry Accidents, Bhopal, Flix borough, SEVESO. Cost of accidents. Material safety data sheet (MSDS). Work permit system, Personal Protective Equipments (PPE). Regulatory Bodies: -National safety council, OSHA, Loss prevention association, PESO, NEBOSH.	9
2	Chemical Hazards: - Fire triangle, LFL, UFL, Types of fire- Pool fire, jet fire, Flash fire and Fire ball. Fire prevention techniques in process industry. Flammability diagram-construction and application. Explosion: - Deflagration, Detonation, UVCE, BLEVE and Dust explosion. Prevention techniques for explosion	9
3	Process and Plant hazards: Physical hazards- Atmospheric contaminants, Sound, Light, Radiation, Pressure, and Temperature.	9

	Electrical hazards-electric shock, flashover, lightning Strokes. Mechanical hazards. Environmental hazards. Human Errors-People Oriented Errors, Situation Oriented errors, System oriented errors.	
4	Prevention techniques for hazards. Hazardous area classification. Safety in transportation of hazardous chemicals by road-HAZCHEM CODE, TREM CARD Relief system and Detectors. Flame arrester sand Flare system. Hazard rating of chemical plants- Dow fire, explosion and Toxicity index. Chemical exposure index. Emergency planning-onsite and offsite emergency planning, Mock drill. Hazard identification and Risk: HAZOP, Bow tie diagram, Fault tree analysis, failure mode and effect analysis, Eventtree analysis ,Layer of protection analysis. Examples.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Recall the basic concepts of safety in Process Industries.	K1
CO2	Apply the chemical engineering fundamentals in the chemical hazards prevention techniques.	K3
CO3	Analyze the physical hazards and human factors in process safety.	K4
CO4	Develop an understanding about safety practices in industries and emergency planning.	K6
CO5	Identify various hazards associated with chemical process industries using various techniques and knowledge of risk.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											
CO5	3								3			

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Safety in Chemical Plants/Industry and Its Management	B. K. BhaskaraRao, Er .R. K. Jain, Vineet Kumar	Khanna Publishers	First edition,2010
2	Industrial Safety, Health and Environment Management Systems	R. K. Jain & Sunil S Rao,	Khanna Publishers,	FourthEdition,2000
3	Chemical Process Industrial safety	K. S. NRaju,	M c Graw Hill,	2014
4	Guidelines for Hazard Evaluation Procedure.	Centre for Chemical Process Safety.	Centre for Chemical Process Safety. AICHE	1990

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Methodologies in Hazard Identification and assessment Manual	K.V. Raghavan and A. A. Khan	CLRI	1990
2	Loss Prevention in Process Industries	Frank P. Lees	Butterworth-Heinemann.	Vol.1,2&3 Second Edn, 1996

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/n7oUOUCIblg?si=VlfrhMcV1kL3WCQQ
2	https://youtu.be/dlhXg7KkD0M?si=cO9B_RF4M3bjb5CW
3	https://youtu.be/WMPodFzWsSs?si=uGjBsf6XUO8uzglc
4	https://youtu.be/K42sGq0vmO8?si=1XINF1rdVPyMrfcD

SEMESTER S7

MOULD AND DIE MANUFACTURE

Course Code	PEPOT756	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Polymer Processing, Polymer Products Design.	Course Type	Theory

COURSE OBJECTIVE:

1. Students will be able to understand design aspects to design tools and dies for polymer product manufacture

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	General introduction to core cavity design, parting surface, sprue, sprue pullers, mould cooling, Ejection system, mould daylight Feed system design, cooling system and Ejection system design for Injection moulds, split, under cuts, side core threads. Runnerless, hot and cold runne rmoulds, working of various ejection actuators, cooling of insert moulds, types and features.	9
2	Extrusion die design, flow through a die, die land, die body, straight through die, Cross head, Offset die design, Coat hanger, fishtail, annular, pipe die design, wire coating die, cold feed and hot feed die design.	9
3	Mould materials, Nitriding steels, case hardened steels, corrosion resistant steels, through hardened steels, maraging steels, hard material alloys, steel castings, physical properties, Aluminium mould making Electro forming, EDM, Hobbing, polishing, surface structuring, hard chrome plating, chemical and physical vapour deposition.	9

4	CAM mould making, EDP systems, Data preparation for machining, Machining process, steps involved Standard components of a mould base, Standards for gate area, standards for ejection, standards for mould heating, Mould maintenance, Servicing and Inspection, Repairs, Storage.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Manufacturing processes for engineering materials.	Kalpakjian, Serope	Pearson Education India	1984
2	Machining And Metal Working Handbook	Walsh, Ronald A.	McGraw- Hill Education	2006
3	Plastics Product Design & Process Engineering	Belofsky, Harold		

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mould making Hand book-	Gunter Menning	Hanser Publications	1998
2	Manufacturing engineering handbook.	Geng, Hwaiyu, and P. E. CMfgE.	McGraw-Hill Education	2016
3	How to make injection moulds.	Menges, Georg, Walter Michaeli ,and Paul Mohren.	Carl Hanser Verlag	2013
4	Design of Plastic Moulds and Die-	Laszlo Sors	Elsevier ScienceLtd	1989
5	Die Design for Extrusion of Plastic Tubes and Pipes: A Practical Guide	Sushil Kainth	Hanser Publications- 2017	2017

Module No.	Link ID
1	https://youtu.be/QxZ54WgYhnA?si=61XyknGEvPcAab9m
2	https://youtu.be/x4C1hXDROZM?si=5DRMrk8gDICewIue
3	https://youtu.be/SoTfSOFj6q0?si=mJUx9J6QkpEEDaGs
4	https://youtu.be/49RET0N-ITY?si=fLJ0wYKlmkSF3lo0

SEMESTER: S7**POLYMERS & PHOTOVOLTAIC TECHNOLOGY**

Course Code	PEPOT757	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Electrical and Electronics Engineering, Plastic Materials	Course Type	Theory

Course Objectives:

1. Students will be able to understand manufacture methods for polymer and organic based photovoltaic cells

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Solar energy, solar neutrinos, radiant power, types of solar cells, organic solar cells, band gap and efficiency, photo generated current density, absorption coefficient, power conversion efficiency, auger and SRH lifetime, multiple junctions, hot carrier cells.	9
2	Photo voltaic modules, Modelling, electrical characteristic of PV module, double diode model, single diode model, model parameters, circuit parameters, short circuit current, open circuit voltage, loading conditions, photovoltaic generator current-voltage characteristics, peak power, fill factor.	9
3	Materials and processing –substrate, front electrode, first intermediate layer, materials used, processing, active layer, materials and processing, polymer-polymer solar cells, polymer-fullerene solar cells, organic-inorganic hybrid solar cells, p type and n type polymers, second intermediate layer, materials used, processing	9

4	Stability and characterisation of devices, stability measurement, manufacture of polymer solar cells, photoactive wide band gap donor, small band gap donor, polymer tandem cells, small molecule tandem cells, semi-transparent tandem solar cells. Solution processed organic photovoltaics, slot-die coating process, inkjet printing process, roll to roll printing process, materials for functional layers, flexible substrates, silver back electrodes, interfacial layer, hole transport layer, electron transport layer, issues in scalable OPVs.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Solar Cell Device Physics.	Fonash, Stephen	Elsevier	2012.
2	Thin Film Solar Cells: Fabrication, Characterization And Applications.	Poortmans, Jef, and Vladimir Arkhipov, eds	Vol. 18. John Wiley & Sons	2006.
3	Fuel Cell Technology Handbook.	Hoogers, Gregor, ed	CRC press	2002.
4	Solar cell array design handbook: the principles and technology of photovoltaic energy conversion.	Rauschenbach, Hans S	Springer Science & Business Media	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Polymer solar cells." Photo responsive Polymers II	Hoppe, Harald, and N. Serdar Sariciftci.		2007):
2	Stability and degradation of organic and polymer solar cells	Krebs, Frederik C., ed.	John Wiley & Sons,	2012.
3	Molecular devices for solar energy conversion and storage.	Tian, Haining, Gerrit Boschloo and Anders Hagfeldt, eds.	Singapore: Springer	2018
4	Nanostructured materials for solar energy conversion.	Soga, Tetsuo, ed.	Elsevier	2006.
5	Carbon-containing polymer composites	Rahaman, Mostafizur, Dipak Khastgir, and Ali KanakhirAldalbahi, eds	Singapore: Springer,	2019.
6	Polymer photovoltaics: materials, physics, and device engineering.	Huang, Fei, Hin-Lap Yip, and Yong Cao, eds	Royal Society of Chemistry	2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1 - 4	Prof Soumitra Satapathi, IIT Roorkee https://www.youtube.com/watch?v=5zAQot4pKgU&list=PLLy_2iUCG87Dxsmc322YcSuNI_KCEbqPl https://www.youtube.com/watch?v=F2sZgWgvTAg

SEMESTER: S7

POLYMERS IN CONSTRUCTION

Course Code	PEPOT758	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart the basic concepts of polymers in construction.
2. Develop knowledge on recent advances in construction using polymers.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Plastics in Building and Construction, structural, secondary structural and Non-structural uses, Sand witch panel, piping, cables, cladding and profiles, insulation, sealant, gaskets and adhesives. Roofing and flooring systems, glazing and wall covering. Polymer coating, Materials, solvent based coatings, water-based coatings, curing techniques, Powder coatings, In tumescent coatings.	9
2	Polymers in civil engineering, geotechnical, roadways, pavements, drainage and erosion control systems, polymer concrete, FRP composites, bond strength models, strengthening of RC beams, RCslabs, RC columns, masonry wall and infills strengthening	9
3	Polymer foams, foam manufacturing technology, Thermoplastic foams, thermosetting foams, Special applications, ageing studies.	9
4	Polymer in repair, Injection grouting, patching, coating, sustainable construction, eco-design, recycling of scraps, flammability of polymer foams and composites.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Define the concepts of powder coating, sand witch panels, pipes and cables.	K1
CO2	Illustrate the properties of polymer sealants, gaskets and adhesives	K2
CO3	Develop the design of FRP composites and to strengthen RC slabs.	K3, K6
CO4	Analyse the role of polymers in sustainable development and ecology.	K4
CO5	Determine the polymer systems for various constructional requirements.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1									
CO2	3	3		1								
CO3	3	3			2							
CO4	3	2				2	3	3		2		1
CO5	1	2			2				1			

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymers in construction	Guneri Akovali	Rapra	2005
2	Text book of polymer Science	F W Billmeyer	Wiley international	1984
3	Polymers and Polymers in construction	R L Hollowa Thomas Telford	Rapra	1990

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer composites in construction. 2000.	Halliwell, Sue M.	CRC,	2000.
2	Construction materials: Types, uses and applications	Hornbostel, Caleb.	John Wiley & Sons,	1991.
3	Polymers in Construction	Guneri Akoveli	Rapra review report	2004.

Video Links (NPTEL, SWAYAM...)	
Module	Link ID
1	https://www.humix.com/video/Heq3mzXBGju
2	https://www.humix.com/video/ac4dc5a933df62144aa840626a97908daf1e846b10cb55b884bc21d39253efd6
3	https://www.humix.com/video/TO4UejWBbWu
4	https://youtu.be/Y-B6g7nz4wU

SEMESTER: S7

MODELLING & SIMULATION OF POLYMER MATERIALS

Course Code	PEPOT755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Polymer Materials Polymer Physics.	Course Type	Theory

Course Objectives:

1. Study of polymer flow properties and types of modelling systems.
2. Understand the methodology of preparing simulations based on macromolecular structures.
3. Trouble shooting in polymerisation and processing with simulators.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Use and scope of mathematical modelling, Principles of model formulation, Role and importance of steady-state and dynamic simulation, Classification of models, Model building, Modelling difficulties, Degree-of-freedom analysis, Selection of design variables, Review of numerical techniques, Model simulation. Equations of continuity, energy, momentum, and state, Transport properties, Equilibrium and chemical kinetics, Review of thermodynamic correlations for the estimation of physical properties like phase equilibria, bubble and dew points.	9
2	Constant and variable holdup CSTRs under isothermal and non-isothermal conditions, Stability analysis, Gas phase pressurized CSTR, two phase CSTR, Non-isothermal PFR, Batch and semi-batch reactors, Heat conduction in a bar, Laminar flow of Newtonian liquid in a pipe, Gravity flow tank, Single	9

	component vaporizer, multi-component flash drum, Absorption column, Ideal binary distillation column and non-ideal multi-component distillation column, Batch distillation with holdup etc. Simulation: Simulation of the models, Sequential modular approach, Equation oriented approach, Partitioning and tearing, Introduction and use of process simulation software for flow sheet simulation.	
3	Transporting of polymer particulate solid, flow instabilities, mechanical displacements, mixing mechanisms, disk pump, multiphase systems, mixing equipments and operations for rubbers, multicomponent and multiphase systems, Characterisation of mixers, computational analysis.	9
4	Simulation of reactive polymer processing, polymer compounding, capillary flow, sheet forming, wire coating, profile extrusion, injection moulding, compression moulding, mould flow analysis, fibre spinning, finite element analysis, stress- strain and mechanical evaluation.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Exam	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. Problem Definition (2 Points)

- Check the clarity and completeness of the problem definition.
- Check for clear objectives and divide into sub objectives.

2. **Problem Analysis (4 Points)**
 - Examine the Problem and Identify the variables and responses.
 - Divide the problem to manageable parts.
3. **Implementation of software (2 points)**
 - Use suitable software for getting the solution.
 - Check the appropriateness of the tool used.
4. **Evaluate (4 Points)**
 - Evaluate the proposed solution.
 - Assess the correctness of the solution.
5. **Validation of the Results (4 Points)**
 - Validate the results using available data.
 - Do the error analysis of the results obtained.

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modeling and simulations of polymers: a roadmap	Gartner III, Thomas E., and Arthi Jayaraman	Macromolecules	2019
2	Mechanics of solid polymers: theory and computational modeling.	Bergstrom, Jorgen S	William Andrew.	2015
3	Models for polymeric and anisotropic liquids.	Kröger, Martin.	Springer Science & Business Media.	2005
4	Polymer Simulations Companion: An Introduction to Brownian Dynamics.	Sunthar, P	Springer	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Simulation and Modelling of Polymers	Gedde, Ulf W., et al.	Applied Polymer Science	2021
2	The physics of polymers	Strobl, Gert R.,	Springer	1997
3	The physics of glassy polymers.	Haward, Robert Nobbs, ed.	Springer Science & Business Media	2012
4	Field-theoretic computer simulation methods for polymers and complex fluids.	Fredrickson, Glenn H., Venkat Ganesan, and François Drolet.	Macromolecules	2002

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Science Talks Lecture 60: Molecular modeling, theory, and simulation studies of polymeric materials (youtube.com)
2	https://youtu.be/BuOH8dIxq-o?si=yOz5XG--tzEeO5um
3	https://youtu.be/kzlcmlV9Pgw?si=w-Bj7geOYLdvyRcX
4	Molecular Dynamics Simulation of the Mechanical Properties of Photopolymers used in 3D-Printing (youtube.com)

SEMESTER 8

POLYMER ENGINEERING

SEMESTER: S8

ADHESIVE SCIENCE & TECHNOLOGY

Course Code	PEPOT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Understanding of Chemistry, Particularly Organic Chemistry and Polymer Chemistry	Course Type	Theory

Course Objectives:

1. Explain the Industrial manufacturing of adhesives with its development processes.
2. Introduces students to theoretical and practical aspects in formulating adhesives for commercial and speciality uses.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definition of adhesives and adhesive bonding, functions of adhesives, classification of adhesives, advantages and disadvantages of joining using adhesives, requirements of a good bond, thermodynamics of adhesive, contact angle, work of adhesion, wetting, strength of adhesion, rheology of adhesion	9
2	Surface treatments for metals and plastics, solvent cleaning, mechanical abrasion, chemical treatments, primers, plasma treatments, Methods for Evaluating the Effectiveness of Surface Preparation, theories of adhesion, Mechanical Theory, Electrostatic (Electronic) Theory, Diffusion Theory, Wetting Theory, Chemical Bonding, Acid–Base Theory, Weak Boundary Layer Theory, diffusion theory.	9
3	Classification of adhesives, adhesives from natural sources, rubber based adhesives, TPE based adhesives, poly vinyl acetates, poly vinyl alcohols, di isocyanates, cyano acrylate, anaerobic, acrylic, hot melt adhesives, pressure sensitive adhesives, structural adhesives, Phenolic resins, epoxides, wood	9

	adhesives. Adhesive applications and bonding processes, methods of adhesive application, adhesives in the electronics Industry, adhesives in the automotive and aerospace, adhesives in building construction, adhesives in electrical industry.	
4	Stresses, types of joints, selection of joint detail, joint design criteria effect of joint geometry, effect of temperature, service life prediction. Standard test methods destructive and non-destructive testing, fracture mechanics of adhesives.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Outline the basics and theories of adhesion	K2
CO2	Summarize the different types of resins used as adhesives	K2
CO3	Analyze adhesive formulation for various applications	K4
CO4	Explain applications of adhesives	K2
CO5	Design different adhesive joints	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		1						1	1
CO2	2	1	2	3		1						1
CO3	2		2		2							1
CO4	2	1		2							2	1
CO5	1	3		1		1	2					1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Adhesives and Sealants	Edward M. Petrie	McGraw Hill Handbook	2nd Edn. (2007)
2	Handbook of Adhesives	I. Skeist	Chapman and Hall,	3rd Edn. (1990)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Handbook of Adhesive Technology	A. Pizzi, K.L. Mittal	Marcel and Dekker Inc.,	2nd Edition 2003
2	Adhesive and Coating Technology	P. Ghosh	Tata McGraw-Hill	2008
3	Handbook of Adhesives	Shields	Butterworths	1984

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/VFn83lMJ7Cg?si=RqSUDNh8-csqN9C4
2	https://youtu.be/KrUBlycxmk0?si=mtglirscAtFlbJ7k https://youtu.be/GJQKYq7fFHg?si=yCxxkt5Qatk6swTg
3	https://youtu.be/FTrFKPaOpm8?si=b8hN0666RV-vjJFX
4	https://youtu.be/MEhuwjMH7FM?si=rJCyk8B1kmohU5y3

SEMESTER: S8

POLYMERS FOR ELECTRONIC APPLICATIONS

Course Code	PEPOT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Understanding of Polymer Science and Engineering, Including Polymer Chemistry, Properties, And Processing	Course Type	Theory

Course Objectives:

1. After the completion of this course, students should achieve sufficient knowledge about the basics of polymers for electronic application.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics on Inorganic and Organic Semiconductors – Classification of materials based on electrical conductivity. Basic laws on electrical conductivity: Ohms law and Coulomb's law. Valance band theory-basic concept of band model. Concept of doping. P-type and n-type doping and doping mechanism. P-n junctions. Oxidative and reductive dopants. Inorganic and organic dopants. Type of doping technique. Charge carriers polarons, bipolarons and solitons. Effect of doping on the properties of organic and organic semi-conductors. Effect of temperature on conductivity of inorganic and organic semiconductors	9
2	Historical Development and Synthesis of Important Conducting Polymers – Historical development of organic conductors. Basic structural characteristics of organic conductors. Methods of preparation of conducting polymers: chain growth polymerization, step growth polymerization, electrochemical polymerization. Synthesis of organic conductors: polyacetylene, poly (para phenylene), polyaniline, polypyrrole,	9

	poly(phenylene vinylene) and poly(vinyl carbazole). Conduction mechanism in organic conductors. Interchain and intrachain conduction.	
3	Properties and Processing of Conducting Polymers – Important properties of conducting polymers: electrical conductivity, photoconductivity, charge storage capacity, photoluminescence and electroluminescence. Dielectric constant, dielectric loss and absorption properties. Processing of conducting polymers. Methods to enhance the process ability of conducting polymers. Advantages and disadvantages of conducting polymers	9
4	Applications of Conducting Polymers and Device Fabrication Methods – Applications of conducting polymers. Electroactive applications polymer rechargeable batteries, supercapacitors, sensors, electrochemical actuators. Conductivity applications: antistatic coatings and conductive adhesives. Electronic applications: EMI shielding and Frequency selective surface.	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze the structure, theory and properties of organic and inorganic semiconductors	K4
CO2	Explain the different preparation methods of conducting polymers	K2
CO3	Outline the properties and processing of conducting polymers	K2
CO4	Identify the application and device fabrication of conducting polymers	K3
CO5	Design and develop polymer-based materials for electronic applications.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		1						1	1
CO2	2	1	2	3		1						1
CO3	2		2		2							1
CO4	2	1		2							2	1
CO5	1	3		1		1	2					1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electrical Properties of Polymers	Tony Blythe, David Bloor	Cambridge University Press	2nd Edn 2005
2	J. Margolis	Conducting Polymers and Plastics	Chapman and Hal	(1993).
3	Organic conductive molecules and polymers	H.S. Naiwa	John wiley and sons	vol. 2, (1977)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Conjugated Polymers	J. L. Bredas, R. Silbey	Kluwer, Dordrecht	1991
2	Doping In Conjugated Polymers	Pradip Kar	Scivener Publishing	2013
3	Hand Book Of Conducting Polymers	T.A. Skotheim, R.L. Elsenbaumer, J.R. Reynolds	Marcel Dekker, New york, vol.1-2.,	2ed 1998

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/1TGTvQbMIlc?si=cW__GWQqkgdQP4rG https://youtu.be/tFZmOEwjEPE?si=2-PibHy3GKIAB70y
2	https://youtu.be/UjMbws0LOkU?si=ziqAJEC396bDACkm
3	https://youtu.be/9J9OVISUol4?si=dQYQEfRBYS8Yrw9H
4	https://youtu.be/HE2zRQVhImw?si=CNllkyzHnsX30Qes

SEMESTER: S8

CONDUCTING POLYMERS

Course Code	PEPOT863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Polymer Materials & Polymer Physics	Course Type	Theory

Course Objectives:

1. Students will be able to demonstrate knowledge of mechanism of electrical conduction in electro active polymers
2. Analyse the properties of conducting polymers with respect to the structure and choose conducting polymers for specific applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Electronic states, structural disorder, tight-binding theory, band structure of c-c bonds, band gap, pendant groups and electronic states, electron-hole interaction, molecular- ion model, photoemission, intrinsic bulk states, extrinsic bulk state, energy level diagram.	9
2	Charge carriers in semiconductors, dopant atoms and energy levels, donors and acceptors, probability function, charge neutrality, position of fermi energy level, fermi-dirac integral. Charging techniques, charge measuring techniques, interpretation of real charge and dipole polarization, concepts of electronic energy transfer, interpretation and measurement of electronic energy transfer.	9
3	Piezoelectricity of polypeptides, formation of electrets, structure of PVDF electrets, piezo and pyroelectric properties of polar crystals, properties of ceramic composites, piezoelectric relaxation. Applications of conducting polymers as batteries, light emitting diodes, sensors, electro optic devices,	9

	microwave and electrochromic devices, membranes, corrosion protection, imaging and lithography	
4	Solubility of conducting polymers, solvents, soluble and sulfonated conducting polymers, processing and coating by insitu polymerisation, melt and heat processability, colloidal solution and precursor methods, L-B films, direct vapour deposition, other deposition methods	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the principles of electrical conduction in polymer materials.	K2
CO2	Explain the significance of electrical charge storage and energy transfer.	K2
CO3	Identify the applications of conducting polymers	K3
CO4	Determine the properties and synthesis of electroactive polymers	K5
CO5	Illustrate different techniques to impart conductivity in polymers	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											
CO5	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Springer Handbook Of Electronic And Photonic materials.	Kasap, Safa, and Peter Capper	Springer	2017.
2	Handbook of polymers in electronics.	Malhotra, Banshi D	ISmithers Rapra Publishing	2001
3	Handbook of Advanced Electronic and Photonic Materials and Devices, Ten-Volume Set.	Nalwa, Hari Singh, ed	Vol. 1. Academic Press,	2000.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Conducting polymers	Yilmaz, Faris.		2016
2	Electrical Conductivity in Polymer-Based Composites: Experiments, Modelling, and Applications	Taherian, Reza, and Ayesha Kausar	William Andrew	2018.
3	Conjugated conducting polymers.	Kiess, Helmut G., and Dionys Baeriswyl	Vol. 102. Berlin: Springer-Verlag	1992
4	Conductive polymers and plastics.	Margolis, James	Springer Science & Business Media	2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://www.youtube.com/watch?v=nSAvyQajVzE&t=23s https://www.youtube.com/watch?v=ygtbo5KDXeI https://www.youtube.com/watch?v=9J9OVISUol4 Prof.Abhijit P Deshpande, IIT Madras

SEMESTER: S8

PLASTICS & DECORATION

Course Code	PEPOT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Impart the basic concepts of decoration
2. Develop knowledge on recent advances in decoration

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Surface treatment for plastics and steel, thermodynamics of wetting, methods of wetting of surfaces, chemical analysis of polymer surfaces, decorating plastics with colour, colour selection criteria, colorant classification, pigment dispersion, colour concentrators Painting on plastics, modifications on plastic part surfaces, plastic surface conductivity aerosol paints, application of multiple paint films, plastic film -substrate interactions, measurement of quality of painted films	9
2	Plating on plastics, electro less copper or nickel plating, electroplating, straight through electroplating, trivalent chromium electroplating, microporous chrome plating, plating chemical removal methods, palatable plastics, individual plastic character to plating, design factors	9
3	Thermal spray coating on plastics, flame spraying, arc spraying, plasma, cosmetics, priming, priming with sprayed zinc, steps in plating, finishing methods, sealing surfaces and applications.	9
4	Vacuum metallizing, metallizing equipment, coating application, drying, equipment for drying, process control of metallizing, plastic hot stamping,	9

	foils, silicone dies for stamping, hot stamping machines, vertical presses, linear contact machines.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

At the end of the course students should be able to:

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

[illegible]

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automotive paint hand book: paint technology for auto enthusiasts& body shop professionals.	P fanstiehl, John	Penguin,	1998
2	Paint hand book	Weismantel GuyE.	McGraw-Hil	1981
3	In-Mould decoration of plastics	Love, J. C.,and Vannessa Goodship	Rapra	2002.
4	Decoration and Assembly of Plastic Parts	Muccio, Edward A	ASM International	1999

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Plastics: Surface and finish	Pinner, Solomon Hand Wilfred GordonS impsn	Elsevier,	2013
2	Principles of manufacturing materials and processes	Campbell, James S	McGraw-Hill,	1961
3	Innovation Trends in Plastics Decoration and Surface Treatment	Crutchley, Edward B	Smithers Rapra,	2014
4	Decorating and coating of plastics	Robinson,P.J	Rapra	1993

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.machinemfg.com/types-of-surface-treatment-process/
2	https://youtu.be/Ny6LcgV5qJY
3	https://youtu.be/kOoRuIl03YU
4	https://youtu.be/zoBJHLLBFek

SEMESTER: S8

TECHNOLOGY OF PRINTING INKS

Course Code	PEPOT866	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Knowledge About Paints	Course Type	Theory

Course Objectives

1. The student will be able to explain the manufacturing steps of printing inks with each specialisation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Nature of printing Inks, drying characteristics, adhesive nature, resistance properties, printing processes, letterpress, lithographic, gravure, screen printing, non-impact printing, substrate selection, print recognition.	9
2	Raw materials for printing Inks, pigments, dyestuffs, Oils, resins, solvents, plasticisers, driers, chelating agents, antioxidants, deodorants and reodorants, deforming agents, laking agents Formulating principles of letterpress inks, lithographic inks, gravure inks, flexographic inks, screen inks, radiation curable systems, inks for special purposes.	9
3	Manufacture of inks and varnishes, processes, mixing equipment, milling equipment, handling, storage and manufacture of UV inks, manufacture of newspaper inks, handling and storage of inks, modern production trends.	9
4	Rheology of printing inks, viscosity, powerlaw, Newtonian and Non Newtonian fluids, tack and tack measurements, dry paint performance tests,	9

	analysis of printing inks, safety measures in handling printing chemicals, mechanical and operational aspects, international standards and national standards in ink manufacturing area, environmental aspects.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the nature of printing Inks.	K2
CO2	Distinguish the composition of different inks.	K4
CO3	Identify the formulation of inks of different applications	K3
CO4	Inspect the techniques for manufacture of inks.	K4
CO5	Evaluate inks based on environmental and standard tests aspects.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											
CO2	3											
CO3	3											
CO4	3											
CO5	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Printing materials: Science and Technology.	Thompson, Robert	Pira International	2004.
2	The chemistry of inkjet inks	Magdassi, Shlomo,	. World scientific,	2009.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nanomaterials for 2D and 3D Printing	Magdassi, Shlomo, and Alexander Kamyshny,	John Wiley & Sons	2017
2	Printing on polymers: fundamentals and applications	Izdebska-Podsiadły, Joanna, and Sabu Thomas	. William Andrew,	2015
3	"Optimization of dispersion and viscosity of a ceramic jet printing ink."	Teng, Wan D., Mohan J. Edirisinghe, and Julian RG Evans	Journal of the American Ceramic Society 80, no. 2	1997

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-4	https://www.youtube.com/watch?v=IYo5WGTPahQ IIT Delhi https://www.youtube.com/watch?v=hUEHPGACW4Q IIT Delhi

SEMESTER: S8

FAILURE ANALYSIS OF POLYMERS

Course Code	PEPOT867	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Previous Coursework in Materials Failure Analysis, Fracture Mechanics, Or Polymer Degradation.	Course Type	Theory

Course Objectives:

1. After the completion of this course, students should achieve sufficient knowledge about the fundamental principles and mechanisms of polymer failure, including degradation, fracture, and fatigue

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Fracture – molecular failure process. Failure modes: brittle failure, ductile failure, ductile-brittle transitions, toughened polymers, mechanisms of rubber toughening, failure of highly filled systems/composites. Fracture mechanics: linear elastic fracture mechanics (LEFM), fracture predictions based on stress concentration and stress intensity factor, fracture predictions based on an energy balance	9
2	Fracture Mechanics – Short term and long-term test methods and analysis: impact strength, impact failure, creep rupture, fatigue, friction and wear	9
3	Failure analysis and degradation of polymers – Identification of strategic weakness, stability of polymer structures, environmental effect on polymer failure - weather, thermal, photo and oxidation	9
4	Degradation Mechanisms - Ionizing radiation, hydrolysis, environmental stress cracking-crazing and cracking, electrical treeing and water treeing,	9

	chemical degradation, electrochemical degradation, biodegradation, physical ageing.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Illustrate the different modes of failure occurring in polymers and composites	K2
CO2	Explain the fracture mechanics during impact tests and in long-term tests	K2
CO3	Evaluate the effect of various environmental factors on the degradation behaviour of polymers	K5
CO4	Analyze the environmental stress cracking phenomenon in polymers	K4
CO5	Discuss degradation mechanisms in polymers	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1			1						1	1
CO2	2	1		3		1						1
CO3	2	1			2							1
CO4	2	1		2							2	1
CO5	1	3		1		1	1					1

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Failure of Plastics and Rubber Products	David C. Wright	RAPRA Technology Ltd	2001
2	Material Science of Polymers for Engineers	Tim A. Osswald, Georg Menges	Hanser Publications	3rd Edn 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Polymers	R. J. Young, P. A. Lovell	CRC Press	3rd Edn 2011
2	Plastics Engineering	R.J. Crawford	Elsevier Butterworth- Heinmann	3rd Edn 2002
3	Handbook of plastics testing and failure analysis	Shah, V.	John Wiley & Sons	2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/ZoIVHEgV164?si=Sn_73C1xh0tO_Xlw
2	https://youtu.be/WJydAz0g1nM?si=8aXeeKxQgLhmcpCb
3	https://youtu.be/uQuj77gRTbo?si=Ya5c0upxG-v-TJA-
4	https://youtu.be/Db9OCPIgZRY?si=XJznldG1RiZi4I3j

SEMESTER: S8

ENERGY TECHNOLOGY

Course Code	PEPOT 868	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2Hr. 30 Min.
Prerequisites (if any)	Nil	Course Type	Theory

Course Objectives:

1. Understand Energy Fundamentals and Global Energy Dynamics.
2. Explore Advanced Energy Storage Systems
3. Investigate Hydrogen and Fuel Cell Technologies

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Energy: Definition and units of energy, power, Forms of energy, Conservation of energy, Energy flow diagram to the earth. Origin of fossil fuels, time scale of fossil fuels, Renewable Energy Resources, Role of energy in economic development and social transformation. Global Energy Scene Energy consumption in various sectors, projected energy consumption for the next century, exponential increase in energy consumption, energy resources, coal, oil, natural gas, nuclear power and hydroelectricity, impact of exponential rise in energy consumption on global economy, future energy options.	9
2	Energy Storage Systems	9

	Electrochemical Energy Storage Systems: Batteries- Primary, Secondary, Lithium, Solid-state and molten solvent batteries; Lead acid batteries; Nickel Cadmium Batteries; Advanced Batteries. Role of carbon nano-tubes in electrodes. Magnetic and Electric Energy Storage Systems: Superconducting Magnet Energy Storage (SMES) systems; Capacitor and Batteries: Comparison and application; Super capacitor: Electrochemical Double Layer Capacitor(EDLC), principle of working, structure, performance and application, role of activated carbon and carbon nano-tube.	
3	Hydrogen Technology and Fuel Cell Technology Hydrogen Production Methods: Production from fossil fuels, electrolysis, thermal decomposition, photochemical, photocatalytic, hybrid. Hydrogen Storage Methods: Metal hydrides, Metallic alloy hydrides, Carbon nano-tubes; Sea as the source of Deuterium, Its merit as a fuel and Applications Fuel Cell: Basics Fuel cell definition, difference between batteries and fuel cells, components of fuel cells, principle of working of fuel cell, performance characteristics of fuel cells, efficiency of fuel cell, fuel cell stack, fuel cell power plant: fuel processor, fuel cell power section, power conditioner, Advantages and disadvantages of fuel cell power plant.	9
4	The New Energy Technologies Introduction:Development in the field of superconductivity, Basic parameters of superconductivity, Types of superconductors, BCS theory, Meissner Effect, Josephson effect in Superconductors. High Tc Superconductors: Cuprate Superconductors; La, Y, Bi, Tl and Hg based superconductors, Intermetallic MgB₂superconductor crystal structure and superconducting properties, Applications of Superconductors in Energy.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
● 2 Questions from each module. ● Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	● Each question carries 9 marks. ● Two questions will be given from each module, out of which 1 question should be answered. ● Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts of energy, including its forms, units, and conservation principles, and analyze global energy consumption patterns and their economic impact	K4
CO2	Describe various renewable and non-renewable energy resources, their origins, and their roles in economic and social development.	K2
CO3	Evaluate different electrochemical, magnetic, and electric energy storage systems, including advanced batteries and supercapacitors, in terms of their principles, performance, and applications.	K5
CO4	Explain the methods of hydrogen production and storage, and differentiate between batteries and fuel cells in terms of components, working principles, and efficiency.	K4
CO5	Assess the developments and applications of superconductors in energy technology, including their basic parameters, types, and specific properties.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3						2					
CO3			3		2							
CO4		3	2									
CO5					3	2						

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Energy Technology: Non-Conventional, Renewable and Conventional	S. Rao, B. B. Parulekar	Khanna Publishers	4th Edition, 2005
2	Renewable Energy Resources	John Twidell, Tony Weir	Routledge	3rd Edition, 2015
3	Energy Systems Engineering: Evaluation and Implementation	Francis Vanek, Louis Albright, Largus Angenent	McGraw-Hill	3rd Edition, 2016
4	Fuel Cell Technology Handbook	Gregor Hoogers	CRC Press	1st Edition, 2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Energy Storage Systems	A. G. Ter-Gazarian	Institution of Engineering and Technology	1st Edition, 1994
2	Hydrogen and Fuel Cells: Emerging Technologies and Applications	Bent Sørensen	Academic Press	1st Edition, 2005
3	Superconductivity: Physics and Applications	P. G. de Gennes, J. B. Goodenough	Springer	1st Edition, 1999
4	Modern Batteries: An Introduction to Electrochemical Power Sources	Colin A. Vincent, Bruno Scrosati	Butterworth-Heinemann	2nd Edition, 1997

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Link: " https://nptel.ac.in/courses/112/101/112101104/ "
2	Link: " https://nptel.ac.in/courses/108/108/108108034/ "
3	Link: " https://nptel.ac.in/courses/103/106/103106176/ "
4	Link: " https://nptel.ac.in/courses/115/108/115108121/ "

SEMESTER: S8

FUNCTIONAL POLYMERS & APPLICATIONS

Course Code	PEPOT865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic Knowledge in Polymers	Course Type	Theory

Course Objectives:

1. Students will learn about the different types of functional polymers, their properties, and their applications in various fields

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Functional Polymers: - Definition and classification of functional polymers, Importance of functional polymers, Various synthesis methods- -step-growth polymerisation, chain-growth polymerisation, ring-opening polymerisation	9
2	Polymer Functionalisation: Chemical modification of polymers with suitable examples, Grafting of functional groups with examples, blockcopolymerisation with examples. Properties of functional polymers-mechanical, thermal, electrical and optical	9
3	Applications of Functional Polymers: biomedical applications, drug delivery systems, water treatment, catalysis, electronics, coatings and adhesives. Types of functional polymers with suitable examples- Hydrogels, smart polymers, conductive polymers, biodegradable polymers, shape memory polymers, self-healing polymers, water soluble polymers, stimuli responsive polymers.	9

4	Characterisation of Functional Polymers- using microscopic techniques, spectroscopic techniques, chromatographic techniques and thermal analysis. Modelling and simulation of functional polymers using molecular dynamics simulations, Monte Carlo simulations and finite element modelling.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

1. **Problem Definition (2 points)**
 - Check for clarity and completeness of the problem definition.
 - Check for clear objectives and divide into sub objectives
2. **Problem Analysis (4 points)**
 - Examine the problem and identify the variables and responses
 - Divide the problem to manageable parts
3. **Implementation of software (2 points)**
 - Use suitable software for getting the solution
 - Check for the appropriateness of tools used
4. **Evaluate (4 points)**
 - Evaluate the proposed solution.
 - Assess the correctness of solution.
5. **Validation of the results (4 points)**
 - Validate the results using available data.

- Do the error analysis of the results obtained.

6. **Conclusion and presentation (4 points)**

- Summarize the results obtained
- Do critical thinking of the results obtained and evaluate.
- Submit a technical report

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the importance of functional polymers	K5
CO2	Discuss the various synthetic ways of preparing functional polymers	K6
CO3	Summarize the properties and modifications of functional polymers	K2
CO4	Explain the applications of functional polymers	K2
CO5	Inspect the various characterisation techniques of functional polymers	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3					1						
CO2	3											
CO3	3		2		2							
CO4	3		2									
CO5	3											

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Polymer Science and Engineering	Fred W. Billmeyer Jr		1971
2	"Introduction to Polymer Science and Chemistry" by	Manas Chanda		2013
3	"Chromatography: Fundamentals and Applications of Chromatography and Related Differential Migration Methods"	by Christian G. Fraga	Elsevier	2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Smart Hydrogels: Properties, Synthesis, and Applications"	Liang Guo	Wiley	2020
2	"Thermal Characterization of Polymer Materials"	A. R. Horrocks	Woodhead Publishing	2015
3	"Biomedical Applications of Polymers and Composites" by	V. K. Thakur	Elsevier	2018

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1-2	https://www.youtube.com/watch?v=54urJPOnaeU&list=PLyqSpQzTE6M_KQ5MqUkoOqAxxOrdvFOMB
3	https://www.youtube.com/watch?v=xX5AWl6QcWU
4	https://www.youtube.com/watch?v=zMosmn5Sn-U